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R.M.D. ENGINEERING COLLEGE

22EC909 OPTICAL COMMUNICATION AND NETWORKING

Department	:	**ECE**
Batch/Year	:	**2022-2026/III Year**
Created By	:	**R.M SENTHIL PRIYA**

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1. COURSE OBJECTIVES:

- To acquire the knowledge of optical fiber transmission mechanisms and various fiber types.
- To study the factors which produce signal degradation in fibers.
- To learn the concept of optical sources and optical detectors
- To familiarize and study the power coupling in optical communication
- To gain knowledge on fiber splicing techniques
- To explore the trends of optical fiber measurement systems.
- To enrich the idea of optical fiber networking.

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2. Pre Requisites:

22EC403-Analog and Digital Communication
22EC201-Electronics Devices and Circuits

3. SYLLABUS
22EC909 OPTICAL COMMUNICATION AND NETWORING

L T P C 3 0 0 3

UNIT I INTRODUCTION TO OPTICAL FIBERS 9

Elements of an Optical Fiber Transmission link-Basic Optical Laws and Definitions-Total internal reflection, Acceptance angle, Numerical aperture, Skew rays - Optical fiber modes and Configurations - Single mode fibers-Graded Index fiber structure –Mode theory of Circular waveguides-Overview of modes, Modes in Step-Index fibers, Linearly Polarized modes.

UNIT II SIGNAL DEGRADATION IN OPTICAL FIBERS 9

Attenuation - Absorption, Scattering losses, Bending losses, Core and Cladding losses. Signal distortion in Optical Wave guides- Group delay, Material dispersion, Waveguide dispersion, Signal distortion in SM fibers, Polarization mode dispersion, Intermodal dispersion – Dispersion Optimization of SM Fibers Characteristics of RI profiles and cut-off wavelength.

UNIT III FIBER OPTICAL SOURCES AND COUPLING 9

Direct and indirect band gap materials-LED structures -Light source materials -Quantum efficiency and LED power, Modulation of a LED. Lasers diodes-modes and Threshold condition -Rate equations - External quantum efficiency -Resonant frequencies - Temperature effects. Introduction to Quantumlaser. Power launching and coupling-Lensing schemes-Fiber -to-Fiber joints-Fiber splicing.

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UNIT IV FIBER OPTIC RECEIVER AND MEASUREMENTS 9

Principles of Photodetectors – PIN & APD - Fundamental receiver operation- Receiver configuration– Digital receiver performance- Probability of error – Quantum limit, Pre amplifiers. Fiber attenuation measurements- Dispersion measurements – Fiber refractive index profile measurements– Fiber diameter measurements.

UNIT V OPTICAL NETWORKS AND SYSTEM TRANSMISSION 9

Basic networks – SONET / SDH – Broadcast and select WDM networks – Wavelength routed networks – Link power budget -Rise time budget- Operational principles of WDM and EDFA system – Solitons – Optical CDMA – Ultra high-capacity networks- Introduction to Li-Fi and LIDAR.

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UNIT 3
FIBER OPTICAL SOURCES AND COUPLING

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4. COURSE OUTCOMES

At the end of the course, the student should be able to:

COURSE OUTCOMES	COURSE OUTCOMES	LEVELS IN BLOOMS TAXONOMY
C306.1	Describe basic elements in optical fibers, different modes and configurations	K2
C306.2	Summarize the transmission characteristics associated with dispersion and polarization techniques	K2
C306.3	Illustrate the Characteristics of various fiber optical sources.	K3
C306.4	Comprehend the optical receiver performance and measure various fiber parameters for designing optical fiber.	K3
C306.5	Elucidate the Characteristics of various fiber optical detectors.	K3
C306.6	Estimate the power budget required for optical network design and improve the performance of WDM/EDFA system.	K3

5. CO- PO/PSO Mapping

COURSE OUTCOMES:
After successful completion of the course, the students should be able to

Course Out Comes	Level of CO	Program Outcomes												Program Specific Outcomes		
		K3	K4	K4	K5	K3,K5, K6	A3	A2	A3	A3	A3	A3	A2	K6 PSO-1	K5 PSO-2	K6 PSO-3
		PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12			
C306.1	**K2**	**2**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**1**
C306.2	**K2**	**2**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**1**
C306.3	**K3**	**2**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**1**
C306.4	**K3**	**3**	**2**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**2**
C306.5	**K3**	**3**	**2**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**2**
C306.6	**K3**	**3**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**1**
C306		3	2	2	2									-	1	2

Enter correlation levels as defined below:
1.Slight (Low) 2. Moderate (Medium) 3. Substantial (High)
If there is no correlation, put "-".

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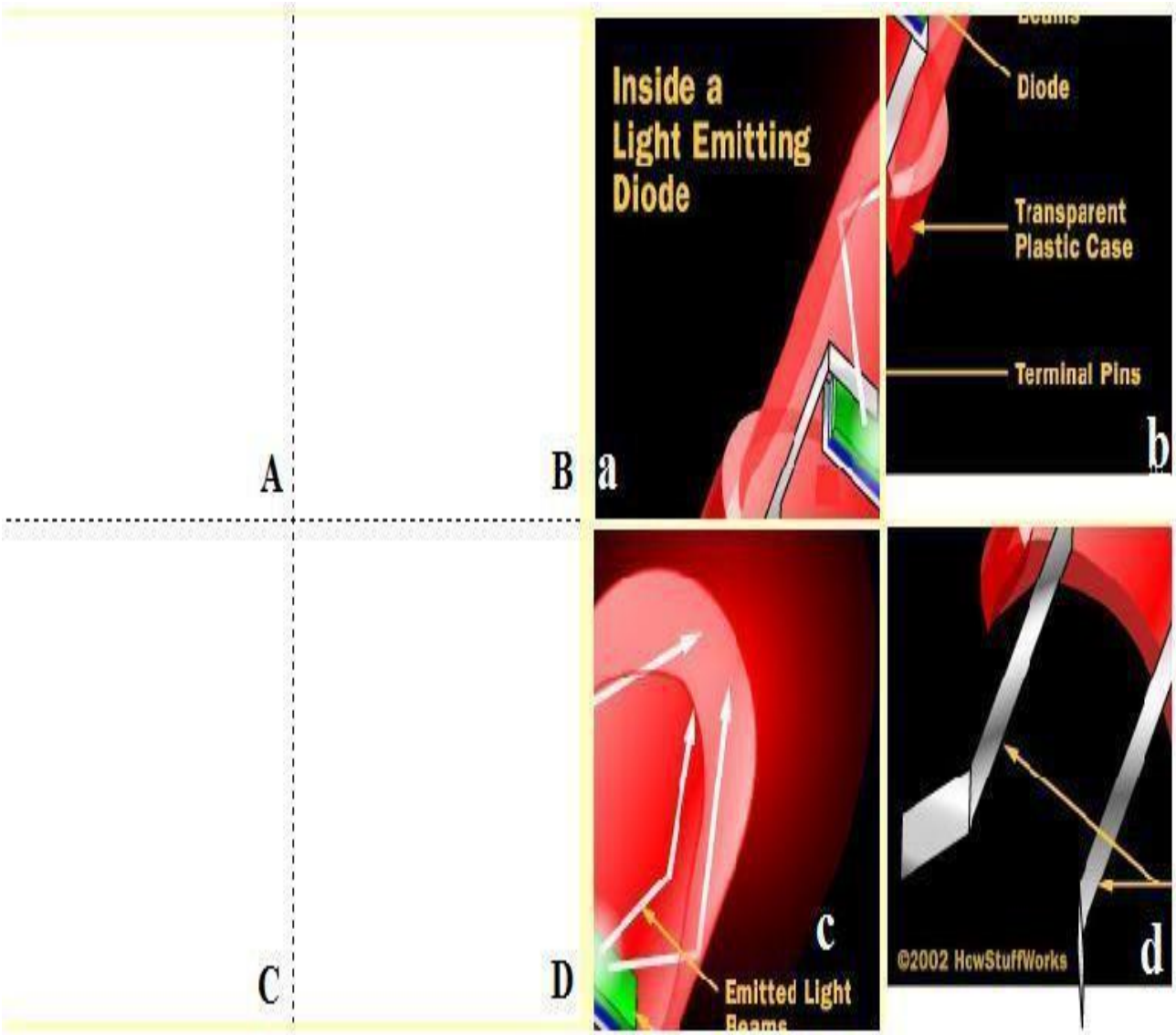
6.1.LECTURE PLAN

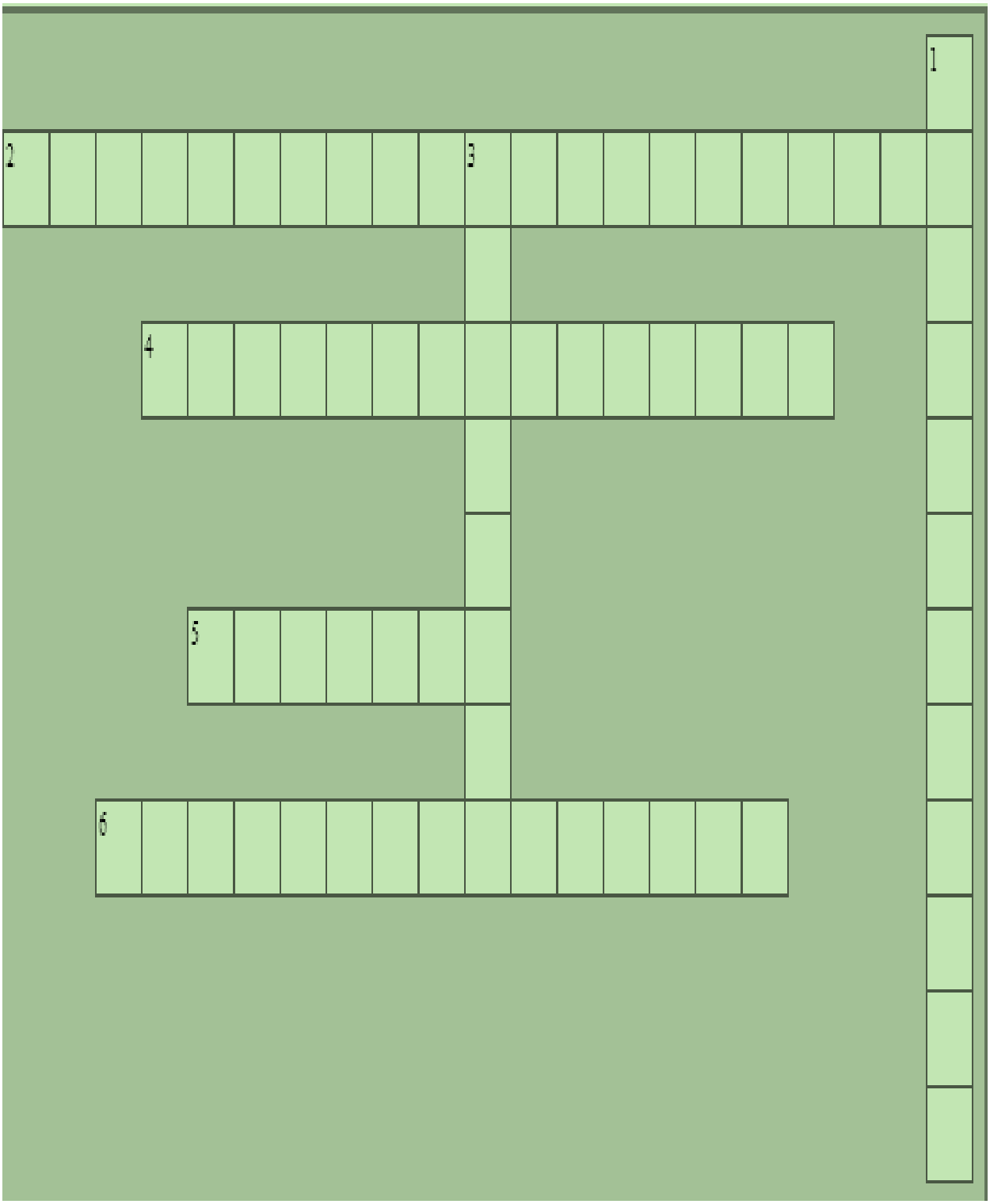
22EC909	OPTICAL COMMUNICATION AND NETWORKING						L T P C 3 0 0 3
	UNIT III OPTICAL SOURCES AND DETECTORS						
S.No.	Topic	No. of Periods	Proposed date	Actual Lecture Date	Pertaining CO	Taxonomy level	Mode of Delivery
1	Sources: Intrinsic and extrinsic material Direct and indirect Band gap materials	1			CO3	K2	MD1
2	LED structures, Surface emitting and Edge emitting LED	1				K2	MD1
3	Light source materials ,Quantum efficiency and LED power, Modulation of a LED	1				K2	MD2
4	lasers Diodes, Modes and Threshold condition, Rate equations	1				K2	MD1
	External Quantum efficiency ,Resonant frequencies	1				K2	MD1
6	Structures and radiation patterns, single mode laser, external modulation, temperature effort.	1				K2	MD1&2
7	Detectors: PIN photo detector, Avalanche photo diodes	1				K2	MD1
8	Photo detector noise, noise sources, SNR, detector response time	1				K2	MD
9	Avalanche multiplication noise, temperature effects, comparisons of photo detectors.	1				K2	MD1

6.2.Activity based learning – Chart-UNIT III

Arrange the scrambled picture:



6.2.Activity based learning –UNIT III cross word



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6.2. Activity Based Learning – UNIT III-CROSS WORD

Across

2. both carrier & optical confinement can be obtained using this structure

4. LED's are best choice

5. An indirect band gap material

6. The plane of the emitting region is oriented perpendicular to the fiber axis.

Down

1. Time delay between the application of current pulse & the onset of optical emission

3. Optical power radiated into a unit solid angle /unit area.

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UNIT III Quiz Questions

1) In the dynamic response of Injection Laser Diode (ILD), the delay which is followed by_____frequency damped oscillations give rise to the generation of relaxation oscillations.

a. Low
b. Medium
c. High
d. All of the above

2) Which type of injection laser involves the use of geometry for fabrication of the multimode injection laser with a single or small number of lateral modes?

a. Gain guided laser
b. Index guided laser
c. Quantum well laser
d. Quantum dot laser

3) Which modes are acknowledged due to their association with electromagnetic field and beam profile in the direction perpendicular to the plane of pn junction?

a. Longitudinal Modes
b. Transverse Modes
c. Lateral Modes
d. All of the above

4) Which among the following characteristics of Laser light specifies the precise movement of all individual light waves together through time and space?

a. Monochromatic
b. Directional
c. Coherent
d. Brightness

5) In a laser structure, the existence of standing waves is possible at frequencies for which the distance between the mirrors is an integral number of _____

a. $\lambda / 2$
b. $\lambda / 4$
c. $\lambda / 6$
d. $\lambda / 8$

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UNIT III Quiz Questions

6)Consider a crystal of ruby laser whose length is 6 cm and the refractive index is 1.8, emits the wavelength of about 0.55 μm. What will be the value of number of longitudinal modes?

a. 3.9×10^5
b. 4.9×10^5
c. 5.6×10^5
d. 7.7×10^5

7) In Stimulated Emission, which among the following parameters of generated photon is/are similar to the photon of incident wave?

a. Phase
b. Frequency
c. Polarization & direction of travel
d. All of the above

8) Which among the following is a key process adopted for the laser beam formation as it undergoes the light amplification?

a. Spontaneous Emission
b. Stimulated Emission
c. Both a and b
d. None of the above

9) In spontaneous emission, the light source in an excited state undergoes the transition to a state with _____

a. Higher energy
b. Moderate energy
c. Lower energy
d. All of the above

10) In the fiber optic link, power transfer from one fiber to another and from fiber to detector must take place with_____coupling efficiency.

a. maximum
b. stable
c. minimum
d. unpredictable

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UNIT III Quiz Questions

11. In Lambertian output pattern of LED, the source is_____bright from all directions.

a. Less

b. Equally

c. More

d. Unpredictably

12. In pyroelectric photodetectors, the consequent increase in dielectric constant due to temperature variation by the photon absorption, is generally measured as change in _____

a. resistance
b. inductance
c. admittance
d. capacitance

13. Which among the following controls the length of Fabry-Perot interferometer so that it can act as a tunable optical filter?

a. Transducer
b. Tachometer
c. Multimeter
d. Phase-meter

14. Which photodiodes are crucially applicable to overcome the bandwidth-quantum efficiency trade-off along with its resemblence to the pyramid structure?

a. Mushroom Waveguide Photodiode
b. Traveling Wave Photodiode
c. Resonant Cavity Photodiode
d. All of the above

15. When an optical signal is incident on a photo-detector, which noise originate/s due to statistical nature of production and collection of photoelectrons?

a. Dark Current Noise
b. Quantum Noise
c. Surface Leakage Current noise
d. All of the above

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UNIT III Quiz Questions

16. On which factor/s do/does the response time of photodiode depend/s?

a. Diffusion time of photo carriers outside the depletion region
b. Diffusion time of photo carriers within the depletion region
c. RC time constant
d. All of the above

17. Which nature of charge carriers give rise to the current fluctuations thereby resulting into the generation of shot noise?

a. Continuous
b. Discrete
c. Sampled
d. All of the above

18. A device which converts electrical energy in the form of a current into optical energy is called as _____

a) Optical source
b) Optical coupler
c) Optical isolator
d) Circulator

19. How many types of sources of optical light are available?
a) One
b) Two
c) Three
d) Four

20. The frequency of the absorbed or emitted radiation is related to difference in energy E between the higher energy state E_2 and the lower energy state E_1. State what h stands for in the given equation?

$E = E_2 - E_1 = hf$

a) Gravitation constant

b) Planck's constant

c) Permittivity

d) Attenuation constant

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UNIT III Quiz Questions

21. The radiation emission process (emission of a proton at frequency) can occur in _____ways.

a) Two

b) Three

c) Four

d) One

22. Which process gives the laser its special properties as an optical source?

a) Dispersion

b) Stimulated absorption

c) Spontaneous emission

d) Stimulated emission

23. An incandescent lamp is operating at a temperature of 1000K at an operating frequency of 5.2×1014 Hz. Calculate the ratio of stimulated emission rate to spontaneous emission rate.

a) 3×10-13

b) 1.47×10-11

c) 2×10-12

d) 1.5×10-13

24. The lower energy level contains more atoms than upper level under the conditions of _____

a) Isothermal packaging

b) Population inversion

c) Thermal equilibrium

d) Pumping

25. _____in the laser occurs when photon colliding with an excited atom causes the stimulated emission of a second photon.

a) Light amplification

b) Attenuation

c) Dispersion

d) Population inversion

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UNIT III Quiz Questions

26.A ruby laser has a crystal of length 3 cm with a refractive index of 1.60, wavelength 0.43 μm. Determine the number of longitudinal modes.

a) 1×10^{2}
b) 3×10^{6}
c) 2.9×10^{5}
d) 2.2×10^{5}

27.A semiconductor laser crystal of length 5 cm, refractive index 1.8 is used as an optical source. Determine the frequency separation of the modes.
a) 2.8 GHz
b) 1.2 GHz
c) 1.6 GHz
d) 2 GHz

28. Doppler broadening is a homogeneous broadening mechanism.
a) True
b) False

29. An injection laser has active cavity losses of 25 cm^{-1} and the reflectivity of each laser facet is 30%. Determine the laser gain coefficient for the cavity it has a length of 500μm.
a) 46 cm^{-1}
b) 51 cm^{-1}
c) 50 cm^{-1}
d) 49.07 cm^{-1}

30. Longitudinal modes contribute only a single spot of light to the laser output.
a) True
b) False

31. Considering the values given below, calculate the mode separation in terms of free space wavelength for a laser. (Frequency separation = 2GHz, Wavelength = 0.5 μm)
a) 1.4×10^{-11}
b) 1.6×10^{-12}
c) 1×10^{-12}
d) 6×10^{-11}

32. P-n photodiode is forward biased.

a) True

b) False

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UNIT III Quiz Questions

33. The depletion region must be_____to allow a large fraction of the incident light to be absorbed in the device(photodiode).
a) Thick
b) Thin
c) Long
d) Inactive

34. The process of excitation of an electron from valence band to conduction band leaves an empty hole in the valence band and is called as _____
a) Detection
b) Absorption
c) Degeneration of an electron-hole pair
d) Regeneration of an electron-hole pair

35._____always leads to the generation of a hole and an electron.
a) Repulsion
b) Dispersion
c) Absorption
d) Attenuation

36. The electron hole pairs generated in a photodiode are separated by the _____
a) Magnetic field
b) Electric field
c) Static field
d) Depletion region

37. Electric field in the depletion region should be high.
a) True
b) False

38. The photocurrent of an optical detector should be _____
a) Less
b) More
c) Linear
d) Non-linear

39. How many types of optical detectors are available?
a) One
b) Four
c) Two
d) Three

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6.3. LECTURE NOTES
UNIT III FIBER OPTICAL SOURCES AND COUPLING

3.1 Introduction

- An optical source is the key component of the optical transmitter unit. The purpose of an optical source is to convert an electrical signal reliably into optical radiation (E/O conversion). There are a variety of optical sources that convert electrical energy to an optical signal (light).
- There are two types of semiconductor optical source e.g. Light Emitting Diode (LED) and Injection Laser Diode (ILD) often referred to as Laser Diode (LD). **An optical source should ideally meet the following requirements for use in the transmitter unit of an optical fiber communication system.**

i. The emitted light should be preferably directive for easy launching of light from the source to the fiber

ii. The size of the emitting optical source must be compatible to the size of the optical fiber.

iii. The light output (optical power) must vary linearly with the electrical input for faithful E/O conversion.

iv. The source must have a reasonably high E/O conversion efficiency.

v. The emission wavelength should match with the attenuation window of the fiber (wavelength at which the fiber offers low attenuation).

vi. The spectral width of the source should be small in order to reduce chromatic dispersion during propagation through the fiber.

vii. The source should have a high modulation capability i.e. a large bandwidth in order to meet the large information carrying capacity of the fiber.

viii. The source must be able to couple sufficient optical power to the fiber so that they can travel a long distance and still deliver the required power to the detector for faithful conversion of optical signal into electrical signal (O/E conversion).

ix. The source should have moderately long life.

- Both LED and ILD meet the basic requirements of an optical source for use in optical fiber communication system. Structurally, both LED and ILD consist of pn-junction p-n made of direct bandgap semiconductor materials.
- When forward-biased electrons and holes are injected into p and n regions respectively where they recombine with the majority carriers. In direct bandgap material the recombination is normally radiative in nature giving rise to emission of photons (light).

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- The principle of operation of a laser diode differs significantly from that of an LED. The light from an LED results from spontaneous emission following the random radiative recombination of the carriers. On the other hand a laser diode works on the principle of stimulated emission which dominates only under special circumstances.
- One of the major differences between an LED and an ILD is that the optical output from the former is incoherent whereas that from the latter is coherent. This is because light in an ILD is produced in an optical resonator that ensures both spatial and temporal coherence of the light emanating from the cavity.
- The spatial coherence ensures that the output light is highly monochromatic whereas temporal coherence means that the output beam is highly directional. Since no cavity resonator is used in the case of LED, the light output generally exhibits a relatively large spectral width.

3.2.Intrinsic and Extrinsic Material

- A perfect material containing no impurities is called an intrinsic material. Because of thermal vibrations of the crystal atoms, some electrons in the valence band gain enough energy to be excited to the conduction band. This thermal generation process produces free electron–hole pairs because every electron that moves to the conduction band leaves behind a hole. Thus for an intrinsic material the number of electrons and holes are both equal to the intrinsic carrier density, In the opposite recombination process, a free electron releases its energy and drops into a free hole in the valence band.
- For an extrinsic semiconductor, the increase of one type of carrier reduces the number of the other type. In this case, the product of the two types of carriers remains constant at a given temperature. This gives rise to the **mass-action law**

$$pn = n_{i_2}$$

- Which is valid for both intrinsic and extrinsic materials under thermal equilibrium. Since the electrical conductivity is proportional to the carrier concentration, **two types of charge carriers are defined for this material:**

I. Majority carriers refer either to electrons in n-type material or to holes in p-type material.

II. Minority carriers refer either to holes in n-type material or to electrons in p-type material.

- The operation of semiconductor devices is essentially based on the injection and extraction of minority carriers.

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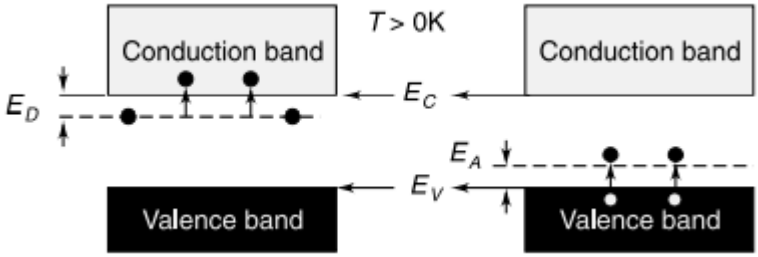
- The conductivity of an intrinsic semiconductor is generally low. The conductivity of the intrinsic semiconductor can be greatly varied by adding tiny and controlled amount of chemical impurity in the semiconductor.
- For example, if a Group-V element (say p) is added to silicon, the p atom containing five electrons in the outermost orbit will substitute an Si atom. Four of its five electrons in the outermost orbit would form covalent bonds with four Si atoms surrounding the p atom and the remaining one electron (called the fifth electron) will remain loosely bound to the nucleus of the p atom. With a small amount of thermal energy, this electron can be easily elevated to conduction band of the semiconductor. This type of impurity donates electron and is therefore known as **Donor impurity**.
- In other words by adding controlled amount of donor impurity it is possible to increase the number of electrons so that the total number of electrons will be much larger than the holes (created by thermal generation). This type of semiconductor in which the electrons are the majority carriers and holes are the minority carriers is called an **n-type semiconductor**.
- On the other hand, if we add a Group-III atom (say Al, in pure silicon), the three outermost electrons will form three complete covalent bonds with adjacent Si atoms when the Al atom occupies one of the Si sites. The covalent bond with the fourth Si atom would remain incomplete with a missing electron. This means that addition of Group-III atoms in Si would result in the generation of holes which would not be accompanied by creation of additional electrons. In this process, the total number of holes in an Si sample can be increased by increasing the concentration of Group-III atoms. The empty position in the incomplete bond is occupied by electrons and the impurity atom becomes negatively charged ions. As the Al dopant accepts electron, this type of impurity is called **Acceptor impurity.**
- Therefore, by adding a controlled amount of acceptor impurity in an intrinsic semiconductor it is possible to make the total number of holes exceed the number of electrons. This type of semiconductor is called **p-type semiconductor** in which holes are the majority carriers. The position of donor and acceptor levels and the formation of electrons and holes are shown in Fig. 3.1.



- **Fig.3.1.The position of donor and acceptor levels and the formation of electrons and holes**

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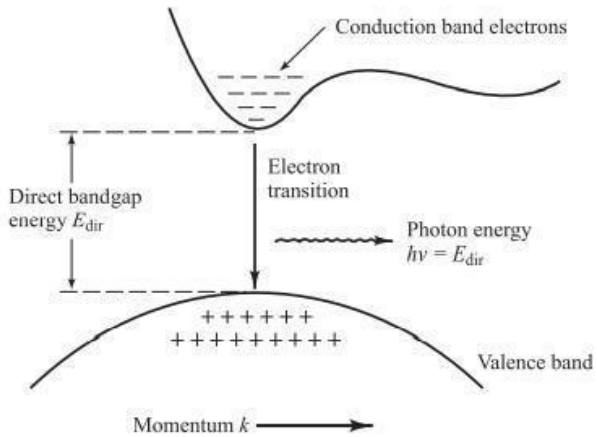
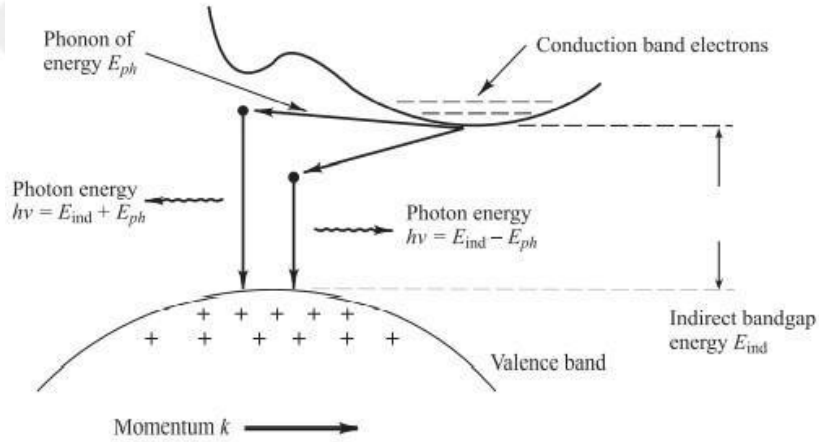
3.3 Direct and Indirect Bandgaps

- In order for electron transitions to take place to or from the conduction band with the absorption or emission of a photon, respectively, both energy and momentum must be conserved. Although a photon can have considerable energy, its momentum hv/c is very small.



(a)



(b)

- **Fig. 3.2 (a) Electron recombination and the associated photon emission for a direct-bandgap material; (b) electron recombination for indirect-bandgap materials requires a phonon of energy E_{ph} and momentum k_{ph}.**
- Semiconductors are classified as either direct-bandgap or indirect-bandgap materials depending on the shape of the bandgap as a function of the momentum k, as shown in Fig 3.2.

- Let us consider recombination of an electron and a hole, accompanied by the emission of a photon. The simplest and most probable recombination process will be that where the electron and hole have the same momentum value (see Fig. 3.2a). This is a direct-bandgap material.

- For indirect-bandgap materials, the conduction-band minimum and the valence-band maximum energy levels occur at different values of momentum, as shown in Fig 3.2b. Here, band-to-band recombination must involve a third particle to conserve momentum because the photon momentum is very small. Phonons (i.e., crystal lattice vibrations) serve this purpose.

4. Light-Emitting Diodes (LEDs)

- For optical communication systems requiring bit rates less than approximately 100–200 Mb/s together with multimode fiber-coupled optical power in the tens of microwatts, semiconductor light-emitting diodes (LEDs) are usually the best light source choice. These LEDs require less complex drive circuitry than laser diodes as no thermal or optical stabilization circuits are needed, and they can be fabricated less expensively with higher yields.

1. LED Structures

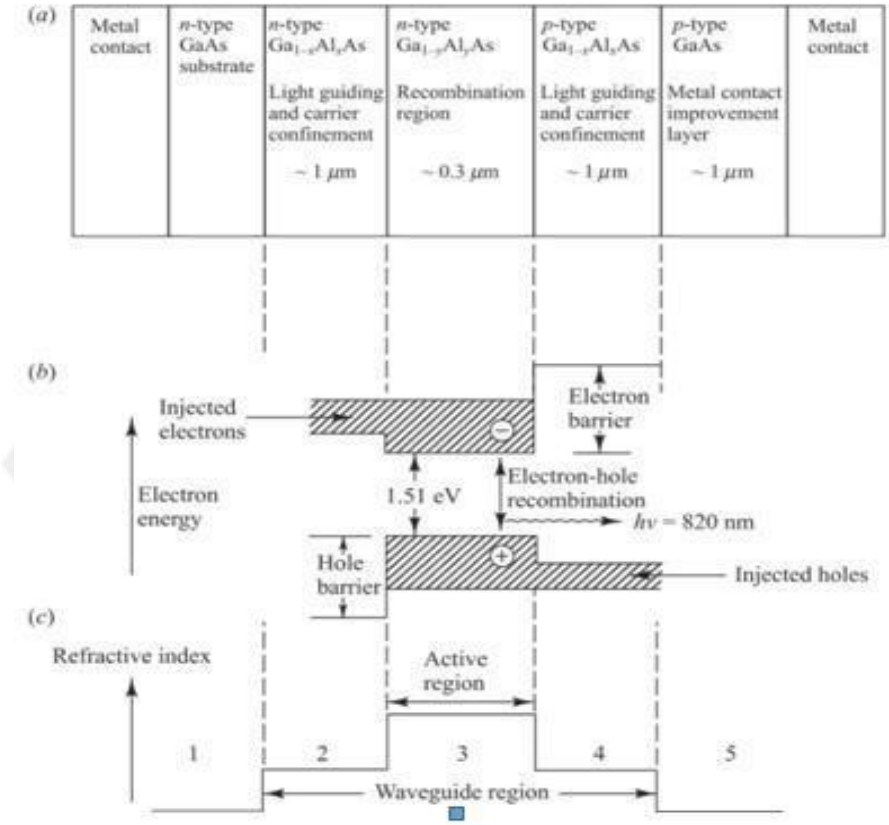
- An LED must have a high radiance output, a fast emission response time, and a high quantum efficiency to be useful in fiber transmission applications. Its radiance (or brightness) is a measure, in watts, of the optical power radiated into a unit solid angle per unit area of the emitting surface .

- High radiances are necessary to couple sufficiently high optical power levels into a fiber. The emission response time is the time delay between the application of a current pulse and the onset of optical emission , this time delay is the factor limiting the bandwidth with which the source can be modulated directly by varying the injected current. The quantum efficiency is related to the fraction of injected electron–hole pairs that recombine radiatively.

- To achieve carrier and optical confinement, LED configurations such as homojunctions and single and double heterojunctions have been widely investigated. The most effective of these structures is the configuration shown in Fig. 3.5. This is referred to as a double-heterostructure (or heterojunction) device because of the two different alloy layers on each side of the active region.

- This configuration evolved from studies on laser diodes. By means of this sandwich structure of differently composed alloy layers, both the carriers and the optical field are confined in the central active layer.

- The bandgap differences of adjacent layers confine the charge carriers (Fig 3.5b), while the differences in the indices of refraction of adjoining layers confine the optical field to the central active layer (Fig.3.5c). This dual confinement leads to both high efficiency and high radiance. Other parameters influencing the device performance include optical absorption in the active region (self-absorption), carrier recombination at the heterostructure interfaces, doping concentration of the active layer, injection carrier density, and active-layer thickness. We shall see the effects of these parameters in the following sections.



- **Fig. 3.3 (a) Cross-sectional drawing (not to scale) of a typical GaAlAs double-heterostructure light emitter. In this structure, x > y to provide for both carrier confinement and optical guiding; (b) energy band diagram showing the active region, and the electron and hole barriers that confine the charge carriers to the active layer; (c) variations in the refractive index; the lower index of refraction of the material in regions 1 and 5 creates an optical barrier around the waveguide region.**

3.4.2 Surface Emitting LED

- The two basic LED configurations being used for fiber optics are surface emitters (also called Burrus or front emitters) and edge emitters .In the surface emitter, the plane of the active light-emitting region is oriented perpendicularly to the axis of the fiber, as shown in Fig. 3.4. In this configuration, a well is etched through the substrate of the device, into which a fiber is then cemented in order to accept the emitted light.

- The circular active area in practical surface emitters is nominally 50 μm in diameter and up to 2.5 μ m thick. The emission pattern is essentially isotropic with a 120° half-power beam width.

- This isotropic pattern from such a surface emitter is called a lambertian pattern. In this pattern, the source is equally bright when viewed from any direction, but the power diminishes as cosΘ, where Θ is the angle between the viewing direction and the normal to the surface (this is because the projected area one sees decreases as cosΘ). Thus, the power is down to 50 percent of its peak when Θ = 60°, so that the total half-power beam width is 120°.

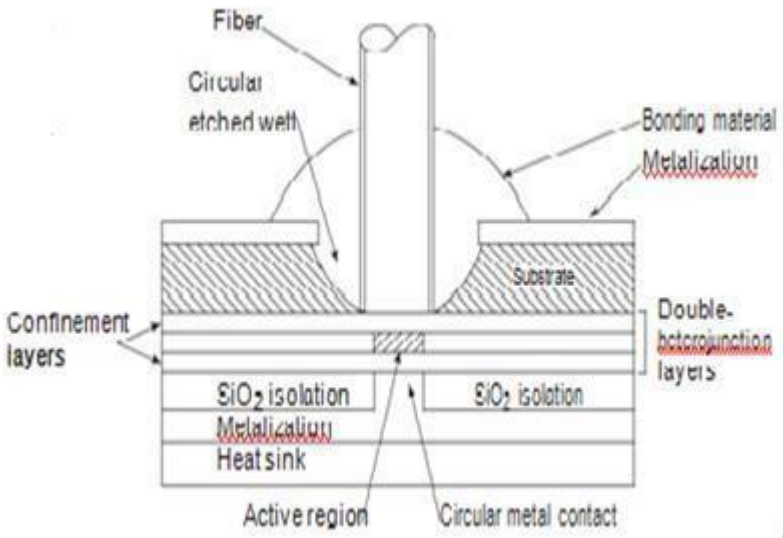


Fig. 3.4. Surface Emitting LED

3.4.3 Edge Emitting LED

- The edge emitter depicted in Fig. 3.5 consists of an active junction region, which is the source of the incoherent light, and two guiding layers. The guiding layers both have a refractive index lower than that of the active region but higher than the index of the surrounding material. This structure forms a waveguide channel that directs the optical radiation toward the fiber core.
- To match the typical fiber-core diameters (50–100 μm), the contact stripes for the edge emitter are 50–70 μ m wide. Lengths of the active regions usually range from 100 to 150 μm. The emission pattern of the edge emitter is more directional than that of the surface emitter, as is illustrated in Fig. 3.5.
- In the plane parallel to the junction, where there is no waveguide effect, the emitted beam is lambertian (varying as cos Θ) with a half-power width of $\Theta_{||}$ = 120°. In the plane perpendicular to the junction, the half-power beam $\Theta_{\perp}$ has been made as small as 25–35° by a proper choice of the waveguide thickness.

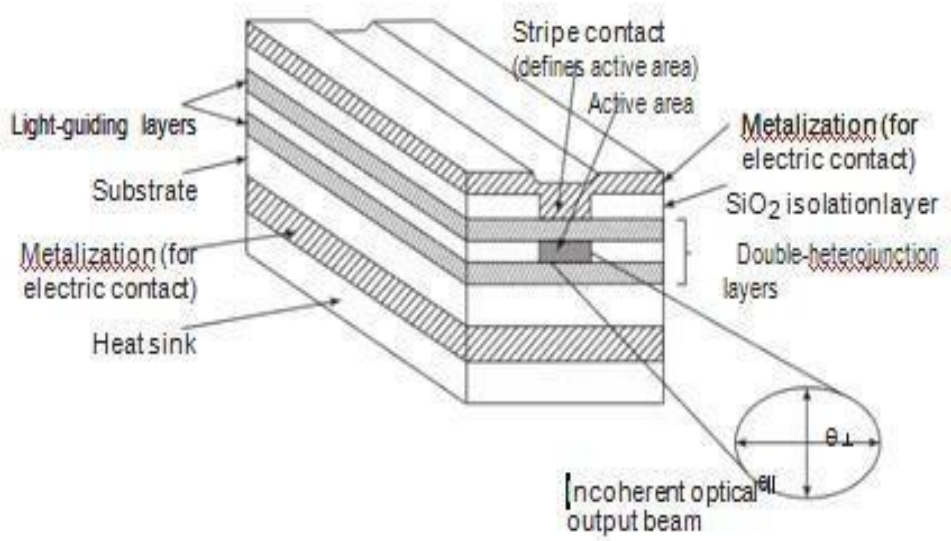


Fig. 3.5 An Edge-emitting Double-Heterojunction LED

3.4.4 Quantum Efficiency and LED Power

- An excess of electrons and holes in p- and n-type material, respectively (referred to as minority carriers) is created in a semiconductor light source by carrier injection at the device contacts.
- The excess densities of electrons n and holes p are equal, since the injected carriers are formed and recombine in pairs in accordance with the requirement for charge neutrality in the crystal. When carrier injection stops, the carrier density returns to the equilibrium value. In general, the excess carrier density decays exponentially with time according to the relation

$$n = n_0 e^{-t/\tau}$$

- Where n_0 is the initial injected excess electron density and the time constant t is the carrier lifetime. This lifetime is one of the most important operating parameters of an electro-optic device. Its value can range from milliseconds to fractions of a nanosecond depending on material composition and device defects.
- The excess carriers can recombine either radiatively or non-radiatively. In radiative recombination a photon of energy hv, which is approximately equal to the bandgap energy, is emitted. Non-radiative recombination effects include optical absorption in the active region (self-absorption), carrier recombination at the heterostructure interfaces, and the Auger process in which the energy released during an electron–hole recombination is transferred to another carrier in the form of kinetic energy.
- When there is a constant current flow into an LED, an equilibrium condition is established. That is, the excess density of electrons n and holes p is equal since the injected carriers are created and recombined in pairs such that charge neutrality is maintained within the device.
- The total rate at which carriers are generated is the sum of the externally supplied and the thermally generated rates.
- The externally supplied rate is given by J/qd, where J is the current density in A/cm^2, q is the electron charge, and d is the thickness of the recombination region. The thermal generation rate is given by n/r.
- Hence, the rate equation for carrier recombination in an LED can be written as

$$\frac{dn}{dt} = \frac{J}{qd} - \frac{n}{\tau} \qquad .....(1)$$

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- The equilibrium condition is found by setting above equation equal to zero, yielding

$$n = \frac{J\tau}{qd} \qquad \text{.....(2)}$$

- This relationship gives the steady-state electron density in the active region when a constant current is flowing through it.

- The internal quantum efficiency in the active region is the fraction of the electron–hole pairs that recombine radiatively. If the radiative recombination rate is R_r and the nonradiative recombination rate is R_{nr}, then the internal quantum efficiency η_{int} is the ratio of the radiative recombination rate to the total recombination rate:

$$\eta_{int} = \frac{R_r}{R_r + R_{nr}} \qquad \text{.....(3)}$$

- For exponential decay of excess carriers, the radiative recombination lifetime is $r_r = n/R_r$ and the nonradiative recombination lifetime is $r_{nr} = n/R_{nr}$. Thus the internal quantum efficiency can be expressed as

$$\eta_{int} = \frac{1}{1 + \tau_r / \tau_{nr}} = \frac{\tau}{\tau_r} \qquad \text{.....(4)}$$

- where the bulk recombination lifetime t is

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} \qquad \text{.....(5)}$$

- In general, r_r and r_{nr} are comparable for direct-bandgap semiconductors, such as GaAlAs and InGaAsP. This also means that R_r and R_{nr} are similar in magnitude, so that the internal quantum efficiency is about 50 percent for simple homo junction LEDs. However, LEDs having double-hetero junction structures can have quantum efficiencies of 60–80 percent. This high efficiency is achieved because the thin active regions of these devices mitigate the self-absorption effects, which reduces the non-radiative recombination rate.

- If the current injected into the LED is I, then the total number of recombination per second is

$$R_r + R_{nr} = I/q \qquad \text{.....(6)}$$

- Then yields $R_r = \eta_{int} I/q$. Noting that R_r is the total number of photons generated per second and that each photon has an energy hv, then the optical power generated internally to the LED is,

$$P_{int} = \eta_{int} \frac{I}{q} h\nu = \eta_{int} \frac{hcI}{q\lambda} \qquad \text{.....(7)}$$

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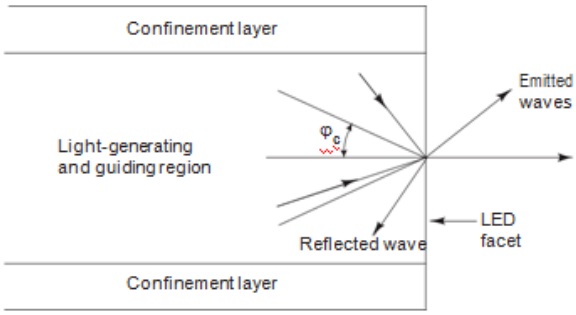

Fig. 3.6. Only light falling within a cone defined by the critical angle f_c will be emitted from an optical source

- Not all internally generated photons will exit the device. To find the emitted power, one needs to consider the external quantum efficiency η_{ext}. This is defined as the ratio of the photons emitted from the LED to the number of internally generated photons. To find the external quantum efficiency, we need to take into account reflection effects at the surface of the LED.
- The external quantum efficiency can then be calculated from the expression

$$\eta_{ext} = \frac{1}{4\pi}\int_0^{\phi_c} T(\phi)(2\pi\sin\phi)\,d\phi \qquad .....(8)$$

- where T(φ) is the Fresnel transmission coefficient or Fresnel transmissivity. This factor depends on the incidence angle φ, but, for simplicity, we can use the expression for normal incidence, which is

$$T(0) = \frac{4n_1n_2}{(n_1+n_2)^2} \qquad .....(9)$$

- Assuming the outside medium is air and letting $n_1 = n$, we have $T(0) = 4n/(n + 1)^2$. The external quantum efficiency is then approximately given by,

$$\eta_{ext} \approx \frac{1}{n(n+1)^2} \qquad .....(10)$$

- From this, it follows that the optical power emitted from the LED is,

$$P = \eta_{ext}P_{int} = \frac{P_{int}}{n(n+1)^2} \qquad .....(11)$$

3.5. Light Source Materials

- The semiconductor material that is used for the active layer of an optical source must have a direct bandgap. In a direct-bandgap semiconductor, electrons and holes can recombine directly across the bandgap without needing a third particle to conserve momentum. Only in direct-bandgap material is the radiative recombination sufficiently high to produce an adequate level of optical emission. Although none of the normal single-element semiconductors are direct-gap materials, many binary compounds are. The most important of these compounds are made from III-V materials. That is, the compounds consist of selections from a group III element (e.g., Al, Ga, or In) and a group V element (e.g., P, As, or Sb). Various ternary and quaternary combinations of binary compounds of these elements are also direct-gap materials and are suitable candidates for optical sources.

- For operation in the 800-to-900-nm spectrum, the principal material used is the ternary alloy Ga1–x Alx As. The ratio x of aluminum arsenide to gallium arsenide determines the bandgap of the alloy and, correspondingly, the wavelength of the peak emitted radiation.At longer wavelengths the quaternary alloy In1–xGaxAsyP1–y is one of the primary material candidates. By varying the mole fractions x and y in the active area, LEDs with peak output powers at any wavelength between 1.0 and 1.7 mm can be constructed.

- The alloys GaAlAs and InGaAsP are chosen to make semiconductor light sources because it is possible to match the lattice parameters of the heterostructure interfaces by using a proper combination of binary, ternary, and quaternary materials. A very close match between the crystal lattice parameters of the two adjoining heterojunctions is required to reduce interfacial defects and to minimize strains in the device as the temperature varies. These factors directly affect the radiative efficiency and lifetime of a light source. Using the fundamental quantum mechanical relationship between energy E and frequency v,

$$E = hv = \frac{hc}{\lambda}$$

- The peak emission wavelength λ in micrometers can be expressed as a function of the bandgap energy Eg in electron volts by the equation,

$$\lambda(\mu\text{m}) = \frac{1.240}{E_g\,(\text{eV})}$$

- One can determine the bandgap of a semiconductor by measuring the energy required to excite electrons from the valence band to the conduction band.

3.5.1. Modulation of LED

- The response time or frequency response of an optical source dictates how fast an electrical input drive signal can vary the light output level. The following three factors largely determine the response time: the doping level in the active region, the injected carrier lifetime t_i in the recombination region, and the parasitic capacitance of the LED. If the drive current is modulated at a frequency w, the optical output power of the device will vary as

$$P(\omega) = P_0[1 + (\omega\tau_i)^2]^{-1/2}$$

- where P_0 is the power emitted at zero modulation frequency. The parasitic capacitance can cause a delay of the carrier injection into the active junction, and, consequently, could delay the optical output. This delay is negligible if a small, constant forward bias is applied to the diode.
- The modulation bandwidth of an LED can be defined in either electrical or optical terms. Normally, electrical terms are used because the bandwidth is actually determined via the associated electrical circuitry. Thus the modulation bandwidth is defined as the point where the electrical signal power, designated by p(w), has dropped to half its constant value resulting from the modulated portion of the optical signal. This is the electrical 3-dB point; that is, the frequency at which the output electrical power is reduced by 3 dB with respect to the input electrical power, as is illustrated in Fig.3.7.

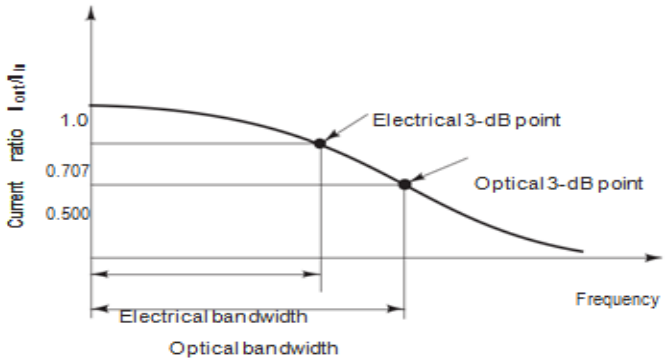

Fig. 3.7 Frequency response of an optical source showing the electrical and optical 3-dB-bandwidth points

- An optical source exhibits a linear relationship between light power and current, so currents rather than voltages (which are used in electrical systems) are compared in optical systems. Thus, since p(w) = I^2(w)/R, the ratio of the output electrical power at the frequency w to the power at zero modulation is

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$$\text{Ratio}_{\text{elec}} = 10 \log\left[\frac{p(\omega)}{p(0)}\right] = 10 \log\left[\frac{I^2(\omega)}{I^2(0)}\right]$$

- where I(w) is the electrical current in the detection circuitry. The electrical 3-dB point occurs at that frequency point where the detected electrical power p(w) = p(0)/2. This happens when

$$\frac{I^2(\omega)}{I^2(0)} = \frac{1}{2}$$

- or I(w)/I(0) = 1/ 2 = 0.707
- Sometimes, the modulation bandwidth of an LED is given in terms of the 3-dB bandwidth of the modulated optical power P(w); that is, it is specified at the frequency where P(w) = $P_0/2$. In this case, the 3-dB bandwidth is determined from the ratio of the optical power at frequency w to the unmodulated value of the optical power. Since the detected current is directly proportional to the optical power, this ratio is

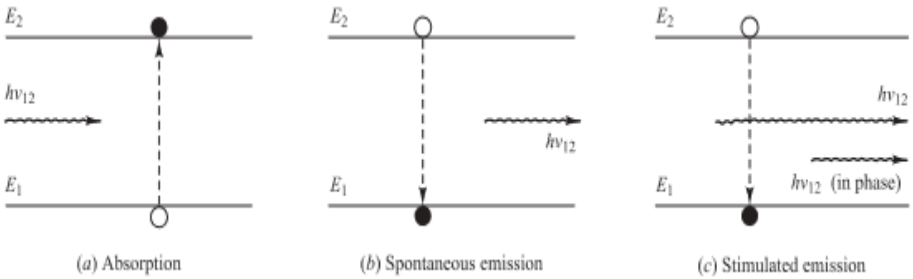
$$\text{Ratio}_{\text{optical}} = 10 \log\left[\frac{P(\omega)}{P(0)}\right] = 10 \log\left[\frac{I(\omega)}{I(0)}\right]$$

- The optical 3-dB point occurs at that frequency where the ratio of the currents is equal to 1/2. As shown in Fig. 3.9, this gives an inflated value of the modulation bandwidth, which corresponds to an electrical power attenuation of 6 dB.

3.6 Laser Diodes

- Lasers come in many forms with dimensions ranging from the size of a grain of salt to one that will occupy an entire room. The lasing medium can be a gas, a liquid, an insulating crystal (solid state), or a semiconductor. For optical fiber systems the laser sources used almost exclusively are semiconductor laser diodes. They are similar to other lasers, such as the conventional solid-state and gas lasers, in that the emitted radiation has spatial and temporal coherence; that is, the output radiation is highly monochromatic and the light beam is very directional.
- Despite their differences, the basic principle of operation is the same for each type of laser. Laser action is the result of three key processes: photon absorption, spontaneous emission, and stimulated emission. These three processes are represented by the simple two-energy-level diagrams in Fig.3.8.

- where E_1 is the ground-state energy and E_2 is the excited-state energy. According to Planck's law, a transition between these two states involves the absorption or emission of a photon of energy $hn_{12} = E_2 - E_1$. Normally, the system is in the ground state. When a photon of energy hn_{12} impinges on the system, an electron in state E_1 can absorb the photon energy and be excited to state E_2, as shown in Fig. 3.8a. Since this is an unstable state, the electron will shortly return to the ground state, thereby emitting a photon of energy hn_{12}.
- This occurs without any external stimulation and is called spontaneous emission. These emissions are isotropic and of random phase, and thus appear as a narrowband gaussian output.



- **Fig. 3.8 The three key transition processes involved in laser action. The open circle represents the initial state of the electron and the heavy dot represents the final state; incident photons are shown on the left of each diagram and emitted photons are shown on the right.**
- The electron can also be induced to make a downward transition from the excited level to the ground-state level by an external stimulation. As shown in Fig. 3.8c, if a photon of energy hn_{12} impinges on the system while the electron is still in its excited state, the electron is immediately stimulated to drop to the ground state and give off a photon of energy hn_{12}. This emitted photon is in phase with the incident photon, and the resultant emission is known as stimulated emission.
- In thermal equilibrium the density of excited electrons is very small. Most photons incident on the system will therefore be absorbed, so that stimulated emission is essentially negligible. Stimulated emission will exceed absorption only if the population of the excited states is greater than that of the ground state. This condition is known as population inversion.
- Since this is not an equilibrium condition, population inversion is achieved by various "pumping" techniques. In a semiconductor laser, population inversion is accomplished by injecting electrons into the material at the device contacts or through an optical absorption method by means of externally injected photons.

3.6.1.Laser Diode Modes and Threshold Conditions

For optical fiber communication systems requiring bandwidths greater than approximately 200 MHz, the semiconductor injection laser diode is preferred over the LED. Laser diodes typically have response times less than 1 ns, can have spectral widths of 2 nm or less, and are capable of coupling from tens to hundreds of milliwatts of useful luminescent power into optical fibers with small cores and small mode-field diameters. Virtually all laser diodes in use are multilayered heterojunction devices.

The double-heterojunction LED configuration evolved from the successful demonstration of both carrier and optical confinement in heterojunction injection laser diodes. The more rapid evolvement and utilization of LEDs as compared with laser diodes lies in the inherently simpler construction, the smaller temperature dependence of the emitted optical power, and the absence of catastrophic degradation in LEDs.

The construction of laser diodes is more complicated, mainly because of the additional requirement of current confinement in a small lasing cavity.

The radiation in one type of laser diode configuration is generated within a Fabry-Perot resonator cavity, shown in Fig.3.9, as in most other types of lasers. Here the cavity is approximately 250–500 mm long, 5–15 mm wide, and 0.1–0.2 μm thick. These dimensions commonly are referred to as the longitudinal, lateral, and transverse dimensions of the cavity, respectively.

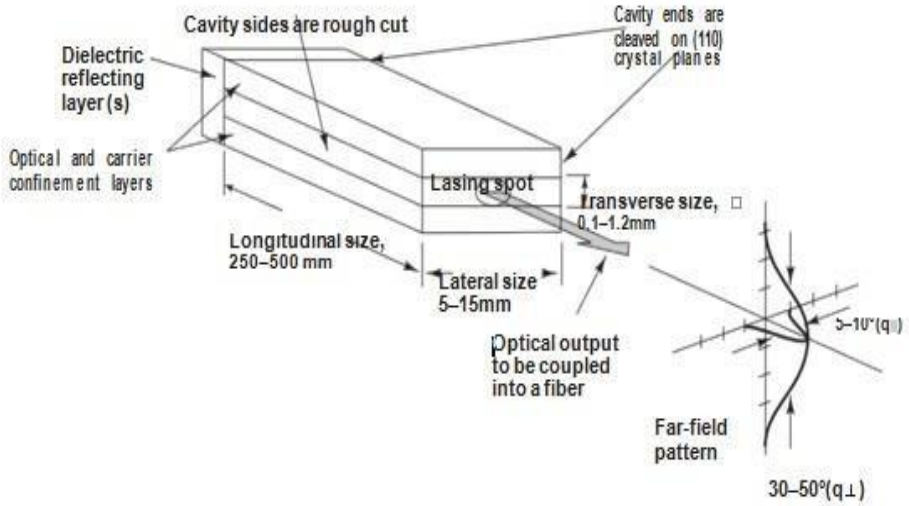


Fig. 3.9 Fabry-Perot resonator cavity for a laser diode. The cleaved crystal ends function as partially reflecting mirrors. The unused end (the rear facet) can be coated with a dielectric reflector to reduce optical loss in the cavity.

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3.6.2.Laser Diode Rate Equations

- The relationship between optical output power and the diode drive current can be determined by examining the rate equations that govern the interaction of photons and electrons in the active region.
- As noted earlier, the total carrier population is determined by carrier injection, spontaneous recombination, and stimulated emission. For a pn junction with a carrier-confinement region of depth d, the rate equations are given by

$$\frac{d\Phi}{dt} = Cn\Phi + R_{sp} - \frac{\Phi}{\tau_{ph}} \qquad .....(1)$$

- =stimulated emission + spontaneous emission + photon loss
- which governs the number of photons F, and

$$\frac{dn}{dt} = \frac{J}{qd} - \frac{n}{\tau_{sp}} - Cn\Phi \qquad .....(2)$$

- =injection + spontaneous recombination + stimulated emission
- which governs the number of electrons n. Here, C is a coefficient describing the strength of the optical absorption and emission interactions, R_{sp} is the rate of spontaneous emission into the lasing mode (which is much smaller than the total spontaneous-emission rate), r_{ph} is the photon lifetime, r_{sp} is the spontaneous-recombination lifetime, and J is the injection-current density.
- The above two Equations may be balanced by considering all the factors that affect the number of carriers in the laser cavity. The first term is a source of photons resulting from stimulated emission.
- The second term, describing the number of photons produced by spontaneous emission, is relatively small compared with the first term.
- The third term in first equation indicates the decay in the number of photons caused by loss mechanisms in the lasing cavity. In second equation, the first term represents the increase in the electron concentration in the conduction band as current flows into the device.
- The second and third terms give the number of electrons lost from the conduction band owing to spontaneous and stimulated transitions, respectively.

- Solving these two equations for a steady-state condition will yield an expression for the output power. The steady state is characterized by the left-hand sides of 1st eqn and 2nd eqn being equal to zero. First, from eqn 1, assuming R_{sp} is negligible and noting that dF/dt must be positive when F is small, we have

$$Cn - \frac{1}{\tau_{ph}} \geq 0 \qquad .....(3)$$

- This shows that n must exceed a threshold value n_{th} in order for F to increase. Using second Eq. , this threshold value can be expressed in terms of the threshold current J_{th} needed to maintain an inversion level n = n_{th} in the steady state when the number of photons Ø = 0:

$$\frac{n_{th}}{\tau_{sp}} = \frac{J_{th}}{qd} \qquad .....(4)$$

- This expression defines the current required to sustain an excess electron density in the laser when spontaneous emission is the only decay mechanism.
- Next, consider the photon and electron rate equations in the steady-state condition at the lasing threshold. Respectively, 1st and 2nd eqns become,

$$0 = Cn_{th}\Phi_s + R_{sp} - \frac{\Phi_s}{\tau_{ph}} \qquad .....(5)$$

$$0 = \frac{J}{qd} - \frac{n_{th}}{\tau_{sp}} - Cn_{th}\Phi_s \qquad .....(6)$$

- where F_s is the steady-state photon density. Adding Eqs 4, 5 and 6th eqns, using 3rd eqn for the term n_{th}/t_{sp}, and solving for F_s yields the number of photons per unit volume:

$$\Phi_s = \frac{\tau_{ph}}{qd}(J - J_{th}) + \tau_{ph} R_{sp} \qquad .....(7)$$

- The number of photons resulting from stimulated emission. The power from these photons is generally concentrated in one or a few modes.
- The second term gives the spontaneously generated photons. The power resulting from these photons is not mode-selective, but is spread over all the possible modes of the volume, which are on the order of 10^8 modes.

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3.6.3. External Quantum Efficiency

- The external differential quantum efficiency $\Cap_{ext}$ is defined as the number of photons emitted per radiative electron–hole pair recombination above threshold. Under the assumption that above threshold the gain coefficient remains fixed at g_{th}, η_{ext} is given by,

$$\eta_{ext} = \frac{\eta_i(g_{th} - \bar{\alpha})}{g_{th}}$$

- Here, η_i is the internal quantum efficiency. This is not a well-defined quantity in laser diodes, but most measurements show that h_i 0.6–0.7 at room temperature. Experimentally, $\Cap_{ext}$ is calculated from the straight-line portion of the curve for the emitted optical power P versus drive current I, which gives

$$\eta_{ext} = \frac{q}{E_g}\frac{dP}{dI} = 0.8065\lambda\ (\mu\text{m})\frac{dP(\text{mW})}{dI\,(\text{mA})}$$

- where E_g is the bandgap energy in electron volts, dP is the incremental change in the emitted optical power in milliwatts for an incremental change dI in the drive current (in milliamperes), and l is the emission wavelength in micrometers. For standard semiconductor lasers, external differential quantum efficiencies of 15–20 percent per facet are typical. High-quality devices have differential quantum efficiencies of 30–40 percent.

3.6.4. Resonant Frequencies

- To examine the resonant frequencies of the laser, consider the equation

$$e^{-j2\beta L} = 1 \qquad \text{.....(1)}$$

This equation gives information concerning the resonant frequencies of the Fabry-Perot cavity. The condition in the above equation holds when,

$$2\beta L = 2\pi m \qquad \text{.....(2)}$$

- Where m is an integer. Using $\beta=2\pi\Cap/\lambda$ for the propagation constant

$$m = \frac{L}{\lambda/2n} = \frac{2Ln}{c}\nu \qquad \text{.....(3)}$$

- Where $c=\nu\lambda$. This states that the cavity resonates (i.e., a standing-wave pattern exists within it) when an integer number m of half-wavelengths spans the region between the mirrors.

- Since in all lasers the gain is a function of frequency (or wavelength, since $c = \nu\lambda$), there will be a range of frequencies (or wavelengths) for which above equation holds. Each of these frequencies corresponds to a mode of oscillation of the laser. The relationship between gain and frequency can be assumed to have the gaussian form.

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$$g(\lambda) = g(0)\exp\left[-\frac{(\lambda-\lambda_0)^2}{2\sigma^2}\right] \qquad .....(4)$$

- where λ_0 is the wavelength at the center of the spectrum, s is the spectral width of the gain, and the maximum gain g(0) is proportional to the population inversion. Let us now look at the frequency, or wavelength, spacing between the modes of a multimode laser. Here, we consider only the longitudinal modes. Note, however, that for each longitudinal mode there may be several transverse modes that arise from one or more reflections of the propagating wave at the sides of the resonator cavity. To find the frequency spacing, consider two successive modes of frequencies n_{m-1} and n_m represented by the integers m – 1 and m. From 2nd eqn. ,we have

$$m-1 = \frac{2Ln}{c}v_{m-1} \qquad .....(5)$$

and

$$m = \frac{2Ln}{c}v_m \qquad .....(6)$$

- Subtracting these two equations yields

$$1 = \frac{2Ln}{c}\left(v_m - v_{m-1}\right) = \frac{2Ln}{c}\Delta v \qquad .....(7)$$

- from which we have the frequency spacing

$$\Delta v = \frac{c}{2Ln} \qquad .....(8)$$

- This can be related to the wavelength spacing Dl through the relationship $\Delta v/v = \Delta\lambda/\lambda$ yielding

$$\Delta\lambda = \frac{\lambda^2}{2Ln} \qquad .....(9)$$

- The output spectrum of a multimode laser follows the typical gain-versus-frequency, where the exact number of modes, their heights, and their spacings depend on the laser construction.

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3.6.5.Laser Diode Structures and Radiation Patterns

- A basic requirement for efficient operation of laser diodes is that, in addition to transverse optical confinement and carrier confinement between heterojunction layers, the current flow must be restricted laterally to a narrow stripe along the length of the laser. Numerous novel methods of achieving this, with varying degrees of success, have been proposed, but all strive for the same goals of limiting the number of lateral modes so that lasing is confined to a single filament, stabilizing the lateral gain, and ensuring a relatively low threshold current.
- Figure 3.11 shows the three basic optical-confinement methods used for bounding laser light in the lateral direction. In the first structure, a narrow electrode stripe (less than 8 mm wide) runs along the length of the diode.
- The injection of electrons and holes into the device alters the refractive index of the active layer directly below the stripe. The profile of these injected carriers creates a weak, complex waveguide that confines the light laterally. This type of device is commonly referred to as a gain-guided laser.

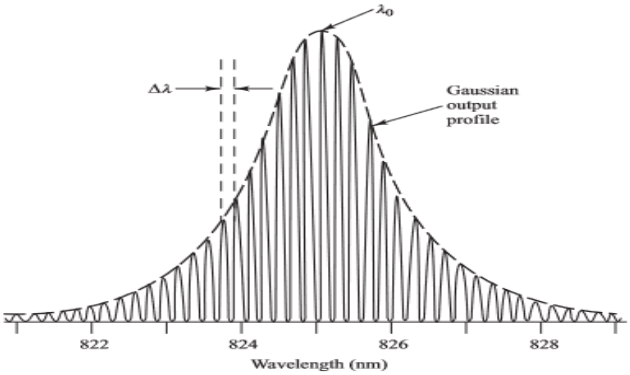

Fig. 3.10 Typical spectrum from a Fabry-Perot GaAlAs/GaAs laser diode.

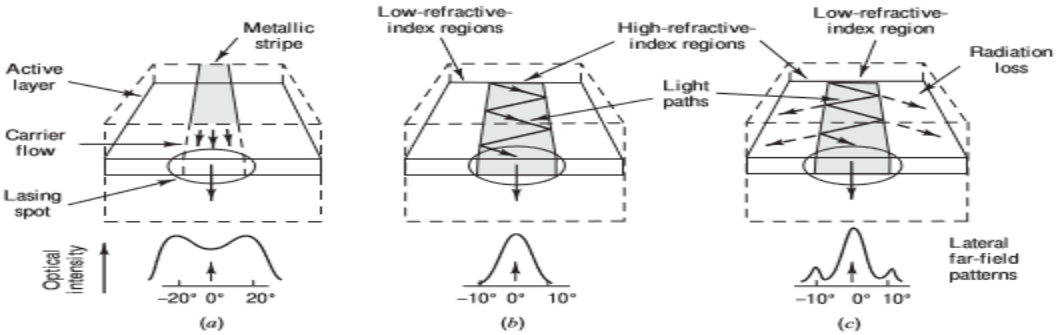

Fig. 3.11 (a)in the gain-induced guide, electrons injected via a metallic stripe contact alter the index of refraction of the active layer; (b) the positive-index waveguide has a higher refractive index in the central portion of the active region; (c) the negative-index waveguide has a lower refractive index in the central portion of the active region.

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- Although these lasers can emit optical powers exceeding 100 μW, they have strong instabilities and can have highly astigmatic, two-peaked beams as shown in Fig. 3.11a.
- More stable structures use the configurations shown in Fig.3.11b and 3.11c. Here, dielectric waveguide structures are fabricated in the lateral direction. The variations in the real refractive index of the various materials in these structures control the lateral modes in the laser.
- Thus, these devices are called index-guided lasers. If a particular index-guided laser supports only the fundamental transverse mode and the fundamental longitudinal mode, it is known as a single-mode laser. Such a device emits a single, well-collimated beam of light that has an intensity profile that is a bell-shaped gaussian curve.
- Index-guided lasers can have either positive-index or negative-index wave-confining structures. In a positive-index waveguide, the central region has a higher refractive index than the outer regions.
- Thus all of the guided light is reflected at the dielectric boundary, just as it is at the core-cladding interface in an optical fiber. By proper choice of the change in refractive index and the width of the higher-index region, one can make a device that supports only the fundamental lateral mode.
- In a negative-index waveguide, the central region of the active layer has a lower refractive index than the outer regions. At the dielectric boundaries, part of the light is reflected and the rest is refracted into the surrounding material and is thus lost.
- This radiation loss appears in the far-field radiation pattern as narrow side lobes to the main beam, as shown in Fig. 3.11c. Since the fundamental mode in this device has less radiation loss than any other mode, it is the first to laser. The positive-index laser is the more popular of these two structures.
- Index-guided lasers can be made using any one of four fundamental structures. These are the buried heterostructure, a selectively diffused construction, a varying-thickness structure, and a bent-layer configuration. To make the buried heterostructure (BH) laser shown in Fig. 3.12, one etches a narrow mesa stripe (1–2 mm wide) in double-heterostructure material.
- The mesa is then embedded in high-resistivity lattice-matched n-type material with an appropriate bandgap and low refractive index.
- This material GaAlAs in 800-to-900-nm lasers with a GaAs active layer, and is InP for 1300-to-1600-nm lasers with an InGaAsP active layer. This configuration thus strongly traps generated light in a lateral waveguide.
- A number of variations of this fundamental structure have been used to fabricate high-performing laser diodes.

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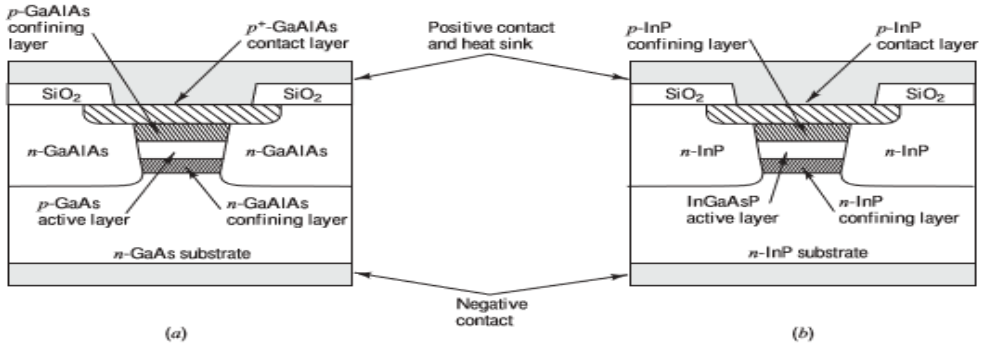
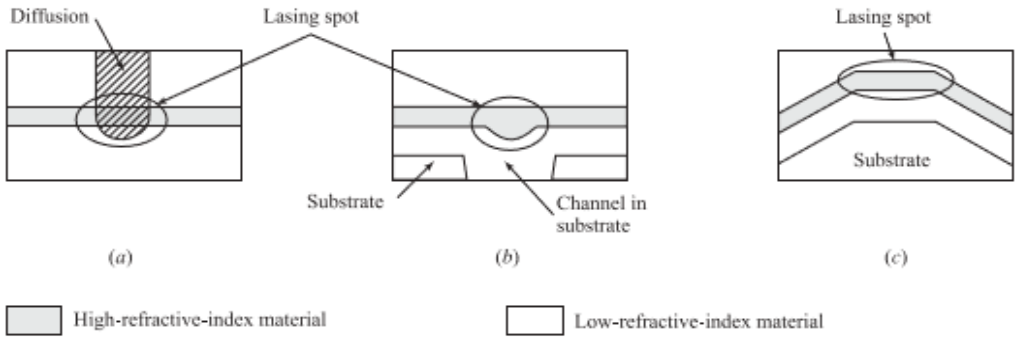
- The selectively diffused construction is shown in Fig. 3.13a. Here, a chemical dopant, such as zinc for GaAlAs lasers and cadmium for InGaAsP lasers, is diffused into the active layer immediately below the metallic contact stripe.



- **Fig. 3.12 (a) Short-wavelength (800–900 nm) GaAlAs and (b) long-wavelength (1300–1600 nm) InGaAsP buried heterostructure laser diodes.**



- **Fig. 3.13 Positive-index optical-wave-confining structure of the (a) selectively diffused, (b) varying-thickness, and (c) bent-layer types.**

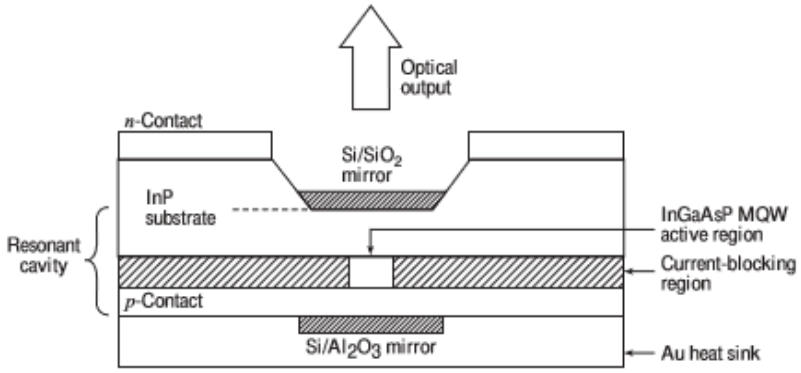
- The dopant changes the refractive index of the active layer to form a lateral waveguide channel. In the varying-thickness structure shown in Fig.3.13b, a channel (or other topological configuration, such as a mesa or terrace) is etched into the substrate. Layers of crystal are then regrown into the channel using liquid-phase epitaxy. This process fills in the depressions and partially dissolves the protrusions, thereby creating variations in the thicknesses of the active and confining layers.

- When an optical wave encounters a local increase in the thickness, the thicker area acts as a positive-index waveguide of higher-index material. In the bent-layer structure, a mesa is etched into the substrate as shown in Fig.3.15c. Semiconductor material layers are grown onto this structure using vapor-phase epitaxy to exactly replicate the mesa configuration.

- The active layer has a constant thickness with lateral bends. As an optical wave travels along the flat top of the mesa in the active area, the lower-index material outside of the bends confines the light along this lateral channel.

3.7.Single-Mode Lasers

- For high-speed long-distance communications one needs single-mode lasers, which must contain only a single longitudinal mode and a single transverse mode. Consequently, the spectral width of the optical emission is very narrow.
- One way of restricting a laser to have only one longitudinal mode is to reduce the length L of the lasing cavity to the point where the frequency separation Δv of the adjacent modes is larger than the laser transition line width; that is, only a single longitudinal mode falls within the gain bandwidth of the device.
- For example, for a Fabry-Perot cavity, all longitudinal modes have nearly equal losses and are spaced by about 1 nm in a 250-mm-long cavity at 1300 nm. By reducing L from 250 μm to 25 μ m, the mode spacing increases from 1 nm to 10 nm. However, these lengths make the device hard to handle, and they are limited to optical output powers of only a few milliwatts.



- **Fig.3.14 Basic architecture of a vertical-cavity surface-emitting laser (VCSEL).**

- Alternative devices were thus developed. Among these are vertical-cavity surface-emitting lasers, structures that have a built-in frequency-selective grating, and tunable lasers. Here, we look at the first two structures. The special feature of a vertical-cavity surface-emitting laser (VCSEL) is that the light emission is perpendicular to the semiconductor surface, as shown in Fig. 3.14.
- This feature facilitates the integration of multiple lasers onto a single chip in one- or two-dimensional arrays, which makes them attractive for wavelength-division-multiplexing applications. The active-region volume of these devices is very small, which leads to very low threshold currents (< 100 mA).
- In addition, for an equivalent output power compared to edge-emitting lasers, the modulation bandwidths are much greater, since the higher photon densities reduce radiative lifetimes.

- The mirror system used in VCSELs to form the resonant cavity is of critical importance, since maximum reflectivity is needed for efficient operation. Figure 3.16 shows one mirror system that consists of a semiconductor material, such as Si/SiO_2, as one material and an oxide layer, such as Si/Al_2O_3, as the other material.

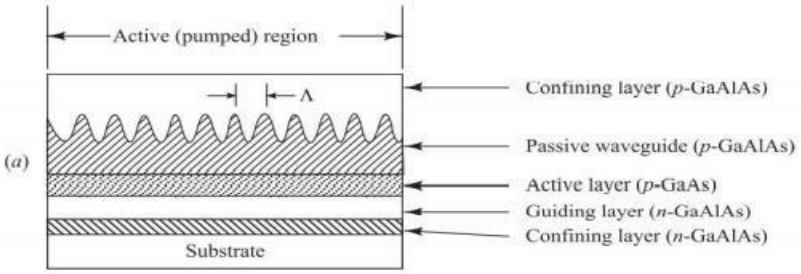
- Three types of laser configurations using a built-in frequency-selective reflector are shown in Fig.3.15. In each case, the frequency-selective reflector is a corrugated grating that is a passive waveguide layer adjacent to the active region. The optical wave propagates parallel to this grating. The operation of these types of lasers is based on the distributed Bragg phase-grating reflector. A phase grating is essentially a region of periodically varying refractive index that causes two counter propagating traveling waves to couple. The coupling is at a maximum for wavelengths close to the Bragg wavelength λ_B, which is related to the period L of the corrugations by

$$\lambda_B = \frac{2n_e\Lambda}{k}$$

- where n_e is the effective refractive index of the mode and k is the order of the grating. First-order gratings (k = 1) provide the strongest coupling, but sometimes second-order gratings are used because their larger corrugation period makes fabrication easier. Lasers based on this architecture exhibit good single-mode longitudinal operation with low sensitivity to drive-current and temperature variations. In the distributed-feedback (DFB) laser, the grating for the wavelength selector is formed over the entire active region.

$$\lambda = \lambda_B \pm \frac{\lambda_B^2}{2n_eL_e}\left(m+\frac{1}{2}\right)$$

- where m = 0, 1, 2, ... is the mode order and L_e is the effective grating length. The amplitudes of successively higher-order lasing modes are greatly reduced from the zero-order amplitude; for example, the first-order mode (m = 1) is usually more than 30 dB down from the zero-order amplitude (m = 0).



- **Fig.3.15 Three types of laser structures using built-in frequency-selective resonator gratings: (a) distributed-feedback**

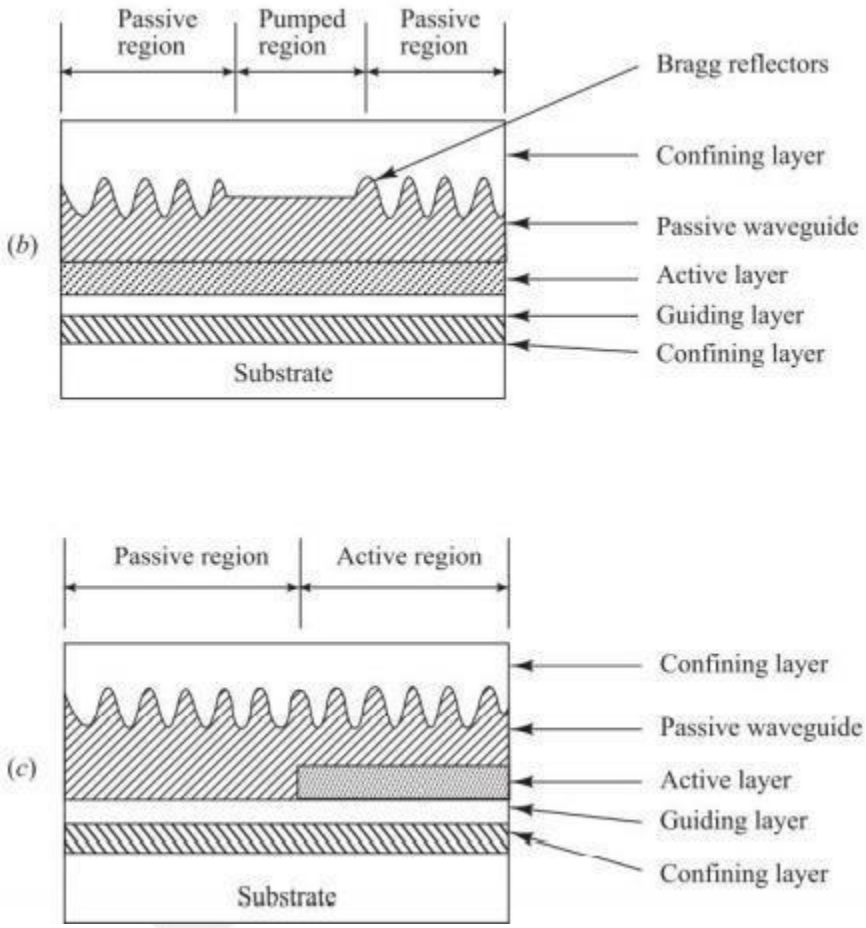
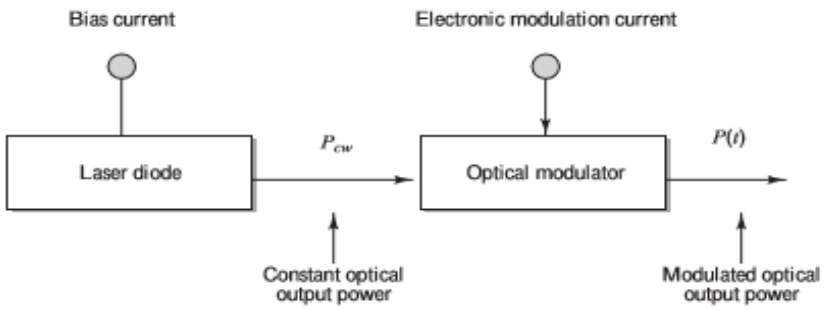


Fig.3.15 Three types of laser structures using built-in frequency-selective resonator gratings: (DFB) laser, (b) distributed-Bragg-reflector (DBR) laser, and (c) distributed-reflector (DR) laser

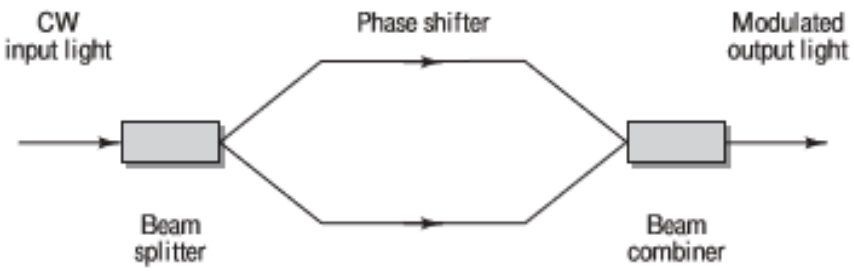
- For the distributed-Bragg-reflector (DBR) laser, the gratings are located at the ends of the normal active layer of the laser to replace the cleaved end mirrors used in the Fabry-Perot optical resonator (Fig. 3.15b). The distributed-reflector laser consists of active and passive distributed reflectors (Fig. 3.15c). This structure improves the lasing properties of conventional DFB and DBR lasers, and has a high efficiency and high output capability.
- Theoretically, in a DFB laser that has both ends antireflection-coated, the two zero-order modes on either side of the Bragg wavelength should experience the same lowest threshold gain and would laser simultaneously in an idealized symmetrical structure.

3.7.1.External Modulation

- When direct modulation is used in a laser transmitter, the process of turning the laser on and off with an electrical drive current produces a widening of the laser linewidth. This phenomenon is referred to as chirp and makes directly modulated lasers undesirable for operation at data rates greater than about 2.5 Gb/s. For these higher-rate applications it is preferable to use an external modulator, as shown in Fig.3.16.
- In such a configuration, the optical source emits a constant-amplitude light signal, which enters the external modulator. In this case, instead of varying the amplitude of the light coming out of the laser, the electrical driving signal dynamically changes the optical power level that exits the external modulator.



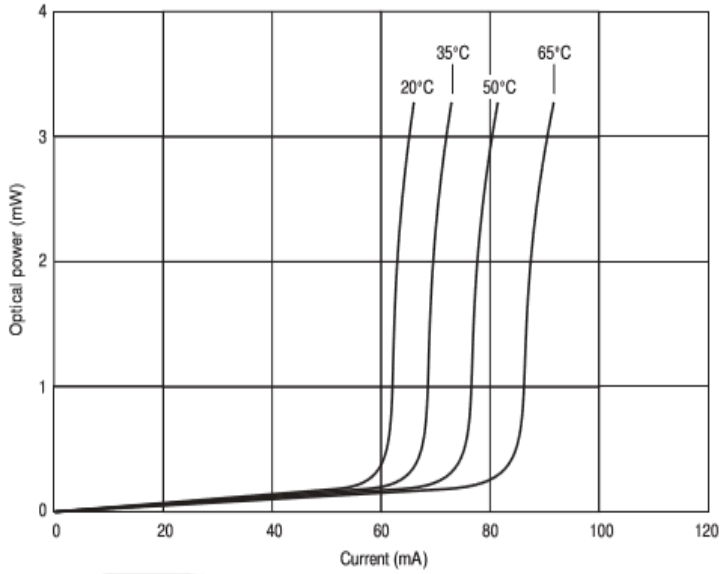
- **Fig. 3.16 Operational concept of a generic external modulator.**



- **Fig. 3.17 Operational concept of an electro-optical lithium-niobate external modulator**

- This process thus produces a time-varying optical signal. The external modulator either can be integrated physically in the same package with the light source or it can be a separate device. The two main device types are the electro-optical phase modulator and the electro-absorption modulator.
- The electro-optical (EO) phase modulator (also called a Mach-Zehnder Modulator or MZM) typically is made of lithium-niobate ($LbNiO_3$).

- In an EO modulator the light beam is split in half and then sent through two separate paths, as shown in Fig.3.17. A high-speed electric signal then changes the phase of the light signal in one of the paths. This is done in such a manner that when the two halves of the signal meet again at the device output, they will recombine either constructively or destructively.
- The constructive recombination produces a bright signal and corresponds to a 1 pulse. On the other hand, destructive recombination results in the two signal halves canceling each other so there is no light at the output of the beam combiner. This corresponds to a 0 pulse. $LbNiO_3$ modulators are separately packaged devices and can be up to 12 cm (about 5 inches) long.



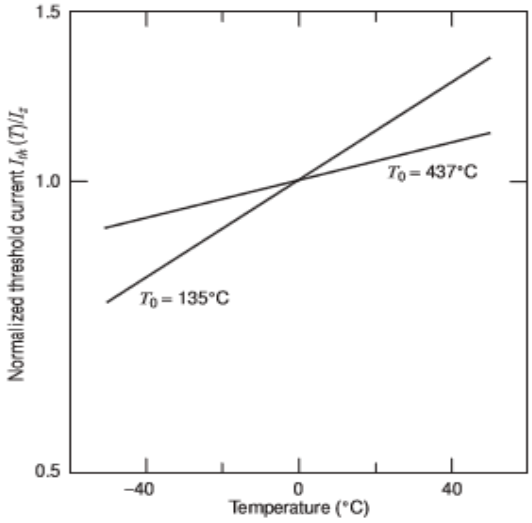
- **Fig. 3.18 Temperature-dependent behavior of the optical output power as a function of the bias current for a particular laser diode with T_0 = 135ºC and I_z = 52 mA.**
- The electro-absorption modulator (EAM) typically is constructed from indium phosphide (InP). It operates by having an electric signal change the transmission properties of the material in the light path to make it either transparent during a 1 pulse or opaque during a 0 pulse.
- Since InP is used as the material for an EAM, it can be integrated onto the same substrate as a DFB laser diode chip.
- The complete laser plus modulator module then can be put in a standard butterfly package, thereby reducing drive voltage, power, and space requirements compared to having separate laser and $LbNiO_3$ modulator packages.

3.7.2. Temperature Effects

- An important factor to consider in the application of laser diodes is the temperature dependence of the threshold current $I_{th}(T)$. This parameter increases with temperature in all types of semiconductor lasers because of various temperature-dependent factors.
- The complexity of these factors prevents the formulation of a single equation that holds for all devices and temperature ranges. However, the temperature variation of I_{th} can be approximated by the empirical expression

$$I_{th}(T) = I_z e^{T/T_0}$$

- where T_0 is a measure of the threshold temperature coefficient and I_z is a constant. For a conventional stripe-geometry GaAlAs laser diode, T_0 is typically 120–165°C in the vicinity of room temperature.
- An example of a laser diode with T_0 = 135°C and I_z = 52 mA. The variation in I_{th} with temperature is 0.8 percent/°C, as is shown in Fig. 3.19. Smaller dependences of I_{th} on temperature have been demonstrated for GaAlAs quantum-well heterostructure lasers.



- **Fig. 3.19. Variation with temperature of the threshold current I_{th} for two types of laser diodes.**
- For these lasers, T_0 can be as high as 437°C. The temperature dependence of $\mathbf{I_{th}}$ for this device is also shown in Fig. 3.19. The threshold variation for this particular laser type is 0.23 percent/°C.

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3.8. Photodetectors

- A photodetector is a key component of an optical receiver in an optical fiber communication system where the optical signal is converted to an electrical signal and subsequently processed by associated electronic circuits. The use of a photodetector in a digital optical receiver is illustrated in Fig. 3.20.
- The photodetector receives the transmitted optical pulses containing information (such as voice, video or computer data) impressed on it and converts it into an electrical signal that is supposed to be a replica of the original information signal. However, in practice the signal received by the receiver is generally weak and distorted depending on the nature of the channel (optical fiber in this case).
- The weak mutilated electrical signal extracted by the detector is further amplified and refined by subsequent stages of the receiver before being delivered at the output.
- From this application point of view, a photodetector is basically a transducer that converts a signal from optical domain to electrical domain. This process is known as Optical-to-Electrical (O/E) conversion. A reverse conversion from Electrical-to-Optical (E/O) domain is generally achieved in an optical fiber communication system by an optical source (e.g. light emit- ting diode or injection laser diode) at the transmitter end.

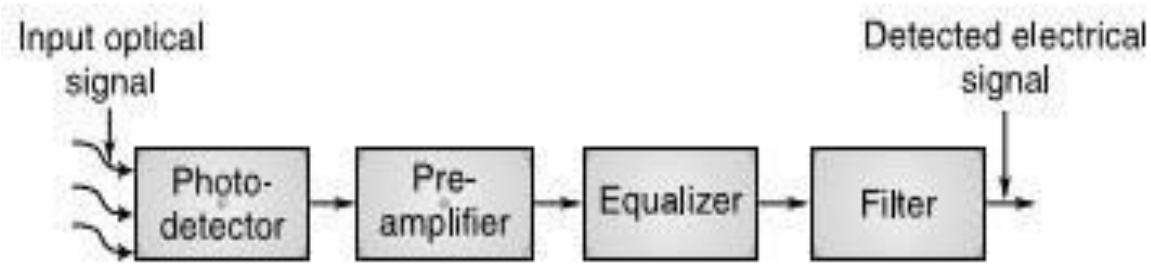

Fig. 3.20 Block diagram of a digital optical receiver

- Semiconductor photodetectors work on the principle of internal photoelectric effect. The photons of the incident optical signal, having energy equal to or larger than the bandgap energy of the semiconductor material used in the detector are absorbed. This results in excitation of electrons from the valence band to the conduction band creating an electron-hole pairs.
- These excess carriers on and above the existing thermally generated carriers are called photo-generated carriers. The photo-generated carriers are subsequently transported over the region of their generation and are finally extracted in the external circuit with the help of a suitable biasing arrangement. The photo-generated carriers ultimately appear in the external circuit in the form of a photo-current or a photo-voltage which is subsequently processed and interpreted.

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3.8.1. Types of Photodetectors

- In general, photoconductive detectors are relatively slow and unsuitable for use in high-speed optical communication systems. Although, all photodetectors work on the general principle discussed earlier, some of the photodetectors involve internal mechanism for multiplying the photo-generated carriers. From this view point, photodetectors can be classified under two heads viz. non-multiplying and multiplying photodetectors. A simple pn-junction photodiode and a p-i-n detector are examples of the former category while an Avalanche Photo Diode (APD) is an example of a multiplying photodetector.
- Depending on whether the energy of the absorbed photon is close to the bandgap or much less than that Photodetectors are further classified into intrinsic and extrinsic categories.
- In intrinsic photodetectors, photo-generated electron and hole pairs are created by direct transition of electrons from valence band to conduction band (band-to-band transition) by consuming the energy of photons larger or equal to the bandgap energy of the semiconductor material.
- In extrinsic photodetectors photons excite electrons from the deep level traps to conduction band or from valence band to deep impurity or defect level leaving holes back in the valence band. These carriers are finally transported and extracted in the external circuit as usual.
- The intrinsic photodetectors are preferred over the extrinsic due to efficient absorption of photons and fast response speed
- There is another variety of extrinsic photodetectors that make use of transitions between energy sub-bands created by a quantum well. Quantum-Well (QW) photodetectors belonging to this category usually absorb photons of energy (~100 meV) which is much less than the bandgap energy of the bulk materials and thereby detects optical signals in the longer wavelength such as Far-Infra Red (FIR) region of the optical spectrum.
- The performance of this detector is limited by the polarization of light that can be absorbed. Photodetectors are often classified as a homojunction (conventional detectors involving same materials on p- and n-side) or a heterojunction photodetectors.
- More often semiconductor photodetectors are realized in the heterojunction form particularly when they are mad alloy semiconductors. Both non multiplying and multiplying photodetectors can be realized in the heterojunction form. The avalanche photodetectors realized in a heterojunction form is often called a Heterojunction Avalanche Photo Diode (HAPD)

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- In intrinsic photodetectors the minimum energy of the photons, corresponding to the frequency or wavelength of the photons that would be absorbed by the semiconducting material constituting the photodetector, depends on the bandgap of the material.
- The longest wavelength corresponding to the shortest frequency (lowest energy photons) that can be absorbed by the semiconducting material can be obtained as

$$h\nu_{\min} = E_g$$

$$\lambda_{\max} = \frac{1.24}{E_g\ (\text{eV})}\ \mu\text{m}$$

3.8.2 Characteristics of a General Photodetector

- The photodetectors used for optical fiber communication are required to meet a number of stringent performance standards. These include
- **(i) Size compatibility**: The size of the detector must be compatible with the size of the fiber carrying the light (radius of the core) so that the optical power carried by the fiber can be efficiently coupled to the photodetector.
- **(ii) High conversion efficiency**: The photodetector must be able to produce maximum electrical signal for a given amount of received optical signal.
- **(iii) High sensitivity at operating wavelength:** The principal material used for making fiber is silica which has several attenuation windows in the infrared region. The photodetectors based on suitable semiconductor materials are required in various operating wavelengths so as to provide a significant electrical output for a small input optical power at room temperature.
- **(iv) High response speed:** The detector must be capable of converting fast-changing optical signals carrying millions of bits of information to the corresponding electrical signal. The present generation optical fibers can support 10Gbps sec of data. In order to handle such a huge bit rate, the response time of the photodetectors must be very low. In other words, the response speed should be extremely high.
- **(v) Minimum noise:** The photodetector must have extremely low noise. The dark current, leakage current and conductance of the photodetector must be low.

3.9. p-i-n (PIN) Photodetector

- The photogenerated carriers in the neutral region take longer time to be extracted in the external circuit as compared to carriers generated in the depletion region. This is because, the carriers generated in the neutral region (zero field region) flow by diffusion mechanism which is a slow process as compared to drift mechanism that is responsible for transportation of the carriers generated in the drift region.
- In order to enhance the speed of response of the photodetector, it is therefore necessary to modify the structure in a way that the photogenerated carriers are mostly created in the depletion region. This can be achieved by making the depletion region much wider as compared to conventional pn-junction. The modified structure is essentially a p-i-n configuration and is most widely used as a photodetector in optical fiber communication system. In this photodetector, the width of the depletion region (i-region) can be tailored to optimize the quantum efficiency and the response speed.
- The schematic of a p-i-n detector is shown in Fig.3.21 along with the energy band diagram. It consists of an intrinsic undoped i-region (usually lightly doped n-type, denoted by n or a lightly doped p-region denoted by p) sandwiched between heavily doped p- and n-regions.

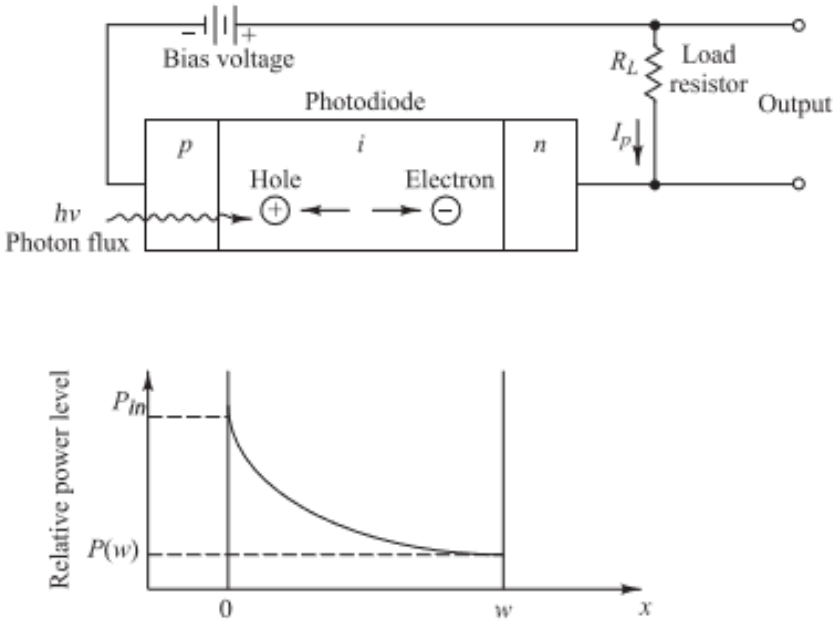

Fig.3.21.Representation of a pin photodiode circuit with an applied reverse bias. An incident optical power level decays exponentially inside the device.

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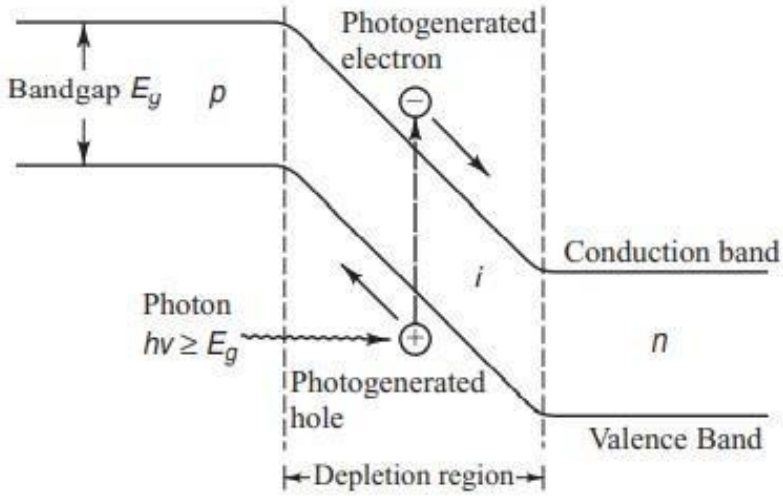
- The device structure consists of p and n regions separated by a very lightly n-doped intrinsic (i) region. In normal operation a sufficiently large reverse-bias voltage is applied across the device so that the intrinsic region is fully depleted of carriers. That is, the intrinsic n and p carrier concentrations are negligibly small in comparison with the impurity concentration in this region.
- When an incident photon has an energy greater than or equal to the bandgap energy of the semiconductor material, the photon can give up its energy and excite an electron from the valence band to the conduction band. This absorption process generates mobile electron–hole pairs, as Fig. 3.22 shows.
- These electrons and holes are known as photocarriers, since they are photon-generated charge carriers that are available to produce a current fl ow when a bias voltage is applied across the device. The number of charge carriers is controlled by the concentration level of impurity elements that are intentionally added to the material



- **Fig. 3.22 Energy band diagram showing the transportation of generated carriers.**
- The photodetector is normally designed so that these carriers are generated mainly in the depletion region (the depleted intrinsic region) where most of the incident light is absorbed. The high electric field present in the depletion region causes the carriers to separate and be collected across the reverse-biased junction. This gives rise to a current flow in an external circuit, with one electron flowing for every carrier pair generated. This current flow is known as the photocurrent.

As the charge carriers flow through the material, some electron–hole pairs will recombine and hence disappear. On the average, the charge carriers move a distance Ln or Lp for electrons and holes, respectively. This distance is known as the diffusion length. The time it takes for an electron or hole to recombine is known as the carrier lifetime and is represented by τn and τp, respectively. The lifetimes and the diffusion lengths are related by the expressions

$$L_n = (D_n \tau_n)^{1/2} \quad \text{and} \quad L_p = (D_p \tau_p)^{1/2}$$

- where Dn and Dp are the electron and hole diffusion coefficients respectively.
- The upper cutoff wavelength λc is determined by the bandgap energy Eg of the material. If Eg is expressed in units of electron volts (eV), then λc is given in units of micrometers (mm) by

$$\lambda_c(\mu\text{m}) = \frac{hc}{E_g} = \frac{1.24}{E_g(\text{eV})}$$

- The cutoff wavelength is about 1.06 mm for Si and 1.6 mm for Ge. For longer wavelengths, the photon energy is not sufficient to excite an electron from the valence to the conduction band. At the lower-wavelength end, the photoresponse cuts off as a result of the very large values of as at the shorter wavelengths.

3.9.1.Key parameters of a p-i-n photodetector

- Two important characteristics of a photodetector are its
- **Quantum Efficiency**
- **Responsivity**
- **1.Quantum Efficiency (η) :**
- The quantum efficiency of a photodetector is defined as the number of electron-hole pairs generated per quanta i.e., per incident photons. Mathematically, the quantum efficiency can be expressed as

$$\eta = \frac{\text{number of electron–hole pairs generated}}{\text{number of incident–absorbed photons}} = \frac{I_p / q}{P_{in} / h\nu}$$

where, Ip is the photocurrent, q is the electronic charge, Pin is the incident optical power, h is Planck's constant and v is the frequency of incident light.

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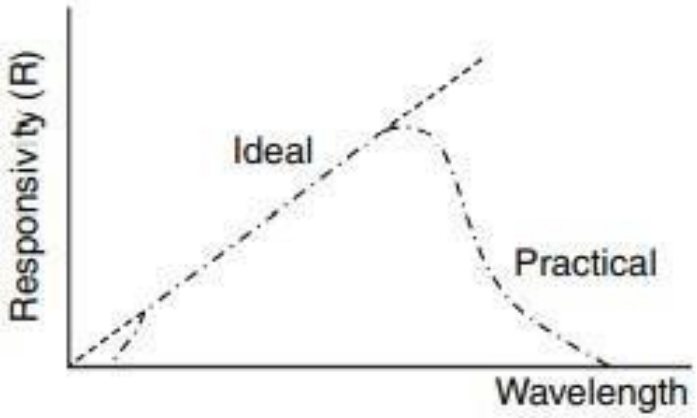
- Optical absorption coefficient(αs) is one of major factor that determines the quantum efficiency. Because, the photon current Ip is related to absorption coefficient αs.

$$I_p = \frac{q}{h\nu} P_{in}(1 - e^{-\alpha_s w})(1 - R_f)$$

- where Pin is the optical power incident on the photodetector, q is the electron charge, and hv is the photon energy, Rf Fresnel reflection coefficient
- The quantum efficiency depends on the absorption coefficient of the material used in the fabrication of the device. The absorption coefficient of a material is a function of wavelength and therefore, the quantum efficiency of a photodetector defined above is a function of wavelength. The quantum efficiency of a practical photodetector is less than unity. The quantum efficiency of a photodetector is generally expressed as percentage quantum efficiency.
- **2.Responsivity(R):**
- The other important parameter used for characterizing a photodetector is the current responsivity, R defined as

$$\mathscr{R} = \frac{I_p}{P_{in}} = \frac{\eta q}{h\nu} \qquad R = \frac{q\eta}{h\nu} = \frac{q\eta\lambda}{hc}$$

- The Variations of responsivity of an ideal photodetector and a practical photodetector with wavelength are shown in Fig. 3.23.



- **Fig.3.23.Variations of responsivity of an ideal photodetector and a practical photodetector with wavelength**

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- Typical pin photodiode responsivities as a function of wave length are shown in Fig. 3.24. The responsivity characteristics of photodetector is based on different materials. This is due to the fact every material has a characteristic band of absorption.
- Representative values are 0.65 A/W for silicon at 900 nm and 0.45 A/W for germanium at 1.3 mm. For InGaAs, typical values are 0.9 A/W at 1.3 μm and 1.0 A/W at 1.55 μ m.

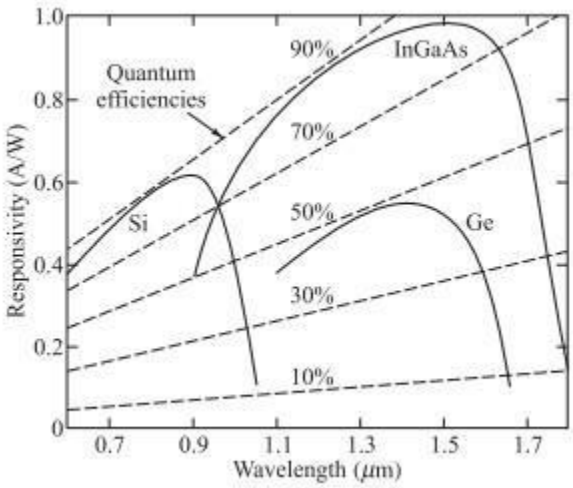

Fig 3.24.Comparison of the responsivity and quantum efficiency as a function of wavelength for pin photodiodes constructed of three different materials

- In most photodiodes the quantum efficiency is independent of the power level falling on the detector at a given photon energy. Thus the responsivity is a linear function of the optical power. That is, the photocurrent Ip is directly proportional to the optical power Pin incident upon the photodetector, so that the responsivity R is constant at a given wavelength (a given value of hv).
- Note, however, that the quantum efficiency is not a constant at all wavelengths because it varies according to the photon energy. Consequently, the responsivity is a function of the wavelength and of the photodiode material (since different mate rials have different bandgap energies).
- For a given material, as the wavelength of the incident photon becomes longer, the photon energy becomes less than that required to excite an electron from the valence band to the conduction band. The responsivity thus falls off rapidly beyond the cutoff wavelength, as can be seen in Fig. 3.24.

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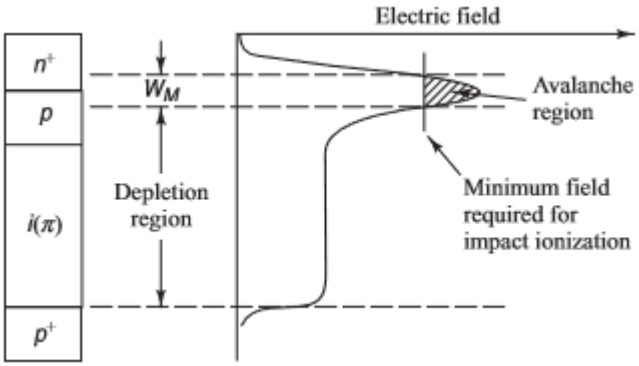
3.10. Avalanche photodiodes

- Avalanche photodiodes (APDs) internally multiply the primary signal photocurrent before it enters the input circuitry of the following amplifier. This increases receiver sensitivity, since the photocurrent is multiplied before encountering the thermal noise associated with the receiver circuit.
- In order for carrier multiplication to take place, the photogenerated carriers must traverse a region where a very high electric field is present. In this high-field region, a photogenerated electron or hole can gain enough energy so that it ionizes bound electrons in the valence band upon colliding with them. This carrier multiplication mechanism is known as impact ionization.
- The newly created carriers are also accelerated by the high electric field, thus gaining enough energy to cause further impact ionization. This phenomenon is the avalanche effect. Below the diode breakdown voltage a finite total number of carriers are created, whereas above breakdown the number can be infinite.



- **Fig. 3.25 Reach-through avalanche photodiode structure and the electric fields in the depletion and multiplication regions.**
- The reach-through avalanche photodiode (RAPD) is composed of a high-resistivity p-type material deposited as an epitaxial layer on a p^+ (heavily doped p-type) substrate. A p-type diffusion or ion implant is then made in the high-resistivity material, followed by the construction of an n^+ (heavily doped n-type) layer. For silicon, the dopants used to form these layers are normally boron and phosphorus, respectively. This configuration is referred to as p^+ppn^+ reach-through structure.
- The p layer is basically an intrinsic material that inadvertently has some p doping because of imperfect purification. It has more complex structures used for InGaAs APDs. The term "reach-through" arises from the photodiode operation. When a low reverse-bias voltage is applied, most of the potential drop is across the pn^+ junction.

- The depletion layer widens with increasing bias until a certain voltage is reached at which the peak electric field at the pn^+ junction is about 5–10 percent below that needed to cause avalanche breakdown. At this point, the depletion layer just "reaches through" to the nearly intrinsic p region.

- In normal usage, the RAPD is operated in the fully depleted mode. Light enters the device through the p^+ region and is absorbed in the p material, which acts as the collection region for the photogenerated carriers. Upon being absorbed, the photon gives up its energy, thereby creating electron–hole pairs, which are then separated by the electric field in the p region. The photogenerated electrons drift through the p region in the pn^+ junction, where a high electric field exists. It is in this high-field region that carrier multiplication takes place. The average number of electron–hole pairs created by a carrier per unit distance traveled is called the ionization rate.

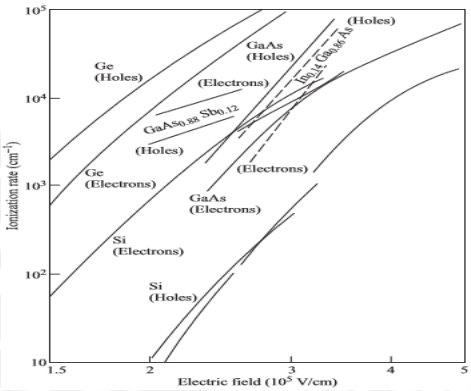


Fig.3.26 Carrier ionization rates obtained experimentally for silicon, germanium, gallium arsenide, gallium arsenide antimonide, and indium gallium arsenide.

- The multiplication M for all carriers generated in the photodiode is defined by

$$M = \frac{I_M}{I_D}$$

- where I_M is the average value of the total multiplied output current and I_p is the primary unmultiplied photocurrent. In practice, the avalanche mechanism is a statistical process, since not every carrier pair generated in the diode experiences the same multiplication. Thus, the measured value of M is expressed as an average quantity.

- The dependence of the gain on the excitation wavelength is attributable to mixed initiation of the avalanche process by electrons and holes when most of the light is absorbed in the n^+p region close to the detector surface.

- This is especially noticeable at short wavelengths, where a larger portion of the optical power is absorbed closer to the surface than at longer wavelengths. In silicon, since the ionization coefficient for holes is smaller than that for electrons, the total current gain is reduced at the short wavelengths. APD is characterized by its responsivity R_{APD}, which is given by

$$\mathcal{R}_{APD} = \frac{nq}{h\nu} M = \mathcal{R}M$$

- where R is the unity gain responsivity.

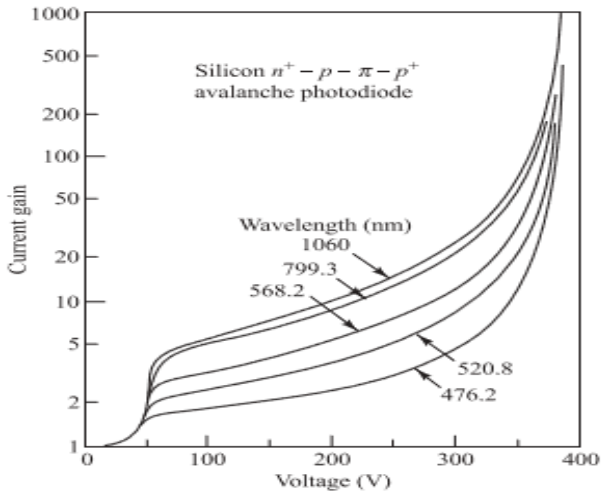


Fig. 3.27 Typical room-temperature current gains of a silicon reach-through avalanche photodiode for different wavelengths as a function of bias voltage.

- **Advantages of Avalanche Photodiode:**
- Excellent Linearity
- Detection of very low light levels
- Increase in sensitivity (5 to 15 dB)
- Wide range of gain variation with response time and reverse bias
- Pin Photodiode receiver is 10 to 12dB less sensitive than APD

- **Disadvantages:**
- Fabrication difficulty due to their more complex structure
- High cost
- High bias voltage is required (50 to 400V) which are wavelength dependent
- Internal gain is dependent on the temperature, thus temperature Compensation is necessary to stabilize the device operation.

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3.11. Photo Detector Noise

- In fiber optic communication systems, the photodiode is generally required to detect very weak optical signals. Detection of the weakest possible optical signals requires that the photodetector and its following amplification circuitry be optimized so that a given signal-to-noise ratio is maintained.
- The power signal-to-noise ratio S/N (also designated by SNR) at the output of an optical receiver is defined by

$$SNR = \frac{S}{N} = \frac{\text{signal power from photocurrent}}{\text{photodetector noise power} + \text{amplifier noise power}}$$

- The noise sources in the receiver arise from the photodetector noises resulting from the statistical nature of the photon-to-electron conversion process and the thermal noises associated with the amplifier circuitry.
- **To achieve a high signal-to-noise ratio, the following conditions should be met:**
- 1. The photodetector must have a high quantum efficiency to generate a large signal power.
- 2. The photodetector and amplifier noises should be kept as low as possible.
- In most applications, it is the noise currents that determine the minimum optical power level that can be detected, since the photodiode quantum efficiency is normally close to its maximum possible value.
- The sensitivity of a photodetector in an optical fiber communication system is describable in terms of the minimum detectable optical power.
- This is the optical power necessary to produce a photocurrent of the same magnitude as the root mean square (rms) of the total noise current, or equivalently, a signal-to-noise ratio of 1.
- A thorough understanding of the source, characteristics, and interrelationships of the various noises in a photodetector is therefore necessary to make a reliable design and to evaluate optical receivers.

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3.11.1. Noise Sources

- To see the interrelationship of the different types of noises affecting the signal-to-noise ratio, let us examine the simple receiver model and its equivalent circuit shown in Fig.3.28. The photodiode has a small series resistance R_s, a total capacitance C_d consisting of junction and packaging capacitances, and a bias (or load) resistor R_L. The amplifier following the photodiode has an input capacitance C_a and a resistance R_a. For practical purposes, R_s is much smaller than the load resistance R_L and can be neglected.

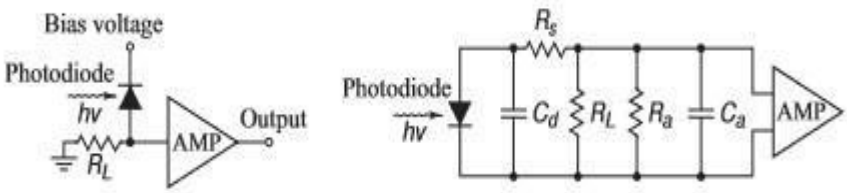

Fig.3.28. (a) Simple model of a photodetector receiver, and (b) its equivalent circuit.

- If a modulated signal of optical power P(t) falls on the detector, the primary photocurrent $i_{ph}(t)$ generated is

$$i_{ph}(t) = \frac{\eta q}{h\nu} P(t) \quad .....(1)$$

- The primary current consists of a dc value I_p, which is the average photocurrent due to the signal power, and a signal component $i_p(t)$. For pin photodiodes the mean-square signal current $< \mathbf{i_s^2} >$ **is**

$$\left\langle i_s^2 \right\rangle = \sigma_{s,\,pin}^2 = \left\langle i_p^2(t) \right\rangle \quad .....(2)$$

- where σ is the variance. For avalanche photodetectors,

$$\left\langle i_s^2 \right\rangle = \sigma_{s,\,\mathrm{APD}}^2 = \left\langle i_p^2(t) \right\rangle M^2 \quad .....(3)$$

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- Where M is the average of the statistically varying avalanche gain. For a sinusoidally varying input signal of modulation index m , the signal component $< I_p(t) >$ is of the form

$$\left\langle i_p^2(t) \right\rangle = \sigma_p^2 = \frac{m^2}{2} I_p^2 \qquad .....(4)$$

- The principal noise sources associated with photodetectors that have no internal gain are quantum noise, dark-current noise generated in the bulk material of the photodiode, and surface leakage current noise. The quantum or shot noise arises from the statistical nature of the production and collection of photoelectrons when an optical signal is incident on a photodetector. The shot noise current has a mean-square value in a receiver bandwidth B_e that is proportional to the average value of the photocurrent I_p.

$$\left\langle i_{\text{shot}}^2 \right\rangle = \sigma_{\text{shot}}^2 = 2qI_p B_e M^2 F(M) \qquad .....(5)$$

- where F(M) is a noise figure associated with the random nature of the avalanche process.
- The photodiode dark current is the current that continues to flow through the bias circuit of the device when no light is incident on the photodiode. This is a combination of bulk and surface currents. The bulk dark current i_{DB} arises from electrons and/or holes that are thermally generated in the pn junction of the photodiode. In an APD, these liberated carriers also get accelerated by the high electric field present at the pn junction, and are therefore multiplied by the avalanche gain mechanism. The mean-square value of this current is given by

$$\left\langle i_{DB}^2 \right\rangle = \sigma_{DB}^2 = 2qI_D M^2 F(M) B_e \qquad .....(6)$$

- where I_D is the primary (unmultiplied) detector bulk dark current.
- The surface dark current is also referred to as a surface leakage current or simply the leakage current. It is dependent on surface defects, cleanliness, bias voltage, and surface area. An effective way of reducing surface dark current is through the use of a guard ring structure that shunts surface leakage currents away from the load resistor. The mean-square value of the surface dark current is given by

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$$\langle i_{DS}^2 \rangle = \sigma_{DS}^2 = 2qI_L B_e \quad .....(7)$$

- where I_L is the surface leakage current. Note that since avalanche multiplication is a bulk effect, the surface dark current is not affected by the avalanche gain.
- A comparison of typical dark currents for Si, Ge, GaAs, and $In_xGa_{1-x}As$ photodiodes is given in Fig. 3.29 as a function of applied voltage normalized to the breakdown voltage V_B. Note that for $In_xGa_{1-x}As$ photodiodes the dark current increases with the composition x. Under a reverse bias, both dark currents also increase with the area. The surface dark current increases in proportion to the square root of the active area, and the bulk dark current is directly proportional to the area.

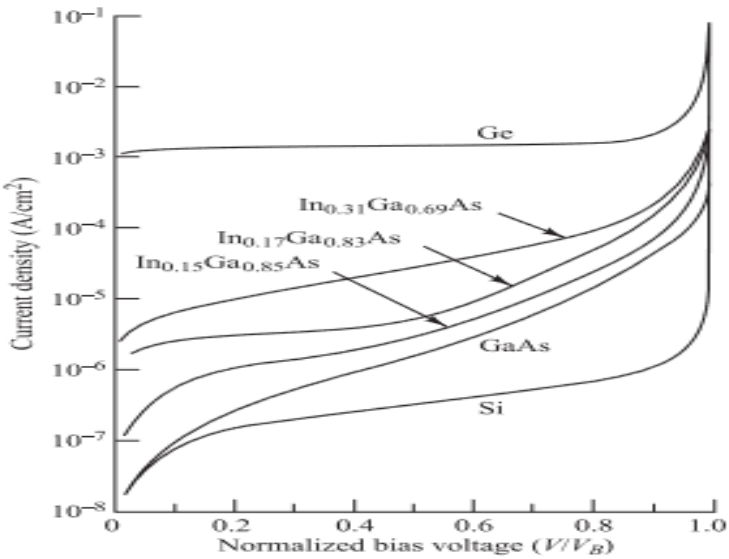

Fig.3.29 A comparison of typical dark currents for Si, Ge, GaAs, and InGaAs photodiodes as a function of normalized bias voltage.

- Photodetector noise current I^2N can be written as

$$\langle i_N^2 \rangle = \sigma_N^2 = \langle i_{shot}^2 \rangle + \langle i_{DB}^2 \rangle + \langle i_{DS}^2 \rangle = \sigma_{shot}^2 + \sigma_{DB}^2 + \sigma_{DS}^2 \quad .....(8)$$

$$= 2q(I_p + I_D)M^2F(M)B_e + 2qI_L B_e \quad .....(9)$$

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- To simplify the analysis of the receiver circuitry, we assume here that the amplifier input impedance is much greater than the load resistance, so that its thermal noise is much smaller than that of R_L. The photodetector load resistor contributes a mean-square thermal (Johnson) noise current.

$$\langle i_T^2 \rangle = \sigma_T^2 = \frac{4k_B T}{R_L} B_e \quad \text{.....(10)}$$

- where k_B is Boltzmann's constant and T is the absolute temperature. This noise can be reduced by using a load resistor that is large but still consistent with the receiver bandwidth requirements.

3.11.2 Signal-to-Noise Ratio

- The signal-to-noise ratio at the input of the amplifier, we have

$$\frac{S}{N} = \frac{\langle i_p^2 \rangle M^2}{2q(I_p + I_D) M^2 F(M) B_e + 2qI_L B_e + 4k_B T B_e / R_L} \quad \text{.....(11)}$$

- In general, in the expression given by above expression one can ignore the negligible leakage current. Furthermore, the term involving I_D can be dropped when the average signal current is much larger than the dark current. The signal-to-noise ratio then becomes

$$\frac{S}{N} = \frac{\langle i_p^2 \rangle M^2}{2qI_p M^2 F(M) B_e + 4k_B T B_e / R_L} \quad \text{.....(12)}$$

- When the optical signal power is relatively high, then the shot noise power is much greater than the thermal noise power. In this case the SNR is called shot-noise limited or quantum limited. When the optical signal power is low, then thermal noise usually dominates over the shot noise. In this case the SNR is referred to as being thermal-noise limited.

- The signal power is multiplied by M^2 and the shot noise plus bulk dark current is multiplied by $M^2F(M)$. The surface-leakage current is not altered by the avalanche gain mechanism. Since the noise figure F(M) increases with M, there always exists an optimum value of M that maximizes the signal-to-noise ratio.

- The optimum gain at the maximum signal-to-noise ratio can be found by differentiating above equation with respect to M, setting the result equal to zero, and solving for M. Doing so for a sinusoidally modulated signal, with m = 1 and F(M) approximated by M^x, yields

$$M_{opt}^{x+2} = \frac{2qI_L + 4k_B T/R_L}{xq(I_p + I_D)} \qquad .....(13)$$

3.12. Avalanche Multiplication Noise

- The avalanche process is statistical in nature, since not every photogenerated carrier pair undergoes the same multiplication. The probability distribution of possible gains that any particular electron–hole pair might experience is sufficiently wide so that the mean-square gain is greater than the average gain squared. That is, if m denotes the statistically varying gain, then

$$\langle m^2 \rangle > \langle m \rangle^2 = M^2 \qquad .....(1)$$

- where the symbols denote an ensemble average and m = M is the average carrier gain. Since the noise created by the avalanche process depends on the mean-square gain m^2 , the noise in an avalanche photodiode can be relatively high. From experimental observations it has been found that, in general, m^2 can be approximated by

$$\langle m^2 \rangle = M^{2+x} \qquad .....(2)$$

- where the exponent x varies between 0 and 1.0 depending on the photodiode material and structure.
- The ratio of the actual noise generated in an avalanche photodiode to the noise that would exist if all carrier pairs were multiplied by exactly M is called the excess noise factor F and is defined by

$$F = \frac{\langle m^2 \rangle}{\langle m \rangle^2} = \frac{\langle m^2 \rangle}{M^2} \qquad .....(3)$$

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- This excess noise factor is a measure of the increase in detector noise resulting from the randomness of the multiplication process. It depends on the ratio of the electron and hole ionization rates and on the carrier multiplication.
- The derivation of an expression for F is complex, since the electric field in the avalanche region (of width W_M,) is not uniform, and both holes and electrons produce impact ionization. For injected electrons and holes, the excess noise factors are

$$F_e = \frac{k_2 - k_1^2}{1 - k_2} M_e + 2\left[1 - \frac{k_1(1-k_1)}{1-k_2}\right] - \frac{(1-k_1)^2}{M_e(1-k_2)} \quad .....(4)$$

$$F_h = \frac{k_2 - k_1^2}{k_1^2(1-k_2)} M_h - 2\left[\frac{k_2(1-k_1)}{k_1^2(1-k_2)} - 1\right] + \frac{(1-k_1)^2 k_2}{k_1^2(1-k_2) M_h} \quad .....(5)$$

- where the subscripts e and h refer to electrons and holes, respectively. The weighted ionization rate ratios k_1 and k_2 take into account the non uniformity of the gain and the carrier ionization rates in the avalanche region. They are given by

$$k_1 = \frac{\int_0^{W_M} \beta(x) M(x)\, dx}{\int_0^{W_M} \alpha(x) M(x)\, dx} \quad .....(6)$$

$$k_2 = \frac{\int_0^{W_M} \beta(x) M^2(x)\, dx}{\int_0^{W_M} \alpha(x) M^2(x)\, dx} \quad .....(7)$$

- where α (x) and β (x) are the electron and hole ionization rates, respectively.
- Normally, to a first approximation k_1 and k_2 do not change much with variations in gain and can be considered as constant and equal.

$$F_e = M_e\left[1 - (1 - k_{eff})\left(1 - \frac{1}{M_e}\right)^2\right] \quad .....(8)$$

$$= k_{eff} M_e + \left(2 - \frac{1}{M_e}\right)(1 - k_{eff}) \quad .....(9)$$

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- for electron injection, and

$$F_h = M_h\left[1-\left(1-\frac{1}{k'_{\text{eff}}}\right)\left(1-\frac{1}{M_h}\right)^2\right] \quad .....(10)$$

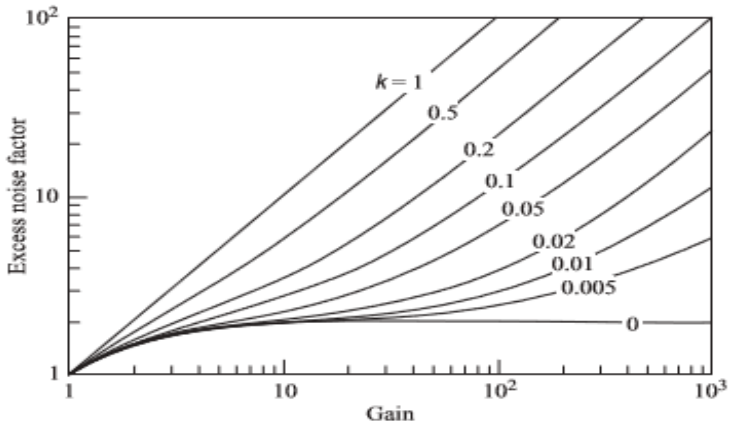
$$= k'_{\text{eff}} M_h - \left(2-\frac{1}{M_h}\right)(k'_{\text{eff}}-1) \quad .....(11)$$

- for hole injection, where the effective ionization rate ratios are

$$k_{\text{eff}} = \frac{k_2 - k_1^2}{1-k_2} \simeq k_2 \quad .....(12)$$

$$k'_{\text{eff}} = \frac{k_{\text{eff}}}{k_1^2} \simeq \frac{k_2}{k_1^2} \quad .....(13)$$

- F_e as a function of the average electron gain M_e for various values of the effective ionization rate ratio k_{eff}. If the ionization rates are equal, the excess noise is at its maximum so that F_e is at its upper limit of M_e. As the ratio β/α decreases from unity, the electron ionization rate starts to be the dominant contributor to impact ionization, and the excess noise factor becomes smaller. If only electrons cause ionization, $\beta = 0$ and F_e reaches its lower limit of 2.



- **Fig. 3.30 Variation of the electron excess noise factor F_e as a function of the electron gain for various values of the effective ionization rate ratio k_{eff}.**

- This shows that to keep the excess noise factor at a minimum, it is desirable to have small values of k_{eff}, we thus see the superiority of silicon over other materials for making avalanche photodiodes. The effective ionization rate ratio k_{eff} varies between 0.015 and 0.035 for silicon, between 0.3 and 0.5 for indium gallium arsenide, and between 0.6 and 1.0 for germanium.
- The excess noise factor can be approximated by $$F = M^x \quad \text{.....(14)}$$
- The parameter x takes on values of 0.3 for Si, 0.7 for InGaAs, and 1.0 for Ge avalanche photodiodes.

3.13. Temperature Effects on Avalanche Gain

- The gain mechanism of an avalanche photodiode is very temperature sensitive because of the temperature dependence of the electron and hole ionization rates. This temperature dependence is particularly critical at high bias voltages, where small changes in temperature can cause large variations in gain. An example of this is shown in Fig. 3.31 for a silicon avalanche photodiode.
- For example, if the operating temperature decreases and the applied bias voltage is kept constant, the ionization rates for electrons and holes will increase and so will the avalanche gain.
- To maintain a constant gain as the temperature changes, the electric field in the multiplying region of the pn junction must also be changed. This requires that the receiver incorporate a compensation circuit that adjusts the applied bias voltage on the photodetector when the temperature changes.
- A simple temperature-dependent expression for the gain can be obtained from the empirical relationship is

$$M = \frac{1}{1 - (V/V_B)^n}$$

- where V_B is the breakdown voltage at which M goes to infinity; the parameter n varies between 2.5 and 7, depending on the material; and $V = V_a - I_M R_M$, with V_a being the reverse-bias voltage applied to the detector, I_M is the multiplied photocurrent, and R_M accounts for the photodiode series resistance and the detector load resistance. Since the breakdown voltage is known to vary with temperatures as

$$V_B(T) = V_B(T_0)[1 + a(T - T_0)]$$

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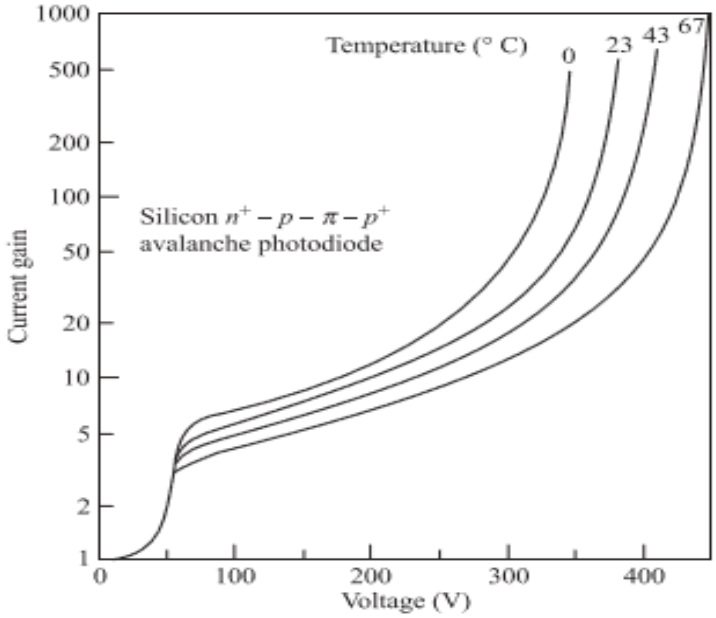

Fig. 3.31 Example of how the gain mechanism of a silicon avalanche photodiode depends on temperature. The measurements for this device were carried out at 825 nm.

- The temperature dependence of the avalanche gain can be approximated by substituting the above two equations together with the expression

$$n(T) = n(T_0)[1 + b(T - T_0)]$$

- The constants a and b are positive for reach-through avalanche photodiodes and can be determined from experimental curves of gain versus temperature.

3.13.1. Noise-Equivalent Power

- The sensitivity of a photodetector in an optical fiber communication system is describable in terms of the minimum detectable optical power.
- This is the optical power necessary to produce a photocurrent of the same magnitude as the root mean-square (rms) of the total noise current, or equivalently, a signal to-noise ratio of 1. This optical signal power is referred to as the noise equivalent power or NEP, which is designated in units of W/√ Hz.
- As an example, consider the thermal-noise limited case for a pin photodiode.

$$\mathrm{SNR} = \mathscr{R}^2 P^2 / (4k_B T B_e / R_L)$$

To find the NEP we set the SNR equal to 1 and solve for P to obtain

$$\mathrm{NEP} = \frac{P_{\min}}{\sqrt{B_e}} = \sqrt{4k_B T / R_L} / \mathscr{R}$$

3.14. Detector Response Time

1. Depletion Layer Photocurrent

- To understand the frequency response of photodiodes, let us first consider the schematic representation of a reverse-biased pin photodiode shown in Fig. 6.10. Light enters the device through the p layer and produces electron–hole pairs as it is absorbed in the semiconductor material. Those electron–hole pairs that are generated in the depletion region or within a diffusion length of it will be separated by the reverse-bias-voltage-induced electric field, thereby leading to a current flow in the external circuit as the carriers drift across the depletion layer.
- Under steady-state conditions, the total current density J_{tot} flowing through the reverse-biased depletion layer is

$$J_{tot} = J_{dr} + J_{diff} \qquad .....(1)$$

- Here, J_{dr} is the drift current density resulting from carriers generated inside the depletion region, and J_{diff} is the diffusion current density arising from the carriers that are produced outside of the depletion layer in the bulk of the semiconductor (i.e., in the n and p regions) and diffuse into the reverse-biased junction. The drift current density can be

$$J_{dr} = \frac{I_p}{A} = q\Phi_0(1 - e^{-\alpha_s w}) \qquad .....(2)$$

- where A is the photodiode area and φ_0 is the incident photon flux per unit area given by

$$\Phi_0 = \frac{P_{in}(1 - R_f)}{Ah\nu} \qquad .....(3)$$

- The surface p layer of a pin photodiode is normally very thin. The diffusion current is thus principally determined by hole diffusion from the bulk n region. The hole diffusion in this material can be determined by the one-dimensional diffusion equation.

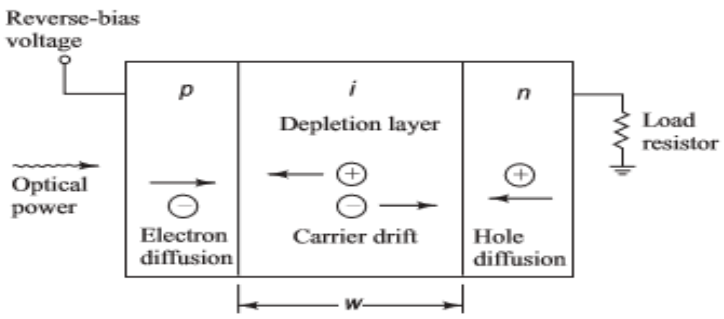

Fig.3.32. Schematic Representation of a Reverse Biased Pin Photodiode

$$D_P \frac{\partial^2 p_n}{\partial x^2} - \frac{p_n - p_{n0}}{\tau_p} + G(x) = 0 \qquad .....(4)$$

- where D_p is the hole diffusion coefficient, p_n is the hole concentration in the n-type material, τ_p is the excess hole lifetime, p_{n0} is the equilibrium hole density, and G(x) is the electron–hole generation rate given by

$$G(x) = \Phi_0 \alpha_s e^{-\alpha_s x} \qquad .....(5)$$

- The diffusion current density is found to be

$$J_{\text{diff}} = q\Phi_0 \frac{\alpha_s L_P}{1 + \alpha_s L_p} e^{-\alpha_s w} + q p_{n0} \frac{D_p}{L_p} \qquad .....(6)$$

- we have that the total current density through the reverse-biased depletion layer is

$$J_{\text{tot}} = q\Phi_0 \left(1 - \frac{e^{-\alpha_s w}}{1 + \alpha_s L_p} \right) + q p_{n0} \frac{D_p}{L_p} \qquad .....(7)$$

- The term involving p_{n0} is normally small, so that the total photogenerated current is proportional to the photon flux φ_0.

2. Response Time

- The response time of a photodiode together with its output circuit depends mainly on the following three factors
  1. The transit time of the photocarriers in the depletion region.
  2. The diffusion time of the photocarriers generated outside the depletion region.
  3. The RC time constant of the photodiode and its associated circuit.
- The photodiode parameters responsible for these three factors are the absorption coefficient a_s, the depletion region width w, the photodiode junction and package capacitances, the amplifier capacitance, the detector load resistance, the amplifier input resistance, and the photodiode series resistance. The photodiode series resistance is generally only a few ohms and can be neglected in comparison with the large load resistance and the amplifier input resistance.
- Let us first look at the transit time of the photocarriers in the depletion region. The response speed of a photodiode is fundamentally limited by the time it takes photogenerated carriers to travel across the depletion region.

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- This transit time t_d depends on the carrier drift velocity v_d and the depletion layer width w, and is given by

$$t_d = \frac{w}{v_d} \qquad .....(8)$$

- In general, the electric field in the depletion region is large enough so that the carriers have reached their scattering-limited velocity. For silicon, the maximum velocities for electrons and holes are 8.4×10^6 and 4.4×10^6 cm/s, respectively, when the field strength is on the order of 2×10^4 V/cm. A typical high-speed silicon photodiode with a 10μm depletion layer width thus has a response time limit of about 0.1 ns.

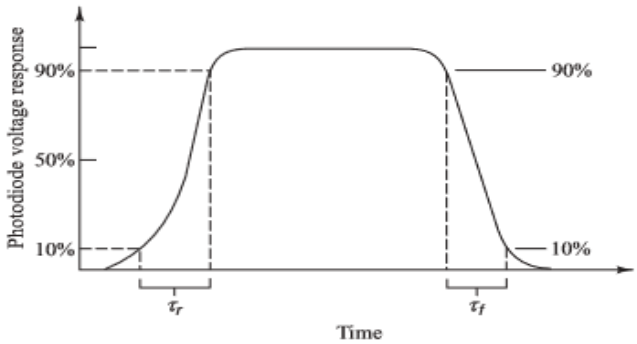

Fig. 3.33. Photodiode response to an optical input pulse showing the 10- to 90-percent rise time and the 10- to 90-percent fall time.

- The diffusion processes are slow compared with the drift of carriers in the high-field region. Therefore, to have a high-speed photodiode, the photocarriers should be generated in the depletion region or so close to it that the diffusion times are less than or equal to the carrier drift times. The effect of long diffusion times can be seen by considering the photodiode response time.

- This response time is described by the rise time and fall time of the detector output when the detector is illuminated by a step input of optical radiation. The rise time τ_r is typically measured from the 10- to 90-percent points of the leading edge of the output pulse, as is shown in Fig. 3.33.

- For fully depleted photodiodes the rise time τ_r and fall time τ_f are generally the same. However, they can be different at low bias levels where the photodiode is not fully depleted, since the photon collection time then starts to become a significant contributor to the rise time.

- In this case, charge carriers produced in the depletion region are separated and collected quickly. On the other hand, electron–hole pairs generated in the n and p regions must slowly diffuse to the depletion region before they can be separated and collected. A typical response time of a partially depleted photodiode is shown in Fig.3.34. The fast carriers allow the device output to rise to 50 percent of its maximum value in approximately 1 ns, but the slow carriers cause a relatively long delay before the output reaches its maximum value.

- To achieve a high quantum efficiency, the depletion layer width must be much larger than $1/\alpha_s$ (the inverse of the absorption coefficient), so that most of the light will be absorbed.

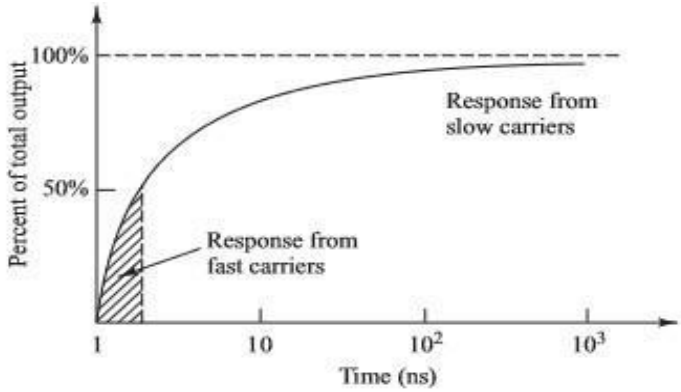

Fig.3.34. Typical response time of a photodiode that is not fully depleted

- The response to a rectangular input pulse of a low-capacitance photodiode having $w >> 1/\alpha_s$ is shown in Fig. 3.35b. The rise and fall times of the photodiode follow the input pulse quite well.
- If the photodiode capacitance is larger, the response time becomes limited by the RC time constant of the load resistor R_L and the photodiode capacitance. The photodetector response then begins to appear as that shown in Fig. 3.35c.
- If the depletion layer is too narrow, any carriers created in the undepleted material would have to diffuse back into the depletion region before they could be collected. Devices with very thin depletion regions thus tend to show distinct slow- and fast-response components, as shown in Fig.3.35d.

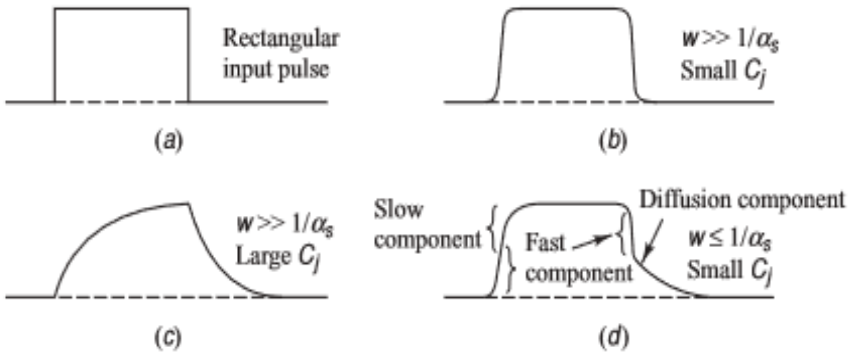

Fig. 3.35. Photodiode pulse Responses Under various Detector Parameter

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- The fast component in the rise time is due to carriers generated in the depletion region, whereas the slow component arises from the diffusion of carriers that are created with a distance L_n from the edge of the depletion region.
- At the end of the optical pulse, the carriers in the depletion region are collected quickly, which results in the fast-detector-response component in the fall time.
- The diffusion of carriers that are within a distance L_n of the depletion region edge appears as the slowly decaying tail at the end of the pulse. Also, if w is too thin, the junction capacitance will become excessive.
- The junction capacitance C_j is

$$C_j = \frac{\epsilon_s A}{w} \qquad .....(9)$$

- where $\boldsymbol{\varepsilon_s}$ = the permittivity of the semiconductor material = $\varepsilon_0 K_s$
- $\mathbf{K_s}$ = the semiconductor dielectric constant
- $\boldsymbol{\varepsilon_0}$ = 8.8542 ¥ 10^{-12} F/m is the free-space permittivity
- **A** = the diffusion layer area

- This excessiveness will then give rise to a large RC time constant, which limits the detector response time. A reasonable compromise between high-frequency response and high quantum efficiency is found for absorption region thicknesses between $1/\alpha_s$ and $2/\alpha_s$.
- If R_T is the combination of the load and amplifier input resistances and C_T is the sum of the photodiode and amplifier capacitances, the detector behaves approximately like a simple RC low-pass filter with a passband given by

$$B_c = \frac{1}{2\pi R_T C_T} \qquad .....(10)$$

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3.15 Comparisons Of Photodetectors

- This section summarizes some generic operating characteristics of Si, Ge, and InGaAs photodiodes. Tables 3.1 and 3.2 list the performance values for pin and avalanche photodiodes, respectively. The values were derived from various vendor data sheets and from performance numbers reported in the literature. They are given as guidelines for comparison purposes. Detailed values on specific devices for particular applications can be obtained from photodetector and receiver module suppliers.

- For short-distance applications, Si devices operating around 850 nm provide relatively inexpensive solutions for most links. Longer links usually require operation in the 1300-nm and 1550-nm windows; here, one normally uses InGaAs-based devices.

Table.3.1Generic operating parameters of Si, Ge, and InGaAs pin photodiodes

Parameter	*Symbol*	*Unit*	*Si*	*Ge*	*InGaAs*
Wavelength range	λ	nm	400–1100	800–1650	1100–1700
Responsivity	$\mathscr{R}$	A/W	0.4–0.6	0.4–0.5	0.75–0.95
Dark current	I_D	nA	1–10	50–500	0.5–2.0
Rise time	τ_r	ns	0.5–1	0.1–0.5	0.05–0.5
Modulation (bandwidth)	B_m	GHz	0.3–0.7	0.5–3	1–2
Bias voltage	V_B	V	5	5–10	5

Table.3.1Generic operating parameters of Si, Ge, and InGaAs avalanche photodiodes

Parameter	*Symbol*	*Unit*	*Si*	*Ge*	*InGaAs*
Wavelength range	λ	nm	400–1100	800–1650	1100–1700
Avalanche gain	M	—	20–400	50–200	10–40
Dark current	I_D	nA	0.1–1	50–500	10–50 @ $M = 10$
Rise time	τ_r	ns	0.1–2	0.5–0.8	0.1–0.5
Gain · bandwidth	$M \cdot B_m$	GHz	100–400	2–10	20–250
Bias voltage	V_B	V	150–400	20–40	20–30

3.16. PROBLEMS WITH SOLUTION

1. **Calculate the intrinsic carrier concentration of GaAs at 300K using the following parameters. Eg=1.43eV, Nc=4.4 × 10^{23} m^3 and Nv = 4 × 10^{23} m^3.**

Solution

At 300 K, the value of the product (kT) in eV becomes

$$kT = 1.38 \times 10^{-23} \times 300\text{J} = \frac{4.14 \times 10^{-21}}{1.6 \times 10^{-19}} = 0.0259 \text{ eV}$$

The intrinsic carrier concentration of GaAs at 300K can be

$$n_i = \sqrt{4.4 \times 10^{23} \times 8.2 \times 10^{24}} \exp\left(-\frac{1.43}{2 \times 0.0259}\right) = \approx 1.96 \times 10^{12} \text{ m}^{-3}$$

2. **The carrier recombination lifetime for an LED operating at 50mA DC drive current is 1nS. Estimate the values of electrical and optical bandwidth of the LED.**

Solution

The electrical bandwidth of the LED can be calculated as

$$f_{3dB-el} = \frac{1}{2\pi \times 1 \times 10^{-9}} = \frac{10^8}{6.28} = 159.23 \text{ MHz}$$

The optical bandwidth of the LED is equivalently,

$$f_{3dB-op} = \sqrt{3} f_{3dB-el} = 275.8 \text{ MHz}$$

3. **A Fabry perot laser diode with a 400μm long cavity uses GaAs as the material in the active region with uncoated facets. The cavity offers an average loss of 1000 m^1 at the operating wavelength. Estimate the value of the threshold gain assuming the refractive index of GaAs to be 3.6.**

Solution

The reflectivities of the uncoated facets can be estimated as

$$R_1 = R_2 = \left(\frac{3.6 - 1}{3.6 + 1}\right)^2 = 0.32$$

The threshold gain can be estimated as

$$\Gamma g_{th} = \overline{\alpha} + \frac{1}{2L} \ln\left(\frac{1}{R_1 R_2}\right)$$

$$= 1000 + \frac{1}{400 \times 10^{-6}} \ln\left(\frac{1}{0.32 \times 0.32}\right) = 6697.17 \text{m}^{-1}$$

4. A Si P-i-n photodetector is used in the front end of an optical receiver. The width 0f the i- region is 5 μm and the device area is 0.5 × 10^7 m^2. The load resistance and the input resistance of the amplifier are 1KΩ and 3KΩ respectively. The input capacitance of the amplifier is 5pf. Estimate the bandwidth of the photodetector circuit.

Solution The junction capacitance of the photodetector can be obtained as

$$C_j = \frac{\epsilon A}{W} = \frac{8.854 \times 10^{-12} \times 11.9 \times 0.5 \times 10^{-7}}{5 \times 10^{-6}} = 1 \text{ pF}$$

The effective capacitance is given by

$$C_T = 1 \text{ pF} + 5 \text{ pF} = 6 \text{ pF}$$

The effective value of resistance can be estimated as

$$R_T = \frac{1 \times 3}{1 + 3} \text{ k}\Omega = 0.75 \text{ k}\Omega$$

The bandwidth of the photodetector circuit is

$$B = \frac{1}{2 \times 3.14 \times 0.75 \times 10^3 \times 6 \times 10^{-12}} = 35.38 \text{ MHz}$$

5. Consider a Si APD operating at 300°K and with a load resistor R_L = 1000 Ω. For this APD assume the responsivity R = 0.65 A/W and let x = 0.3. (a) If dark current is neglected and 100 nW of optical power falls on the photodetector, what is the optimum avalanche gain? (b) What is the SNR if B_e = 100 MHz? (c) How does the SNR of this APD compare with the corresponding SNR of a Si pin photodiode? Assume the leakage current is negligible.

Solution: (*a*) Neglecting dark current and with $I_p = \mathscr{R}P$, we have

$$M_{opt} = \left(\frac{4k_BT}{xqR_L\mathscr{R}P}\right)^{1/(x+2)}$$

$$= \left[\frac{4(1.38 \times 10^{-23})(300)}{0.3(1.60 \times 10^{-19})(1000)(0.65)(100 \times 10^{-9})}\right]^{1/2.3} = 42$$

(*b*) Neglecting dark current and with $F(M) = M^x = (42)^{0.3}$, we have

$$\text{SNR} = \frac{(\mathscr{R}PM)^2}{\left[2q\mathscr{R}PM^{2.3} + \left(\frac{4k_BT}{R_L}\right)\right]B_e}$$

$$= \frac{[(0.65)(100 \times 10^{-9})(42)]^2}{\left[2(1.6 \times 10^{-19})(0.65)(100 \times 10^{-9})42^{2.3} + \left(\frac{4(1.38 \times 10^{-23})(300)}{(1000)}\right)\right](100 \times 10^6)} = 659$$

or in decibels, SNR = 10 log 659 = 28.2 dB

(*c*) For a *pin* photodiode with $M = 1$, the above equation yields SNR(*pin*) = 2.3 = 3.5 dB. Thus, compared to a *pin* photodiode, the APD improves the SNR by 24.7 dB.

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6. **If the photodiode capacitance is 3pf, the amplifier capacitance is 4pf, the load resistance is 1kΩ and the amplifier input resistance is 1MΩ, the Cr = 7pf and RT = 1kΩ. What is the circuit bandwidth.**

solution:

$$B_c = \frac{1}{2\pi R_T C_T} = 23\,\text{MHz}$$

If we reduce the photodector load resistance to 50 Ω, then the circuit bandwidth becomes $B_c = 455$ MHz.

7. **A Si P-i-n photodetector has a 5μm thick intrinsic region. Assuming, that the field in the depleted intrinsic region is high enough to cause velocity saturation of the carriers, estimate the 3- dB bandwidth of the photodetector. Assume the saturation velocity of the carrier to be 10 to the power of 5 m/s.**

solution:

The 3-dB bandwidth of the *p-i-n* detector can be obtained as

$$f_{3dB} = \frac{0.4\,v_s}{W} = \frac{0.4 \times 10^5}{5 \times 10^{-6}} = 8\ \text{GHz}$$

The 3-dB bandwidth of the photodetector is 8 GHz.

8. **A photodiode is constructed of GaAs, which has a bandgap energy of 1.43 eV at 300 K. What is the cutoff wavelength of this device?**

Solution:

$$\lambda_c = \frac{hc}{E_g} = \frac{(6.625 \times 10^{-34}\ \text{J·s})(3 \times 10^8\ \text{m/s})}{(1.43\ \text{eV})(1.6 \times 10^{-19}\ \text{J/ev})} = 869\ \text{nm}$$

This GaAs photodiode will not operate for photons of wavelength greater than 869 nm.

9. A particular Ga1–x Alx As laser is constructed with a material ratio x = 0.07. Find (a) the band gap of this material; (b) the peak emission wavelength.

(a) we have Eg = 1.424 + 1.266x + 0.266x2

Eg = 1.424 + 1.266(0.07) + 0.66(0.07)2 = 1.51 eV

(b) Using this value of the bandgap energy in

$$\lambda(\mu m) = \frac{1.240}{E_g(\text{eV})}$$

λ(μm) = 1.240/1.51 = 0.82 μm = 820 nm

10. A double-heterojunction InGaAsP LED emitting at a peak wavelength of 1310 nm has radiative and non radiative recombination times of 30 and 100 ns, respectively. The drive current is 40 mA. Find (a) the bulk recombination time; (b) the internal quantum efficiency; and (c) the internal power level.

(a)The bulk recombination lifetime is

$$\tau = \frac{\tau_r \tau_{nr}}{\tau_r + \tau_{nr}} = \frac{30 \times 100}{30 + 100} \text{ns} = 23 .1 \text{ ns}$$

(b)The internal quantum efficiency is

$$\eta_{int} = \frac{1}{1 + \tau_r / \tau_{nr}} = \frac{\tau}{\tau_r}$$

=0.77

(c) Internal power level of $P_{int} = \eta_{int} \frac{I}{q} h\nu = \eta_{int} \frac{hcI}{q\lambda}$

$$P_{int} = \eta_{int} \frac{hcI}{q\lambda}$$

$$= 0.77 \frac{(6.6256 \times 10^{-34} \text{J} \cdot \text{s})(3 \times 10^8 \text{m/s})(0.040\text{A})}{(1.602 \times 10^{-19} \text{C})(1.31 \times 10^{-6} \text{m})}$$

$$= 29.2\text{m W}$$

11. Assume a typical value of n = 3.5 for the refractive index of an LED material. What percent of the internally generated optical power is emitted into an air medium?

$$\eta_{ext} = \frac{1}{n(n+1)^2} = \frac{1}{3.5(3.5+1)^2} = 1.41 \text{ percent}$$

This shows that only a small fraction of the internally generated optical power is emitted from the device.

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12. A particular LED has a 5-ns injected carrier lifetime. When no modulation current is applied to the device, the optical output power is 0.250 mW for a specifi ed dc bias. Assuming parasitic capacitances are negligible, what are the optical outputs at modulation frequencies of (a) 10 MHz and (b) 100 MHz?

(a) The optical output at 10 MHz is

$$P(\omega) = P_0[1 + (\omega\tau_i)^2]^{-1/2}$$

$$P(\omega) = \frac{0.250}{\sqrt{1 + [2\pi(10 \times 10^6)(5 \times 10^{-9})]^2}}$$
$$= 0.239 \text{ mW} = 239\ \mu\text{W}$$

(b) Similarly, the optical output at 10 MHz is

$$P(\omega) = \frac{P_0}{\sqrt{1 + (\omega\tau_i)^2}}$$
$$= \frac{0.250}{\sqrt{1 + [2\pi(100 \times 10^6)(5 \times 10^{-9})]^2}}$$
$$= 0.076 \text{ mW} = 76\ \mu\text{W}$$

Thus the output of this particular device decreases at higher modulation rates.

13. A given silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900 nm. Suppose 0.5 mW of optical power produces a multiplied photocurrent of 10 mA. What is the multiplication M?

$$I_p = \mathscr{R} P_{in} = \frac{\eta q}{h\nu} P_{in} = \frac{\eta q \lambda}{hc} P_{in}$$
$$= \frac{(0.65)(1.6 \times 10^{-19}\text{C})(9 \times 10^{-7}\text{m})}{(6.625 \times 10^{-34}\text{ J} \cdot \text{s})(3 \times 10^8 \text{ m/s})} 5 \times 10^{-7} \text{ W}$$
$$= 0.235\ \mu\text{A}$$

$$M = \frac{I_M}{I_p} = \frac{10\ \mu\text{A}}{0.235\ \mu\text{A}} = 43$$

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3.17. PROBLEMS FOR PRACTICE

1. Consider the following parameter values for GaAs at 300 K:

- Electron rest mass $m = 9.11 \times 10^{-31}$ kg
- Effective electron mass $m_e = 0.068\ m = 6.19 \times 10^{-32}$ kg
- Effective hole mass $m_h = 0.56\ m = 5.10 \times 10^{-31}$ kg
- Bandgap energy $E_g = 1.42$ eV

What is the intrinsic carrier concentration?

2. A particular $Ga_{1-x}Al_xAs$ laser is constructed with a material ratio $x = 0.07$. Find (a) the bandgap of this material; (b) the peak emission wavelength?

3. A double-hetero-junction InGaAsP LED emitting at a peak wavelength of 1310 nm has radiative and nonradiative recombination times of 30 and 100 ns, respectively. The drive current is 40 mA. Find (a) the bulk recombination time; (b) the internal quantum efficiency; and (c) the internal power level?

4. Assume a typical value of $n = 3.5$ for the refractive index of an LED material. What percent of the internally generated optical power is emitted into an air medium?

5. A particular LED has a 5-ns injected carrier lifetime. When no modulation current is applied to the device, the optical output power is 0.250 mW for a specified dc bias. Assuming parasitic capacitances are negligible, what are the optical outputs at modulation frequencies of (a) 10 MHz and (b) 100 MHz?

6. Assume that the cleaved mirror end faces of a GaAs laser are uncoated and that the outside medium is air. What is the reflectivity for normal incidence of a plane wave on the GaAs-air interface if the GaAs refractive index is 3.6?

7. A GaAs laser operating at 850 nm has a 500-mm length and a refractive index $n = 3.7$. (a) What are the frequency spacing and the wavelength spacing? (b) If, at the half-power point, $l - l_0 = 2$ nm, what is the spectral width s of the gain?

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8. A high speed In 0.53 Ga 0.47 As pin photodetector is made with a depletion layer thickness of 0.15μm. what percent of incident photons are absorbed in this photodetector at 1310 nm if the absorption coefficient is 1.5 μm at this wavelength?

9. The energy bandgap of InSb at 300K is 0.17 eV . The ratio of the effective mass of hole 10 that of electron for this material is 40. Estimate the values of ionization threshold of electrons for InSb.

10. For an InGaAs photodetector operating at 1550 nm, R = 0.90 A/W. What is the Noise Equivalent Power in the thermal noise limited case if the load resistor is RL = 1000Ω and T=300°K ?

11.A given silicon APD has a quantum efficiency of 65 percent at a wavelength of 900nm.Suppose 0.5μw of optical power produces a multiplied photo current of 10μA.What is the multiplication M?

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Reference video links for students

UNIT 3

Operation of LED

https://youtu.be/kaKLmKhUrf4

Working Principle of LED

https://youtu.be/2_YS3DZQIbI

LED Strutures

https://youtu.be/WxiOQvq2P-k

LASER Diode Modes and Strcutures

https://youtu.be/5dFTg-uRjJk

Noise in Photodetectors

https://youtu.be/Y-dbh5UdNqk

Photodiode

https://youtu.be/BtQ7qY-uqs8?list=PLX2ySd0Fawh97q8EUfbr22sBnlRs-vt3P

Photodiode-Basic principle

https://youtu.be/SFc673lEyQA?list=PLX2ySd0Fawh97q8EUfbr22sBnlR_s-vt3P

Avalanche Phtodiode

https://youtu.be/jFV9yJnPBIw?list=PLX2ySd0Fawh97q8EUfbr22sBnlR_s-vt3P

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6.4.Assignments

UNIT III

1. Write assignment on the topic by explaining how optical light is used to design a computing chip based on brain function-case study (K2,CO3)

2. An Engineer has two GaAlAs LEDs: one has band-gap energy of 1.540 eV and the other has x=0.015.Find the aluminum mole fraction x and the emission wavelength for the first LED.Find the band-gap energy and the emission wavelength of the other LED. (K2,CO3)

3. An Engineer has GaAlAs LED of band-gap energy 1.540 eV. Estimate the aluminum mole fraction x and the emission wavelength for the LED. (K2,CO3)

4. A GaAs Laser operating at 850nm, has a 500um length and refractive index n=3.7. What are the frequency and wavelength spacings. If at the half power point, $\lambda-\lambda_0$=2nm,Estimate the spectral width σ of the gain. (K2,CO3)

5. The radiative and non-radiative recombination life time of the minority carriers in the active region of a double heterojunction LED are 60ns and 100ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87μm at a drive current of 40mA(K2,CO3)

6. An optical source is selected from a batch characterized as having lifetimes that follow a slow internal degradation mode. The -3dB mean time to failure of these devices at room temperature is specified as 5 x 104h.If the device emits 1mw at room temperature, what is the expected optical output power after 1 month of operation ,after 1 year, and after 5 years?(K2,CO3)

7. Consider a Lambertian LED source with a 50mm diameter emitting area.If this source is connected to an optical fiber that has a 62.5 μm core diameter with NA = 0.18.Show that the coupling efficiency is 3.24%=-14.9dB.If a lens is used to improve the coupling efficiency ,show that the maximum magnification is Mmax = 1.56.

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6.4.Assignments

7. An InGaAs pin photodiode has the following parameters at a wave length of 1300 nm: ID = 4 nA, η = 0.90, RL = 1000 W, and the surface leakage current is negligible. The incident optical power is 300 nW (–35 dBm), and the receiver bandwidth is 20 MHz. Find the various noise terms of the receiver.

8. A given silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900 nm. Suppose 0.5 μW of optical power produces a multiplied photocurrent of 10 μA. What is the multiplication M?

9. In a 100-ns pulse, 6 X 106 photons at a wavelength of 1300 nm fall on an InGaAs photo detector. On the average, 5.4 X 106 electron–hole (e–h) pairs are generated. Determine the quantum efficiency.(K3,CO3)

10. A double-hetero-junction InGaAsP LED emitting at a peak wavelength of 1310 nm has radiative and nonradiative recombination times of 30 and 100 ns, respectively. The drive current is 40 mA. Find (a) the bulk recombination time; (b) the internal quantum efficiency; and (c) the internal power level? (K3,CO3)

11. An InGaAsP/In P LED emits peak power at 1330nm at a drive current of 50mA.The radiative and non-radiative recombination time of 25ns and 100ns respectively. the refractive index of active region is 3.5, calculate external quantum efficiency of the LED. Also calculate the optical power yielded internally in the LED. (K3,CO3)

12. Consider a Si APD operating at 300°K and with a load resistor RL = 1000 Ω. For this APD assume the responsivity R = 0.65 A/W and let x = 0.3. (a) If dark current is neglected and 100 nW of optical power falls on the photo detector, what is the optimum avalanche gain? (b) What is the SNR if Be = 100 MHz? (c) How does the SNR of this APD compare with the corresponding SNR of a Si pin photodiode? Assume the leakage current is negligible. (K3,CO3)

13. A Fabry Perot laser diode with a 400μm long cavity uses GaAs as the material in the active region with uncoated facets. The cavity offers an average loss of 1000 m^{-1} at the operating wavelength. Estimate the value of the threshold gain assuming the refractive index of GaAs to be 3.6. (K3,CO3)

14. A Si PIN photo detector has a 5μm thick intrinsic region. Assuming, that the field in the depleted intrinsic region is high enough to cause velocity saturation of the carriers, estimate the 3- dB bandwidth of the photo detector. Assume the saturation velocity of the carrier to be 10 to the power of 5 m/s. (K3,CO3)

15. A Si PIN photo detector is used in the front end of an optical receiver. The width of the i- region is 5 μm and the device area is 0.5 × 107 m^2. The load resistance and the input resistance of the amplifier are 1KΩ and 3KΩ respectively. The input capacitance of the amplifier is 5pf. Estimate the bandwidth of the photo detector circuit. (K3,CO3)

6.5. PART A Q&A

UNIT III TWO MARKS QUESTIONS AND ANSWERS

1. What are the advantages of LED? (K_1,CO3)

The advantages of LEDs are

- They have long life
- Fabrication is easier
- Used for short distance communication
- Less expensive
- LEDs are less complex circuits than Laser diodes

2. Explain Quantum Efficiency of a photo detector. (K_2,CO3)

Quantum efficiency is defined as the number of the electron-hole carrier pairs generated per incident photon of energy hu, is given by

- η = (number of electron- hole pairs generated) / (number of incident photons)

$$\frac{I_p/q}{\frac{p_o}{hv}}$$

- where I_p is the photon current, q is the charge of the electron
- p_o is the optical output power, v is the frequency of the optical signal.

3. What do you mean by Laser diode? (K_1,CO3)

A laser diode, or **LD**, is an electrically pumped semiconductor laser in which the active medium is formed by a p-n junction of a semiconductor diode similar to that found in a light-emitting diode.

4. What do you mean by Avalanche Photo Diode? (K_1,CO3)

An **avalanche photodiode (APD)** is a highly sensitive semiconductor electronic device that exploits the photoelectric effect to convert light to electricity. APDs can be thought of as photo-detectors that provide a built-in first stage of gain through avalanche multiplication.

5. What do you mean by direct band gap Materials? (K_1,CO3)

In some materials a direct transition is possible from valance band to conduction band. Such type of materials is called as direct band gap materials. Ex. GaAs, InP, InGaAs.

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6. Explain about hetero-junction? List the advantages of hetero- junction? (K_2,CO3)

A hetero-junction is an interface between two adjoining single crystal semiconductors with different band-gap energies. Devices are fabricated with hetero-junctions are said to have hetero-structure.

Advantages of Hetero-junction are

- Carrier and optical confinement
- High output power
- High coherence and stability.

7. What is the principle of operation of LASER? (K_1,CO3)

The principle of operation of LASER is population inversion. The population of the upper energy level is greater than lower energy level i.e. N_2 is > N_1.this condition is known as population inversion.

8. Why is silicon not used to fabricate LED or LASER diode? (K_2,CO3)

Silicon is not used to fabricate LED or LASER diode because

- It is an indirect band-gap semiconductor
- Its energy level is 1.1eV, radiated emission corresponds to infrared but not the visible light.

9. Explain the three modes of the cavity of LASER Diode? (K_2,CO3)

The three modes of the cavity of LASER are

- Longitudinal modes, related to the length L of the cavity
- Lateral modes lie in the plane of the P-N junction. These modes depend upon the side wall preparation and width of the cavity
- These modes determine the radiation pattern of the LASER.

10. Discuss about Photocurrent and Threshold Current? (K_2,CO3)

The threshold current is conventionally defined by extrapolation of the lasing region of the Power Vs Current curve. At high power outputs, the slope of the curve decreases because of junction heating

The high electric field present in the depletion region causes the carriers to separate and be collected across the reverse- biased junction. This gives to a current flow in the external circuit, with one electron flowing for every carrier pair generated. This current flow is known as photocurrent.

11. What are the necessary features of a photo detector? (K_1,CO3)

The designed features of a photo detector are

- High quantum efficiency
- Low rise time or faster response
- Low dark current.

12. Interpret Impact Ionization? (K_2,CO3)

In order for carrier multiplication to take place, the photo-generated carriers must traverse a region where a very high electric field is present. In this high field region, a photo generated electron or hole can gain energy so that it ionizes bound electrons in the valence band upon colliding with them. This current multiplication mechanism is known as impact ionization.

13. Explain Avalanche Effect ? (K_2,CO3)

The newly created carriers are accelerated by the high electric field, thus gaining enough energy to cause further impact ionization. This phenomenon is called avalanche effect.

14. Define – Multiplication M? (K_1,CO3)

The multiplication M for all carriers generated in the photodiode is defined by

$$M = I_M / I_P$$

I_M = average value of the total multiplied output current

I_P = primary unmultiplied output current

15. Summarize the advantages of Quantum Well LASERs? (K_2,CO3)

The advantages of Quantum Well LASERs are

- High threshold current density
- High modulation speed
- High line width of the device.

16.Differentiate LEDs and Laser diodes? (K_2,CO3)

LED	Laser diodes
The output obtained is incoherent.	The output obtained is coherent.
Less expensive and less complex.	More expensive and more complex.
Long life time.	Less life time.
Their response is fast	Their response is faster than LED
Bandwidth of LED is moderate	Bandwidth of Laser diode is higher

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17. Distinguish between direct and indirect band-gap materials? (K_2,CO3)

Direct Band-gap Materials	Indirect Band-gap Materials
The electron and hole have the same momentum value.	The conduction band minimum and the valence band maximum energy level occur at different values of momentum
Direct transmission is possible from valence band to conduction band	Direct transmission is not possible from valence band to conduction band.

18. Compare surface and edge emitting LEDs? (K_2,CO3)

Surface Emitting LED	Edge emitting LED
Wider spectral width (typically 125 nm)	Narrow spectral width (typically 75 nm)
Emission pattern is less directional	Emission pattern is more directional

19.Compare the performance of APD with PIN diode? (K_2,CO3)

APD	PIN
Excellent Linearity	Linearity is less compared to APD
High sensitivity	Less sensitive compared to APD (15 dB less)
High signal to Noise Ratio	Low signal to Noise Ratio

20. Explain Internal Quantum Efficiency? (K_2,CO3)

The internal quantum efficiency is the fraction of the electron-hole pairs that recombine radiatively. If the radiative recombination rate is R and the nonradiative recombination ratio is R_{nr}, then the internal quantum efficiency is the ratio of the radiative recombination rate to the total recombination rate.

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21. Define-External Quantum Efficiency? (K_1,CO3)

The external quantum efficiency is defined as the number of photons emitted per radiative electron-hole pair recombination above threshold.

22. What are the advantages of Quantum Well LASERs? (K_1,CO3)

The advantages of Quantum Well LASERs are

High threshold current density
High modulation speed
High line width of the device.

23. What is Multiplication M. (K_1,CO3)

The multiplication M for all carriers generated in the photodiode is defined by
M = IM / IP
IM = average value of the total multiplied output current
IP = primary unmultiplied output current

24. What are the basic methods of current confinement? (K_1,CO3)

The basic methods of current confinement are

- Inner strip confinement
- Proton confinement
- Preferential dopant diffusion
- RE growth of back biased PN junction.

25. What are the three requirements of Laser action? (K_1,CO3)

The three requirements of Laser action are

- Absorption
- Spontaneous emission
- Stimulated emission

26. What is a DFB LASER? Differentiate DFB LASER from other types of LASERs? (K_1,CO3)

In Distributed Feed Back LASER, the lasing action is obtained by periodic variations of refractive index, which are incorporated into multilayer structure along the length of the diode. DFB LASER does not require optical feedback unlike the other LASERs.

27. What are the fundamental structures of Index guided lasers? (K_1,CO3)

The fundamental structures of index guided lasers are

- Bent layer configuration
- Selectively diffused construction
- Buried hetero structure
- Varying thickness structure

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28. What is a DFB LASER? Differentiate DFB LASER from other types of LASERs? (K_1,CO3)

In Distributed FeedBack LASER, the lasing action is obtained by periodic variations of refractive index, which are incorporated into multilayer structure along the length of the diode. DFB LASER does not require optical feedback unlike the other LASERs.

29. Give some types of Photo-detectors. (K_1,CO3)

The types of Photo-Detectors are

- Photodiodes
- Phototransistors
- Photomultipliers
- Semiconductor based detectors
- Pyroelectric detectors.

30. Define – Photocurrent(K_1,CO3)

The high electric field present in the depletion region causes the carriers to separate and be collected across the reverse- biased junction. This gives to a current flow in the external circuit, with one electron flowing for every carrier pair generated. This current flow is known as photocurrent.

31. What are the necessary features of a photo detector? (K_1,CO3)

The designed features of a photo detector are

- High quantum efficiency
- Low rise time or faster response
- Low dark current.

32. When an LED has 2V applied to its terminals, it draws 100mA and produce 2mW of optical power. Determine conversion efficiency of the LED from electrical to optical power. (K2,CO3)

Given data Vin = 2V, Iin = 100*10-3 A, Pout = 2 *10-3W,

Formula: LED conversion efficiency = (Pout / Pin)
Pin = Vin * Iin = 2 * 100*10-3

conversion efficiency = (Pout / Pin) = (2 *10-3 / 2 * 100*10-3)* 0.01

33. What is the principle of operation of LASER? (K_1,CO3)

The principle of operation of LASER is population inversion. The population of the upper energy level is greater than lower energy level i.e. N2 is > N1.this condition is known as population inversion.

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34. In a 100ns pulse, 6 *106 photons at wavelength of 1300nm fall on an InGaAs photo-detector on the average, 5.4 *106 electron-hole pairs are generated. Find the quantum efficiency. **(K2,CO3)**

Given $\frac{I_p}{q} = 5.4 * 10^6$

$\frac{p_o}{h\upsilon} = 6 * 10^6$

where I_p is the photon current
q is the charge of the electron
p_o is the optical output power
υ is the frequency of the optical signal.

Quantum efficiency = $5.4 * 10^6 / 6 * 10^6 = 0.9$
% Quantum efficiency = 90%

35. Define – Quantum Efficiency of a photo detector (K_1,CO3)

Quantum efficiency is defined as the number of the electron-hole carrier pairs generated per incident photon of energy $h\upsilon$, is given by

η = (number of electron- hole pairs generated) / (number of incident photons)

i.e.

$$\frac{I_p/q}{\frac{p_o}{h\upsilon}}$$

where I_p is the photon current, q is the charge of the electron
p_o is the optical output power, υ is the frequency of the optical signal.

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6.6. PART B Questions

1. Draw the structures of SLED and ELED and explain their principle of operation (K2,CO3)

2. Explain the three key processes involved in the laser action and explain its principle(K2,CO3)

3. Discuss the structure of Fabry-Perot resonator cavity laser diode and DFB laser diode. Explain three modes of LASER(K2,CO3)

4. Derive expressions for the signal to Noise Ratio of photo detector by incorporating all noise sources(K2,CO3)

5. Derive S/N ratio of a photodetector(K2,CO3)

6. Outline the three factors that decide the response time of detector and noise sources of photodetector? (K2,CO3)

7. Discuss about direct and indirect bandgap materials. (K2,CO3)

8. Derive the expression for quantum efficiency of LED and its LED power(K2,CO3)

9. Discuss in detail modes and threshold conditions of LASER(K2,CO3)

10. Describe the characteristics, structure and operation of pin photodetector(K2,CO3)

11. Describe the characteristics, structure and operation APD photodetector. (K2,CO3)

12. Explain the working of n hetero structure LED(K2,CO3)

13. Briefly explain the different noise sources of a photo detector(K2,CO3)

14. Derive the internal quantum efficiency of an LED (K2,CO3)

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6.7.Supportive online Certification courses:

UNIT III

NPTEL/SWAYAM:

Fiber Optic Communication Technology Prof. Deepa Venkitesh, IITM
12 Weeks

COURSERA:

Active optical devices specialization

Light Emitting Diodes and Semiconductor Lasers

Free online certification course on LEARN FIBER OPTICS THE EASY WAY

UDEMY:

Introduction to fiber optic cabling

ASLMS Online Learning Centre:

Introduction to Lasers (Non-CME)

EDX COURSE:

Optical Materials and Devices

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6.8. Real time Applications in day to day life and to Industry

UNIT III:

1. High intensity Laser medical effect -

https://www.youtube.com/watch?v=kn7JXDGpEfI

2. LASER –

https://www.youtube.com/watch?v=_8LNjaWAv9o

3. LASIK surgery –

https://www.youtube.com/watch?v=Bb8bnjnEM00

4. Applications of LED –

https://www.youtube.com/watch?v=Zo4ZyAaBwHI

5. Applications of LASER -

https://www.youtube.com/watch?v=cdRxQjrL744

6. Semiconductor Photodiode –

https://www.youtube.com/watch?v=S9jMMALKKKs

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6.9. Content beyond syllabus

Multispectral Imaging, Detection, and Active Reflectance (MiDAR)

NASA has developed a novel next-generation remote sensing instrument with advanced scientific capabilities for Multispectral Imaging, Detection and Active Reflectance (MiDAR). The MiDAR transmitter and receiver demonstrate a novel and cost-effective solution for simultaneous high-frame-rate, high-signal-to-noise ratio (SNR) multispectral imaging, with hyperspectral potential, high-bandwidth simplex communication, and in-phase radiometric calibration. The use of computational imaging further allows for multispectral data to be fused using Structure from Motion (SfM) and Fluid Lensing algorithms to produce 3D multispectral scenes and high-resolution underwater imagery of benthic systems as part of future scientific airborne field campaigns. THE TECHNOLOGY The MiDAR transmitter emits coded narrowband structured illumination to generate high-frame-rate multispectral video, perform real-time radiometric calibration, and provide a high-bandwidth simplex optical data-link under a range of ambient irradiance conditions, including darkness. A theoretical framework, based on unique color band signatures, is developed for multispectral video reconstruction and optical communications algorithms used on MiDAR transmitters and receivers. Experimental tests demonstrate a 7-channel MiDAR prototype consisting of an active array of multispectral high-intensity light-emitting diodes (MiDAR transmitter) coupled with a state-of-the-art, high-frame-rate NIR computational imager, the NASA FluidCam NIR, which functions as a MiDAR receiver. A 32-channel instrument is currently in development.Preliminary results confirm efficient, radiometrically-calibrated, high signal-to-noise ratio (SNR) active multispectral imaging in 7 channels from 405-940 nm at 2048x2048 pixels and 30 Hz

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6.9. Content beyond syllabus

These results demonstrate a cost-effective and adaptive sensing modality, with the ability to change color bands and relative intensities in real-time, in response to changing science requirements or dynamic scenes. Potential applications of MiDAR include high-resolution nocturnal and diurnal multispectral imaging from air, space and underwater environments as well as long- distance optical communication, bidirectional reflectance distribution function characterization, mineral identification, atmospheric correction, UV/fluorescent imaging, 3D reconstruction using Structure from Motion (SfM), and underwater imaging using Fluid Lensing. Multipurpose sensors, such as MiDAR, which fuse active sensing and communications capabilities, may be particularly well-suited for mass-limited robotic exploration of Earth and the solar system and represent a possible new generation of instruments for active optical remote

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6.9. Content beyond syllabus

APPLICATIONS

- The technology has several potential applications:
- Multispectral Remote Sensing from aircraft, robotic explorers, spacecraft, and underwater environments in both low-light and normal lighting conditions
- Hyper spectral Imaging
- Simultaneous Optical Communications
- Fluid Lensing for cm-scale benthic imaging
- Mineral identification
- UV/fluorescent imaging from UAVs
- 3D imaging using Structure from Motion (SfM)
- Mass-limited robotic exploration of Earth and the solar system
- Noninvasive medical imaging and diagnosis
- Semiconductor imaging and engineering structure analysis

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6.9.Content beyond syllabus - SCHOTTKY PHOTODIODE

The Schottky photodiode utilizes the same basic structure and technology that is used by the standard Schottky barrier diode.

The Schottky photodiode is able to provide capabilities of high speed and long wavelength detection. These unique attributes provide a niche area which the Schottky diode is able to meet in a way that other forms of photodetector are not able.

Schottky photodiode basics

As the name indicates, the Schottky photodiode uses the Schottky diode (or Schottky barrier diode as it is sometimes called) as its basis of the photodetector.

The Schottky photodiode is unique as a photodetector as it is able to operate in two photo-detection modes:

- ***Electron pair generation:*** This occurs from band to band or energy gap excitation in the semiconductor.
- ***Emission of carriers:*** The emission of carriers occurs from the metal to the semiconductor over the Schottky barrier. This often referred to as internal photoemission.

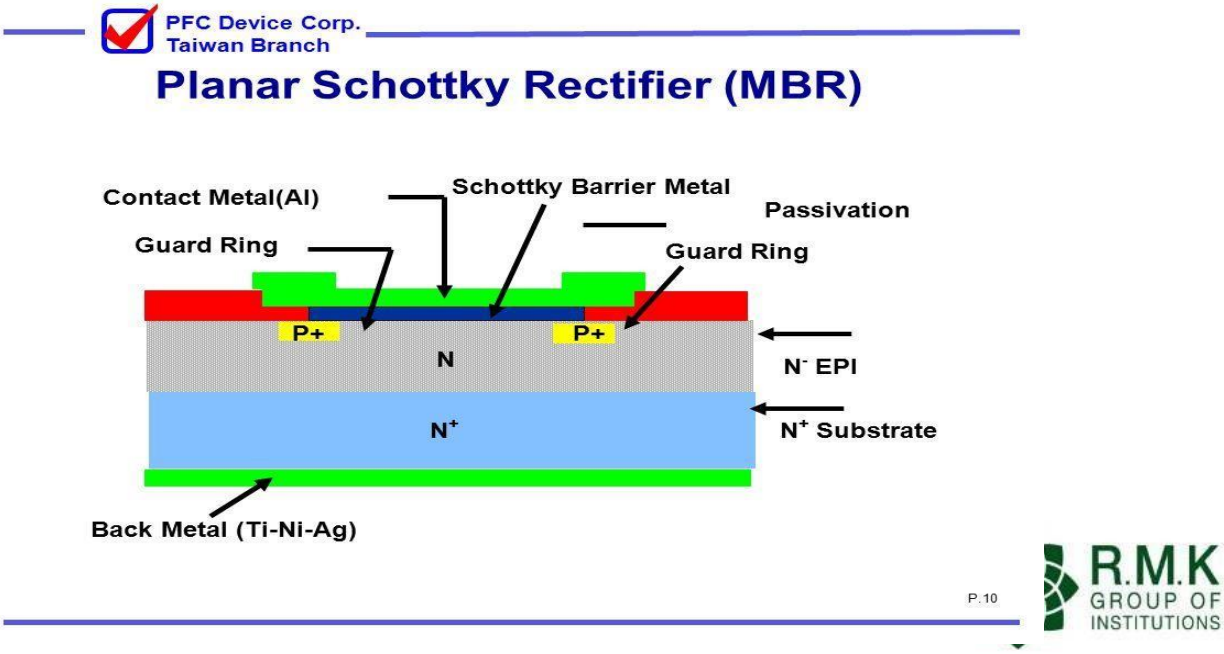
The metal-semiconductor junction provides a similar action to that of the intrinsic layer of the PIN photodiode. Accordingly is provides a larger areas for capture of the photon energy.

Schottky photodiode applications

The Schottky photodiode is particularly compatible with mature silicon and silicide technology. As a result these photodiodes have been widely used in CCD - charge coupled device - as the image sensing photodetector.

The Schottky photodiode can be integrated into a single chip along with CCD transfer gate - these are also compatible with silicon technology. The CCD itself forms a shift register to allow the data from the array of Schottky photodiode detectors to transferred out of the overall chip in a managed way - providing method of extracting the huge data files from the large array of photodetectors with a sensible number of leads on the integrated circuit.

As a result of these advantages the Schottky photodiode detector has been the most widely used technology for focal plane arrays.



7. Assessment Schedule (Proposed Date & Actual Date)

Assessment	Proposed Date	Actual Date
Unit 1 Assignment Assessment		
Unit Test 1		
Unit 2 Assignment Assessment		
Internal Assessment 1		
Unit 3 Assignment Assessment		
Unit Test 2		
Unit 4 Assignment Assessment		
Internal Assessment 2		
Unit 5 Assignment Assessment		
Revision Test 1		
Revision Test 2		
Model Exam		
End Semester Exam		

8. PRESCRIBED TEXT BOOKS AND REFERENCES

TEXT BOOKS:

1. Gerd Keiser, Optical Fiber Communications, McGraw -Hill International, Fourth Edition, 2010.

2. John. M. Senior, Optical Fiber Communications, Principles and Practice, Prentice Hall of India, Third Edition, 2008.

REFERENCES:

1. Ramaswami, Sivarajan and Sasaki, Optical Networks, Morgan Kaufmann, 2009.

2. Govind P. Agrawal, Fiber-Optic Communication Systems, John Wiley & Sons, 5th Edition,2021.

3. Chakrabarti P, Optical Fiber communication, McGraw Hill Education (India) Private Limited,2019.

4. Gower J, Optical Communication System, Prentice Hall of India, 2001.

9. Mini Project Suggestions
Laser Based Transmitter And Receiver

Using this laser based transmitter & receiver circuit you can communicate with your neighbours wirelessly. Instead of RF signals, light from a laser torch is used as the carrier in a circuit. A laser torch can transmit light up to a distance of abo500 metresut. The phototransistor of the receiver must be accurately oriented towards the laser beam from the torch. If there is any obstruction in the path of the laser beam, no sound will be heard from the receiver.

Laser based transmitter circuit

The transmitter circuit (Fig. 1) comprises condenser microphone transistor amplifier BC548 (T1) followed by an op-amp stage built around µA741 (IC1). The gain of the op-amp can be controlled with the help of 1-mega-ohm potmeter VR1. The AF output from IC1 is coupled to the base of transistor BD139 (T2), which, in turn, modulates the laser beam.

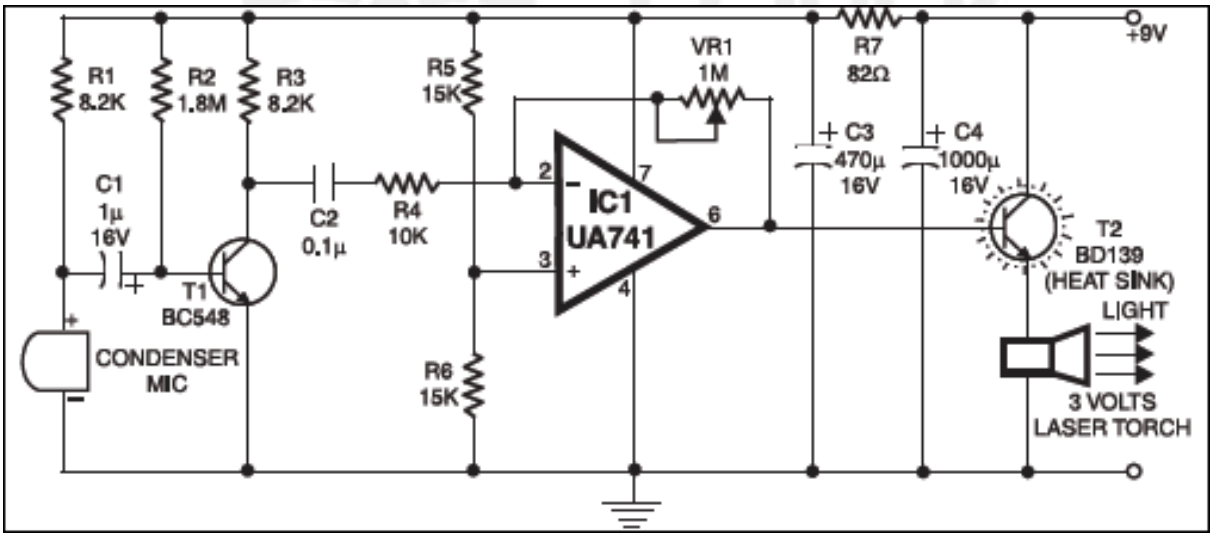


Fig 1. Laser based Transmitter

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9. Mini Project Suggestions
Laser Based Transmitter And Receiver

The transmitter uses 9V power supply. However, the 3-volt laser torch (after removal of its battery) can be directly connected to the circuit—with the body of the torch connected to the emitter of BD139 and the spring-loaded lead protruding from inside the torch to circuit ground.

The mic is the audio input. C1 is a coupling capacitor that connects the audio from the mic to T1. T1 is a transistor that is used to amplify the audio a bit. C2 couples the audio to IC1 which amplifies it some more. The audio from the IC is then routed to T2 which modulates the power to the LED. This puts the audio onto the light beam. VR1 is a variable resistor which adjusts the gain of the IC. C3, C4 and R7 are power filters that remove noise from the power supply. The other resistors Just supply the right voltage to the various parts

Laser based receiver circuit

T3 is a photo sensitive transistor which receives the laser from the LED transmitter. The transistor puts out a signal that is proportional to the light from the LED. This not only receives the signal it also demodulates it. C5 couples this audio to T4 and T5 which amplify the signal. The signal is them coupled through C8 to VR2. VR2 is a volume control for IC2. IC2 amplifies the signal more and drives the speaker, LS1.

To avoid 50Hz hum noise in the speaker, keep the photo-transistor away from AC light sources such as bulbs. The reflected sunlight, however, does not cause any problem. But the sensor should not directly face the sun.

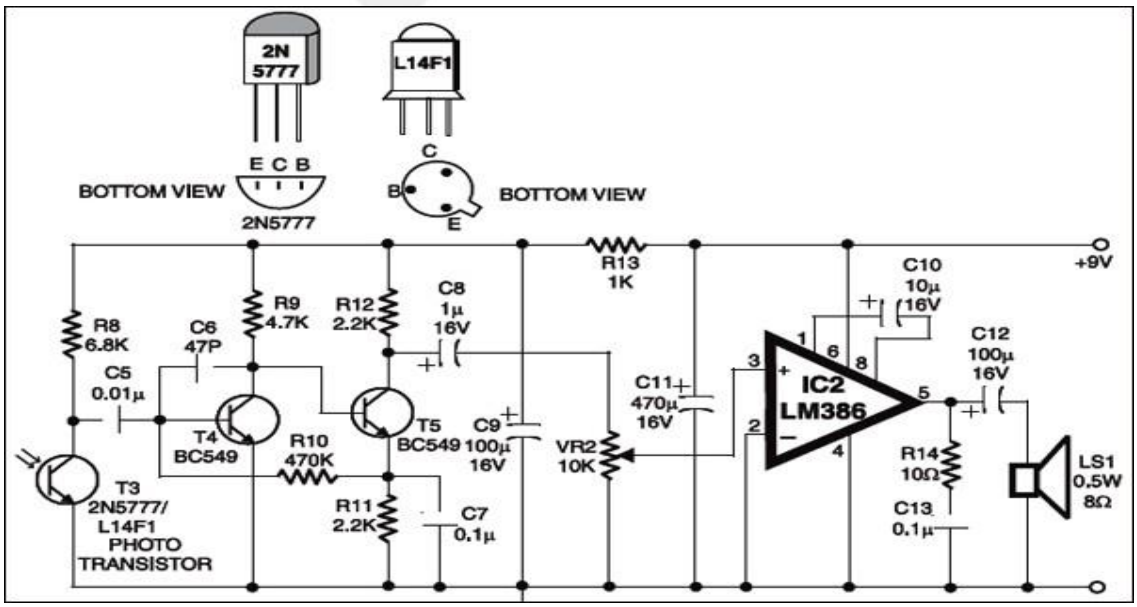

Fig 2. Laser based Receiver

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9. Mini Project Suggestions
PC TO PC FIBER OPTIC COMMUNICATION

INTRODUCTION

DEFINITION

The project PC to PC fiber optic communication deals with data transfer from one computer to another. It uses C programming and the serial ports of the PCs. The ports are programmed in C. We use MAX 232 to convert RS 232 logic to TTL logic and then an optical transmitter circuit to transmit data via fiber optic cable. The optical transmitter circuit has a LED which is matched as far the cable and MAX 232 is concerned. At the receiver we have an optical receiver circuit which receives data using a photo transistor and a MAX 232 again to convert TTL logic to RS 232 for the serial port at the receiving end computer. The desired baud rate can be set using the program. For transmitting data the program is executed once and whatever is present at the serial port is sent to the other
Computer via the fiber optic cable. At the receiver the program is executed to receive data on the serial port.

NEED

Data transfer is the need of the hour. Surely people can use internet for this. But presently it would be time consuming. One can implement data transfer using wireless medium. But then it will be a costly affair. The need, therefore, is felt for fiber optic communication which is cheaper and more suitable for the task. It is cheaper than wireless medium and is prone to lesser loss as compared to wireless medium. Fiber optics has their own advantages; larger bandwidth, high speeds, only to name a few. Therefore, we need to use optic fibers for the internet and LANs.

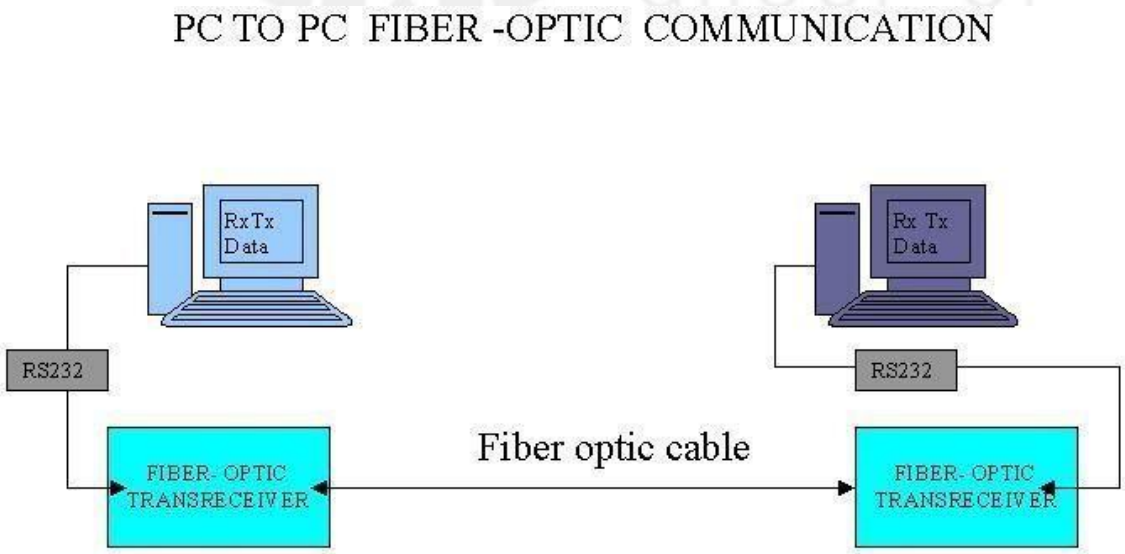
SCOPE

In the very beginning we have tried to send ASCII characters between two computers. But as we proceeded further we realized that this project can help transfer files at desired speed between two PCs. If we expand the idea a bit further, a network of PCs can be connected, like in the cable net available today. The Conventional LAN cables can be replaced by fiber optic cables which can provide much higher speeds and also larger information carrying capacity.

COMPONENTS REQUIRED:

- Voltage Regulator 7805
- Diode IN4007
- MAX 232 Level Converter IC
- DB9 Connectors
- Capacitors 1uf
- optical fiber cable
- IR leds
- Photo diode

BLOCK DIAGRAM



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Description of Microcontroller:

- **89s52:** The AT89S52 is a low-power, high-performance CMOS 8-bit microcontroller with 8K bytes of in-system programmable Flash memory. The device is manufactured using Atmel's high-density nonvolatile memory technology and is compatible with the industry-standard 80C51 instruction set and pinout. The on-chip Flash allows the program memory to be reprogrammed in-system or by a conventional nonvolatile memory programmer. By combining a versatile 8-bit CPU with in-system programmable Flash on a monolithic chip, the Atmel's AT89S52 is a powerful microcontroller which provides a highly-flexible and cost-effective solution to many embedded control applications. The AT89S52 provides the following standard features: 8K bytes of Flash, 256 bytes of RAM, 32 I/O lines, Watchdog timer, two data pointers, three 16-bit timer/counters, a six-vector two-level interrupt architecture, a full duplex serial port, on-chip oscillator, and clock circuitry. In addition, the AT89S52 is designed with static logic for operation down to zero frequency and supports two software selectable power saving modes. The Idle Mode stops the CPU while allowing the RAM, timer/counters, serial port, and interrupt system to continue functioning. The Power-down mode saves the RAM con-tents but freezes the oscillator, disabling all other chip functions until the next interrupt

The hardware is driven by a set of program instructions, or software. Once familiar with hardware and software, the user can then apply the microcontroller to the problems easily.

The following are some of the capabilities of 8051 microcontroller.

- Internal ROM and RAM
- I/O ports with programmable pins
- Timers and counters
- Serial data communication

The 8051 architecture consists of these specific features:

- 16 bit PC &data pointer (DPTR)
- 8 bit program status word (PSW)
- 8 bit stack pointer (SP)
- Internal ROM 4k
- Internal RAM of 128 bytes.
- 4 register banks, each containing 8 registers
- 80 bits of general purpose data memory
- 32 input/output pins arranged as four 8 bit ports:P0-P3
- Two 16 bit timer/counters: T0-T1
- Two external and three internal interrupt sources
- Oscillator and clock circuits

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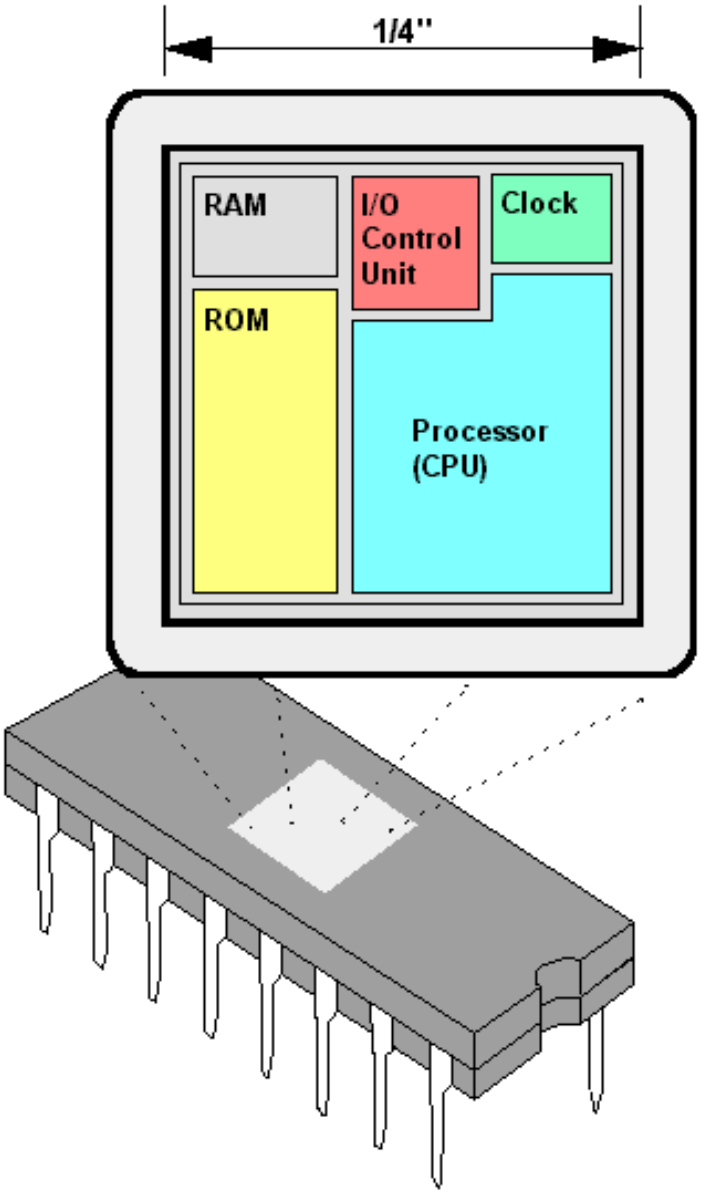

An entire computer on a single chip.

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9.Mini Project Suggestions
LASER/LED BASED VOICE COMMUNICATION SYSTEM

INTRODUCTION-:

This project is about laser based communication system, which is one of the form of optical communications system. It is used for inter and intra-building communication, closed circuit TV's, PC, LAN' etc. It is suitable for the study of optical communication (more specifically laser based communication). This project helps us for the better understanding of optical communication which has become a primary means of communication in the present world and gives us the view about free optics. It enables communication unto several hundred meters and this technology can be extended in future to enable longer distance communication by using parabolic light reflector.

CIRCUIT DESCRIPTION-:

- An optical communication system for such application can be built using the following three basic components:
- A light emmiting element, which could be a laser diode or light emmiting diode (LED).
- Transmission media such as optical fibre or free space.
- A light receiving element which could employ Avalanche Photodiode, PIN-Photodiode or any light sensor.
- Since the communication's performance of the system depends on the overall characteristics of the above elements, the characteristics of the individual elements should match.

BLOCK DIAGRAM

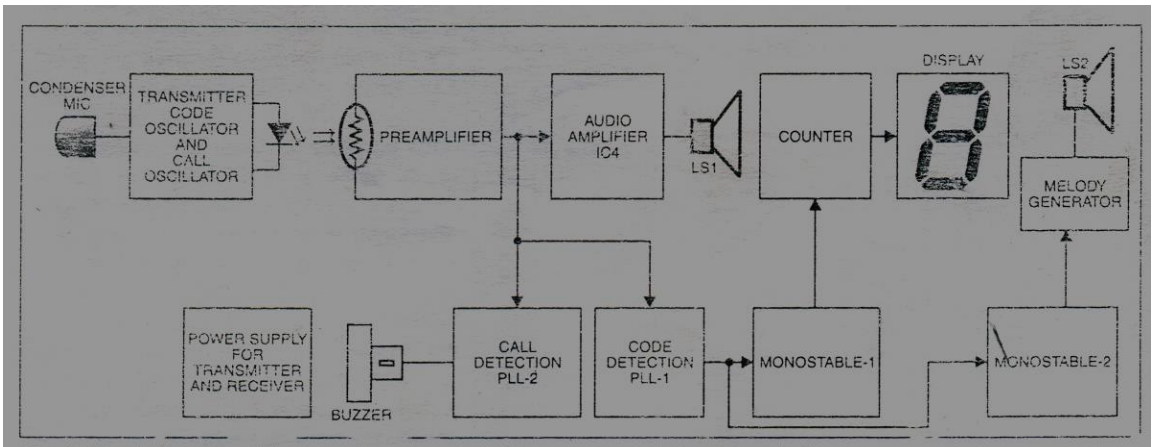

Fig. 2: Block diagram of the laser-based system for one-way speech communication

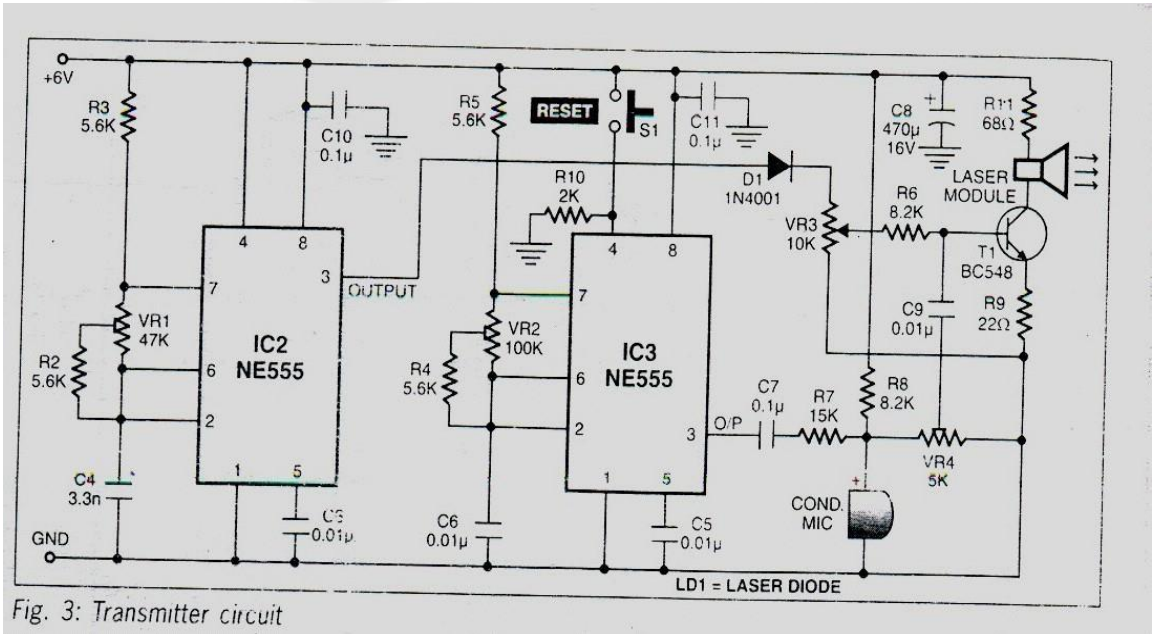

Fig. 3: Transmitter circuit

FEATURES:

The system has the following features:

- Communication is possible up to several hundred meters. The communication range can be extended up to several kilometers by using a parabolic audio.
- It transmits high quality audio.
- It ensures privacy.
- Alignment of transmitter and receiver is easy.
- Audio-visual indication of the interruption.
- The person at receiver is alerted of an impending audio message.
- Voice output from the microphone in transmitter is reproduced through a loudspeaker in receiver section.

ADDITIONAL FACILITIES:

- It also offers the following facilities:
- When someone intercepts the beam, the communication link breaks and the receiver circuit provides an audio-visual indication of the interruption by sounding an alarm and incrementing the count of a seven segment display.
- The person at the receiver end is alerted of an impending audio message through a buzzer sound by depressing a call switch at transmitter end.
- The voice output from the microphone in the transmitter is reproduced through a loudspeaker in the receiver section after suitable amplification.

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9..Mini Project Suggestions
Social Distancing with Automatic Door Control System Using Light

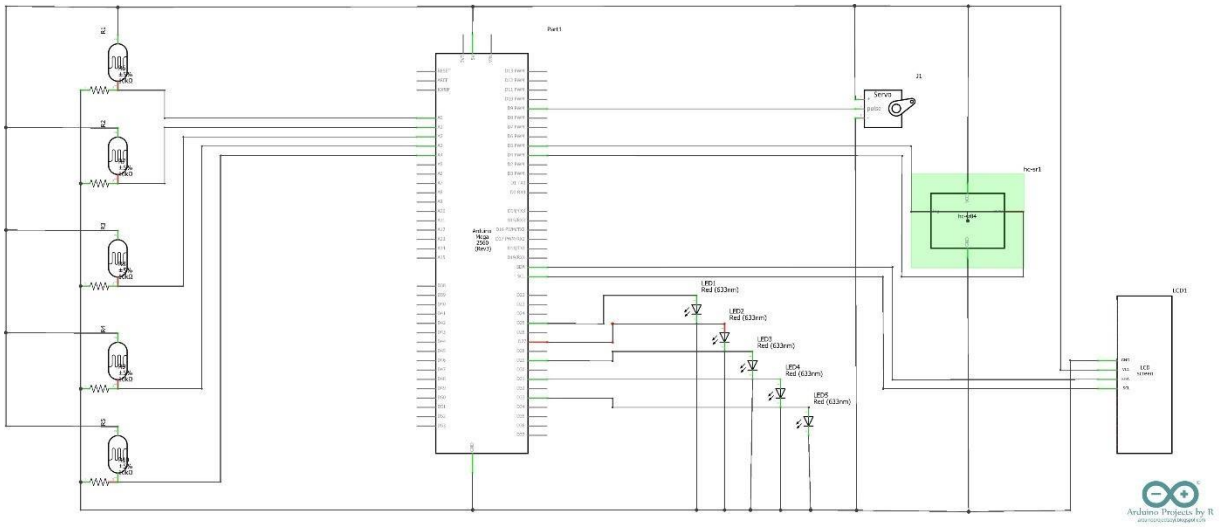
Coronavirus disease (COVID-19) is an infectious disease caused by the severe acute respiratory syndrome coronavirus 2 (SARS-Cov-2). COVID-19 has affected nearly 8.26 million people and has claimed the lives of 446, 135 people from all over the world. This disease affects different people in different ways. Some people may develop mild to moderate illness and recover without special treatment or hospitalization, while some may develop severe illness and even die. The incubation period of this virus is, on average, 5-6 days but it could also be up to 2 weeks. During this period, the person may not experience any symptoms but could still be contagious. The person will be a virus carrier and easily spread it when he/ she does not take any precautionary measures.

Coronavirus has spread rapidly and many countries have been affected by this pandemic. While some countries are recovering, other countries are still in lockdown and some are suffering from the second wave of coronavirus.

In this project, an innovative way of practicing social distancing with an automatic door control system is introduced. This is a prototype but it could also be used in real life situations; it would be helpful in many public places and in reducing the spread of coronavirus. It costed only around $10 (excluding the microcontroller board) to build this prototype.

In this project, people are supposed to stand in the appropriate positions which are at least 1 meter away from each other. This distance can be changed according to your preferences but it has to be more than 1 meter.

If the person does not stand in the proper place, the person will be notified and he/ she can follow social distancing properly. LCD display module is used to display the number of people standing in the queue and the occupied slots. If a slot is not occupied, that slot number will not be displayed in the LCD. An alarm system could also be added to this project to set off an alarm when the appropriate slot is not occupied and the person could then occupy the slot after looking at the LCD.



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9.Mini Project Suggestions
Laser Tripwire Alarm

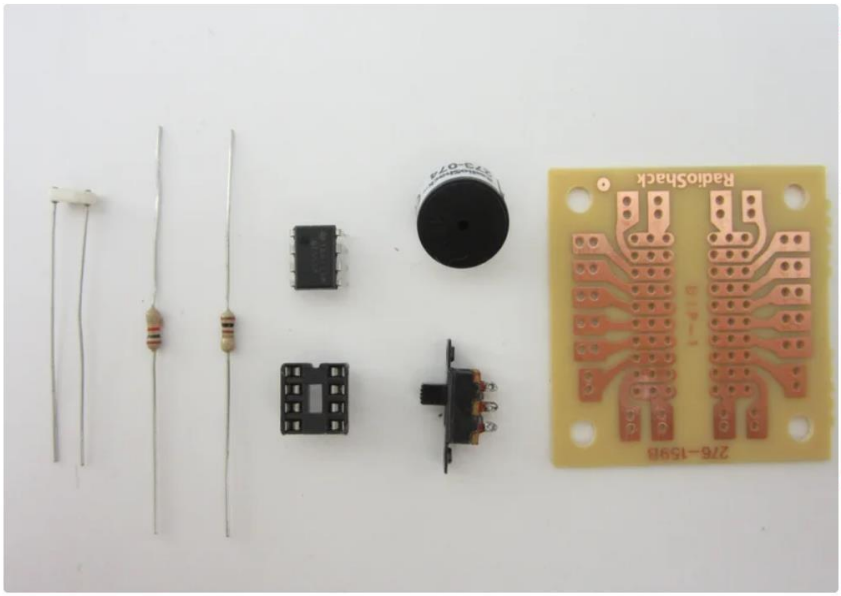
No security system is complete without lasers. So in this project I am going to show you how to build a laser tripwire alarm from a laser point, a couple of mirrors, and a few dollars of electrical parts. With this you can cover an entire house with an array of light beams. If any one of them is crossed it sets off your alarm. It can be a standalone alarm or it can be integrated into a larger DIY security system.

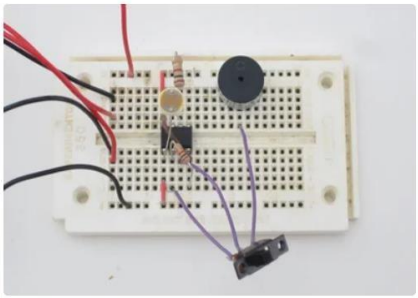
Apparatus Required
Laser Pointer
Printed Circuit Board
555 Timer IC
IC Socket (optional)
3-12 Volt Buzzer
Switch
CdS Photoresistor
2 resistors
3 AA Batteries
3 AA Battery holders
Jumper Wires
Heat Shrink Tubing

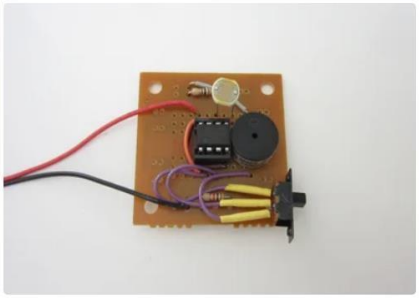
The light sensor that detects the laser is a photo resistor(R3). This is wired in series with standard fixed resistor (R2). These two resistors form a voltage divider that is used to activate the IC. The value of R2 should be approximately the same as the resistance of the photoresistor when you are shining the laser pointer directly at the light sensitive face. Because the output characteristics of photoresistors varies considerably from one to the next, you need to measure it with a multimeter. So connect the photoresistor to the multimeter and shine the laser pointer directly at it. In my case, its resistance was about 100 ohms. So I used a 100 ohm fixed resistor for R2.

When the light beam is interrupted, the resistance of the photoresistor increases dramatically. As a result, the voltage at pin 6 also increases and goes above the reference threshold. This causes the output pin 3 to go LOW and activates the alarm. To turn off the alarm and reset the system, a (single pole double throw) switch disconnects the speaker and sends the LOW signal from the output pin 3 to the trigger pin 2. The system is now deactivated. To reactivate it, flip the switch back to the original position. The alarm will remain off until the next time that the light beam is interrupted. The supply voltage can be anything from 4.5V to 18V. I chose to use 4.5V (three AA batteries) because this is the same voltage that is used by the laser pointer. This gives you the option of powering the laser pointer with the same battery pack as the alarm circuit.The resistor R1 acts as a pull-up resistor for pin 2. It helps to prevent false triggering from static electricity. This can be any value. In many cases it can be left off without causing any problems.

The alarm that I am using is a piezo buzzer. Any buzzer can work as long as it is rated to operate at the appropriate voltage.







The simplest way to set up your alarm is as a single beam tripwire. In this configuration the laser pointer is mounted to one side of the walk way and the alarm circuit is mounted to the other. For the tripwire to work, the laser pointer needs to be constantly on. The easiest way to accomplish this is by tightly wrapping a piece of tape around the button.

To secure the two pieces in place, you can use tape or a temporary adhesive putty such as Sticky Tack. First mount the alarm circuit in place. Then mount the laser pointer to the opposite side. Carefully adjust the position of the laser pointer so that it is pointed directly at the photoresistor.

Once you have the light from the laser pointer centered on the photoresistor, you are ready to arm the alarm. Flip the switch to connect the buzzer and activate the alarm. Whenever someone walks through the beam, the alarm will go off.

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Thank you

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